INTEGRATED SCHOOL SERVICES AND RECORD MANAGEMENT SYSTEM (ISSRMS) FOR SAINT JOSEPH SCHOOL OF MACTAN

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ABSTRACT

Saint Joseph School of Mactan currently uses a chip-based canteen payment process and separate workflows for library services, inventory, notifications, and authorized viewing of institutional identity and logging records. Students convert cash into chips before purchases, while staff maintain service information across distinct processes. The Integrated School Services and Record Management System (ISSRMS) aims to provide a secure, role-controlled platform that digitizes selected workflows while preserving existing institutional boundaries. The system will provide a canteen-only e-wallet, RFID-assisted identification, canteen sales and inventory, library circulation and independently managed fines, notifications, and read-only access to approved identity and logging records. Development will follow a local client-server architecture using a Student/Parent web portal, School Operations desktop application, React, Tailwind CSS, Electron, Node.js, and MySQL. Role-based access control, password hashing, server-side validation, audit logs, immutable double-entry ledger records, and ACID transactions will protect financial data and preserve integrity. The expected result is a tested prototype that reduces the chip-conversion step, streamlines selected school services, and maintains separation from institutional logging, official billing, Statement of Account, and library-fine wallet deductions. The system will be evaluated against approved functional and ISO/IEC 25010 quality requirements.

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CHAPTER 1
INTRODUCTION

1.1 Rationale of the Study

Saint Joseph School of Mactan serves both elementary and high school students, including younger students who are not expected to carry physical money for their daily purchases. To address this, the school currently uses a chip-based payment system in its canteen. Students first convert or purchase physical money into chips and then use these chips to pay for their purchases. While this allows students to make purchases without directly carrying cash, the process introduces an additional step between the student's funds and the actual purchase.

The existing process therefore creates redundancy in canteen transactions, as students must first acquire chips before they can purchase an item. The school seeks a more streamlined cashless process that can retain the purpose of the chip system allowing students, particularly elementary students, to make purchases without carrying physical cash while eliminating the need for physical chips.

The school already has an RFID system that can be used as the basis for student identification. By connecting the existing RFID identification to a digital e-wallet, students can be identified during transactions and use their available digital balance directly for canteen purchases. This removes the intermediate step of converting funds into physical chips while maintaining a cashless purchasing process.

Based on this need, the researchers propose the Integrated School Services and Record Management System (ISSRMS), which uses the school's existing RFID system as part of its student identification and canteen e-wallet functionality. In addition to the streamlined canteen payment process, ISSRMS incorporates selected school services such as library management, inventory

management, notifications, and authorized access to institutional records within a role-controlled platform.

1.2 Statement of the Problem

Saint Joseph School of Mactan currently handles selected school services through separate manual or semi-manual workflows. Canteen payments use physical chips; canteen financial records are maintained separately; and library borrowing, returns, inventory, and fines are recorded through a separate process. The school's RFID system already produces authorized identity and entry/exit logging records, but these institutional sources remain separate read-only references for the proposed platform.

These separate workflows make it harder to track canteen balances and purchase history, verify stock and circulation records, and provide authorized personnel with a consistent view of service information. Physical chips can also be lost or damaged, while manual library records require additional time to maintain and check.

To address these operational gaps, the study proposes ISSRMS: a role-controlled platform that digitizes the canteen e-wallet and inventory workflows, library circulation and fine management, notifications, and authorized read-only viewing of institutional identity and logging records. The platform will keep library fines separate from the canteen financial ledger and will not modify institutional logging, billing, or Statement of Account records.

1.2.1 General Objective

The general objective of this study is to design, develop, and evaluate the Integrated School Services and Record Management System (ISSRMS) for Saint Joseph School of Mactan. The system aims to provide secure, role-controlled management of selected canteen, financial-services, and library workflows, together with authorized read-only access to institutional identity and logging records.

1.2.2 Specific Objectives

Specifically, this study aims to:

1. Gather and analyze the functional and operational requirements needed for the development of the Integrated School Services and Record Management System (ISSRMS).
2. Design an integrated school-service management system that consolidates the school's financial, canteen, and library services into a unified platform.
3. Develop the core modules that constitute the system based on the identified requirements and design:
    1. Financial Module
    2. Canteen Module
    3. Library Module
4. Provide authorized read-only access to RFID identity and existing gate and library logging records for identification and record viewing, without modifying institutional source data.
5. Test and validate the developed ISSRMS to ensure that the financial, canteen, and library modules function accurately and reliably.
6. Evaluate the developed system in terms of functionality, usability, reliability, security, and performance.

7. Deploy the Integrated School Services and Record Management System (ISSRMS) within the school's authorized environment for use by authorized school personnel.

1.3 Significance of the Study

School Administration. The study may provide a clearer and more auditable view of selected school-service transactions while preserving the boundaries of existing institutional systems. Its isolated operational database and role controls can support accountability without granting ISSRMS write access to legacy entry/exit logging, or official accounting records.

Registrar. The Registrar may process confirmed cash top-ups, manage approved canteen credit limits, perform verified manual settlement of outstanding canteen credit, create certified adjustments or reversals, and review detailed wallet and canteen transaction activity. The Registrar may also view unpaid library-fine information and the related circulation activity for reference only. Library-fine assessment, adjustment, reduction, waiver, and settlement remain restricted to authorized Library Personnel.

Canteen Personnel. Authorized Canteen Personnel may process cashier-assisted canteen purchases, manage permitted canteen products and inventory, and perform confirmed over-the-counter cash-only wallet top-ups. They may not change approved credit limits, manually settle outstanding canteen credit, create adjustments or reversals, edit finalized financial records, or manage library fines.

Library Personnel. Separate circulation, inventory, and fine-management functions may improve the monitoring of borrowed materials and penalties. Authorized staff may update, reduce, waive, or settle operational fines without affecting a student’s canteen wallet balance.

Students and Parents. Role-restricted access provides a convenient view of authorized canteen wallet activity, library circulation records, and institutional logging history while keeping sensitive source data read-only.

School IT/Operations Personnel. The documented role controls, audit trails, backup procedures, and approved integration boundaries may support secure deployment, monitoring, and maintenance of ISSRMS.

Future Researchers and Developers. The documented requirements, architecture, and security boundaries may serve as a reference for future improvements without expanding ISSRMS into official billing or Statement of Account processing or write access to institutional sources.

1.4 Scope and Limitations

In-Scope

- RFID-assisted identity verification using existing school-approved cards and compatible readers.
- Read-only reference access to approved institutional identity data and Gate Entry, Gate Exit, Library Entry, and Library Exit logging records.
- The proposed closed-loop canteen e-wallet for confirmed cash top-ups, canteen purchases, approved credit, authorized manual credit settlement, and certified adjustments or reversals.
- Canteen sales, product, and inventory recording.
- Library catalog, circulation, inventory, and independently managed fine amounts and fine statuses, including authorized fine assessment, adjustment, reduction, waiver, and settlement.
- Role-based access for Students/Parents, the Registrar, Canteen Personnel, Library Personnel, and the Administrator.
- A Student/Parent web portal and a School Operations desktop application with controlled local USB RFID-reader communication.
- An isolated ISSRMS operational database with server-side validation, audit logging, ACID transaction handling, immutable balanced ledger entries, and approved backup and recovery controls.

Out-of-Scope

- Editing, deleting, or writing to the school's institutional registry or existing entry/exit logging systems.
- Generating, modifying, querying, consolidating, exporting to, or integrating with the school's official Statement of Account, billing, or accounting systems.
- Processing official school debts through ISSRMS; those debts remain subject to the school's separate process.
- Automatically deducting library fines from the canteen wallet or recording library fines in the canteen financial ledger.
- Using the e-wallet for anything outside approved canteen transactions and their authorized adjustments or reversals.
- External electronic top-up channels, including GCash.
- Replacing the school's existing RFID cards, institutional registry, or logging infrastructure.

Limitations

- Implementation depends on approved read-only access to accurate institutional identity and logging data.
- Deployment depends on compatible RFID cards and USB readers, a functioning local network and server, and ISSRMS-managed account provisioning.
- Portal access depends on the school-approved HTTPS, web-address, firewall, and reverse-proxy configuration.
- Zone 1 integration and Zone 2 operational services must remain within the secured school network and must not be directly exposed to the Internet.
- Final evaluation is limited to the users, records, devices, locations, policies, and evaluation period approved for the capstone deployment.

CHAPTER 2
REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presents the review of related literature and existing systems that provide the theoretical and technological foundation for the proposed Integrated School Services and Record Management System (ISSRMS). The reviewed works cover concepts essential to the platform, including educational management information systems, Radio Frequency Identification (RFID), localized cashless payment architectures, automated library circulation, financial information systems, and integrated information systems.

Furthermore, this chapter contains descriptions of existing software systems and related applications that are relevant to the proposed project. By discussing the specific features of these existing platforms that the proponents intend to replicate, modify, or improve, this review establishes a clear baseline for what is to be expected in the final system design and justifies the technical approach utilized in the succeeding chapters.

School Management Information Systems (SMIS)

The School Management Information System (SMIS) is designed to improve the efficiency and effectiveness of school administration by centralizing and organizing institutional data. According to Telem and Avidov (1995), implementing an SMIS improved the quality of school reports by making them more accurate, current, accessible, and easier to retrieve while reducing reliance on manual paperwork. Similarly, Abulencia et al. (2022) found that SMIS platforms improve report quality by providing accurate data and supporting comparisons and routine administrative reporting. These capabilities enable schools to make informed operational decisions and improve short-term and long-term performance. Pradana et al. (2026) further reported that Management Information Systems improve

administrative efficiency by increasing data accuracy, speeding report generation, and promoting accountability. Integrated platforms also simplify workflows, reduce duplicate documentation, and improve institutional transparency.

Collectively, these studies demonstrate that School Management Information Systems improve administrative efficiency by centralizing institutional data, enhancing report accuracy, streamlining administrative workflows, and supporting evidence-based decision-making. These findings support the development of the proposed system by emphasizing the value of integrating multiple school services into a centralized platform.

Radio Frequency Identification (RFID) Technology

Radio Frequency Identification (RFID) has become an essential technology in educational institutions due to its ability to automate identification, monitoring, and transaction processes. RFID utilizes radio waves to identify individuals through unique tags or cards, allowing schools to improve operational efficiency while reducing manual procedures.

Nivetha et al. (2020) indicates that RFID-based attendance systems significantly reduced the time required for attendance, minimized human errors, and prevented fraudulent attendance recording. RFID technology likewise contributes to the overall improvement of school management. Shaibu et al. (2019) found that RFID cards to authenticate students for identification and academic monitoring improved accuracy while reducing impersonation and simplifying student verification which enhances school security.

Another emerging application of RFID technology is cashless payment within educational institutions. Widhyaestoeti et al. (2023) found

that RFID-based cashless shortened transaction time, reduced payment queues, improved transaction recording, and minimized direct cash handling.

Collectively, these studies demonstrate the versatility of RFID technology in educational institutions. Whether applied to attendance monitoring, student identification, school management, or cashless payment systems, RFID consistently improves efficiency, accuracy, security, and convenience.

School Cashless Payment

Cashless payment systems have become increasingly adopted in educational institutions to streamline financial transactions and improve operational efficiency. Previous studies have consistently reported their benefits in terms of convenience, security, and transaction speed. According to Ya'acob et al. (2019) RFID-based cashless transaction systems enhance financial transparency and reduce the risks associated with manual cash handling. Similarly, Espinosa et al. (2024) emphasizes the practicality of RFID systems in environments with unstable internet connectivity, showing that RFID-based payment solutions can still function efficiently in real-world campus conditions. Expanding the concept further, Abhiram et al. (2024) developed RADIT, a student digital wallet system that integrates RFID technology for financial transactions and campus services. It improves convenience by allowing students to manage payments through a single ID card.

Cashless campus initiatives have become increasingly adopted by educational institutions to support faster, safer, and more efficient financial transactions. Existing literature highlights their positive impact on operational efficiency and the overall campus experience.

According to Sawant et al. (2017) cashless environment reduces administrative burden, improves financial tracking and increases transaction efficiency across campus services.

Collectively, these studies show that RFID-based cashless systems, whether implemented as payment platforms, digital wallets, or full campus ecosystems, consistently enhance efficiency, accuracy, and convenience.

Library Management System

Library Management Systems have significantly evolved with the integration of RFID technology, enabling faster, more accurate, and automated library operations. Across various studies, RFID has been shown to improve borrowing, returning, inventory tracking, and fine management processes in academic libraries.

In terms of borrowing processes, Kaur and Singh (2019) an RFID-based library system reduces queueing time, minimizes human errors, and ensures accurate transaction recording, making borrowing more efficient for both students and library staff. Similarly, RFID technology improves the efficiency of book returns. Patel et al. (2018) found that automated return systems using RFID allow real-time updating of library records as soon as books are returned.

In addition, RFID technology enhances fine management systems. Sharma and Verma (2019) highlights that automation improves transparency, reduces disputes, and ensures consistent enforcement of library policies.

Collectively, these studies show that RFID-based Library Management Systems improve operational efficiency across all major

functions of libraries. From borrowing and returning to inventory tracking and fine computation, RFID technology streamlines processes, reduces human error, and enhances the overall user experience in academic libraries.

Financial Information System

A Financial Information System (FIS) plays a critical role in educational institutions by providing a centralized platform for managing financial transactions, maintaining digital financial records, and generating accurate financial reports. As schools continue to adopt digital technologies, financial information systems have become essential in ensuring transparency, accountability, and efficiency in financial operations.

Arifudin (2019) developed a School Financial Information System that replaced spreadsheet-based financial management with a centralized system that improved data accuracy, simplified transaction monitoring, sped up report generation, and allowed parents and administrators to track financial activities. The system also supported electronic payments and automated billing reminders, showing the benefit of integrating financial processes into one platform.

Wulandari et al. (2025) developed a web-based financial recording system that automated cash inflows and outflows, improved accuracy in financial records, and made report generation faster and more organized. The system also improved transparency and reduced errors in manual financial documentation.

Collectively, these studies show that Financial Information Systems centralize school financial processes such as payments, record-keeping, and reporting. They improve accuracy, reduce manual work, and support real-time monitoring of financial transactions within educational institutions.

Firewall and Demilitarized Zone (DMZ) Security

Firewalls and Demilitarized Zones (DMZ) are widely used network security mechanisms that protect institutional systems by controlling and filtering traffic between trusted internal networks and untrusted external networks such as the internet. A firewall regulates which connections are allowed to pass, while a DMZ places publicly accessible services on an isolated network segment so that, even if compromised, an intruder cannot directly reach the internal network.

Selvi et al. (2014) applied the DMZ concept within firewall technology to secure an online examination system, and found that the architecture improved the monitoring and control of network traffic while restricting unauthorized access from outside candidates. This shows how DMZ-based firewall design can protect systems that handle sensitive academic transactions from external threats. Similarly, Iskandar et al. (2018) implemented a DMZ method on a campus academic information system (Siakad server) using a Mikrotik router, and found that the setup successfully secured internal web-based services from unauthorized crackers while still allowing controlled access for legitimate users.

Collectively, these studies demonstrate that combining firewall technology with DMZ architecture allows educational institutions to expose only the necessary public-facing services while shielding internal databases and records from direct external access. This layered security approach directly supports the secure operational-services component of the proposed system by protecting student, financial, and administrative data from unauthorized network-level intrusion.

Integrated Information Systems

Integrated Information Systems (IIS) are essential in modern educational environments as they unify multiple standalone systems into a single, cohesive platform. These systems enable institutions to manage academic, administrative, financial, and operational data more efficiently, improving both communication and decision-making processes.

According to Laudon and Laudon (2020), Integrated Information Systems consolidate various organizational functions into one system, allowing real-time data sharing across departments. This integration reduces data duplication, improves consistency, and enhances the overall efficiency of institutional operations. In educational settings, this allows schools to manage student records, attendance, and academic data in a more organized and reliable manner.

Similarly, centralized information systems further strengthen data management by storing all institutional data in a single database. Baskerville and Myers (2019) explain that centralized systems improve data accuracy, increase security, and allow easier access to real-time information for decision-making.

In terms of resource management, integrated systems such as Enterprise Resource Planning (ERP) platforms support efficient allocation of institutional resources. Al-Mashari and Zairi (2020) found that integrated resource management systems enhance operational efficiency by unifying academic, financial, and administrative processes into one system. This leads to reduced operational costs, improved planning, and better organizational control.

Collectively, these studies highlight the importance of Integrated Information Systems in unifying institutional processes through centralized data management. By integrating academic, financial, and operational functions into a single platform, these systems improve efficiency, consistency, and organizational decision-making, supporting the integrated architecture proposed in this study.

CHAPTER 3
TECHNICAL BACKGROUND

This chapter presents the technical background of the proposed Integrated School Services and Record Management System (ISSRMS). It discusses the overall technical architecture, major system components, and technologies used to develop the platform. The chapter explains how the modules interact to support secure campus transactions and accurate record keeping.

3.1 Overview of the Proposed System

The proposed Integrated School Services and Record Management System (ISSRMS) is designed to enhance student services and resource management at Saint Joseph School of Mactan. The system provides modules for library management, canteen operations, electronic wallet services, inventory management, and RFID-based identification. The platform retrieves authorized identity information through a school-approved read-only integration mechanism and references authorized institutional logging records. It modernizes campus administration by digitizing workflows currently managed through manual methods. Furthermore, the platform introduces an automated layer for internal campus transactions, including an electronic wallet system, built entirely separate from the school's existing financial and accounting records.

To manage these operations efficiently, user access is divided between two frontend applications that communicate with a shared on-premises backend service. Students and parents use the web-based Student/Parent Portal to monitor wallet balances, track library histories, and view authorized Institutional Logging Records. Campus personnel use the School Operations desktop application on approved school workstations to manage authorized transactions and communicate with locally connected RFID readers. The application reuses the React and Tailwind CSS interface within an Electron wrapper that provides controlled access to the local RFID hardware interface. Role-Based Access

Control regulates the features available to each logged-in account. For example, Canteen Personnel may access point-of-sale functions, while Library Personnel may access circulation and cataloging tools. This dual-client configuration reduces administrative redundancy, separates staff responsibilities, and supports centralized data management.

3.2 System Architecture

The proposed system follows a segmented centralized client-server architecture. Students and parents access the React web portal through a school-approved HTTPS connection. External requests pass through the school's perimeter firewall and a reverse proxy or web server before permitted API requests are forwarded to the internal Node.js backend.

Authorized school personnel use the School Operations desktop application through the internal school network. The application uses React within an Electron wrapper, which provides controlled local access to supported USB RFID readers.

The Node.js backend performs authentication, authorization, business-rule validation, transaction processing, audit logging, and access to the two data zones. Zone 1 contains the institutional identity and logging sources and remains read-only. Zone 2 contains the ISSRMS operational database and is accessed through a separate least-privilege read/write connection.

Client applications shall not connect directly to either data zone. The backend, databases, RFID interfaces, and administration services shall remain inside the secured school network and shall not be directly exposed to the Internet.

3.3 RFID Technology and Hardware Interface

The system uses Radio Frequency Identification (RFID) technology to identify students and facilitate access to various school services. Each student is assigned a unique RFID card linked to their student account within the system. When an RFID card is scanned, the backend validates the student's identity and

authorizes the requested transaction based on the user's role and the selected module.

RFID identity verification is used for student identity verification in library transactions, canteen purchases, and supported electronic-wallet operations. Existing Gate Entry and Gate Exit records from Gate 1 and Gate 2, together with Library Entry and Library Exit records from the Library, are referenced from the school's separate logging infrastructure and are not created or modified by ISSRMS.

All RFID transactions are validated by the centralized backend before any operation is processed. This approach ensures consistent authentication, prevents unauthorized access, and maintains accurate transaction records across all modules.

Communication between the physical RFID reader and the application relies on hardware-data parsing. When an RFID card is tapped, the locally connected reader transmits raw identifier data, which may be formatted as hexadecimal characters. The School Operations desktop application receives the reader output through its controlled Electron hardware interface, normalizes it according to the approved reader protocol, and sends the resulting identifier to the authenticated Node.js backend. The backend then verifies the identifier through the authorized institutional identity source before permitting the requested digital transaction.

3.4 User Authentication and Account Management

The proposed system implements Role-Based Access Control (RBAC) to regulate access to system functions according to user authority. Each account is mapped to a specific role—Student/Parent, Library Personnel, Canteen Personnel, Registrar, or Administrator—which restricts application access to the modules permitted for that role. The Registrar is the authorized actor responsible for operating the Financial Services Module. The Administrator role is limited to platform-administration functions: provisioning and deactivating Zone 2 accounts, assigning approved role permissions, managing announcements and approved

non-financial settings, viewing authorized Institutional Logging Records in read-only form, and reviewing read-only system-wide Audit Logs. The Administrator may not execute financial, canteen, or library operational transactions; every Administrator action remains subject to backend authorization, validation, and auditing.

Accounts utilize the ID number as the username to associate the account profile directly with the existing school record. Since the school's existing database does not maintain authentication credentials, the proposed system stores and manages user credentials independently within its own database layer. Each account record tracks the username, securely hashed credentials, assigned role permissions, and active account status flags.

During initial system deployment, the backend executes an automated batch-import utility to provision user accounts. This utility retrieves authorized active ID numbers and corresponding contact data through the school-approved read-only integration mechanism. For each imported record, the utility automatically generates a temporary password and computes its secure cryptographic hash using algorithms such as Argon2 or bcrypt for storage.

Simultaneously, during initial account provisioning, the system shall generate a temporary password for each authorized account and store only its secure cryptographic hash. The temporary credentials shall be delivered to the registered contact information through a school-approved secure communication method, which may include the school’s existing SMS infrastructure if technical access and authorization are provided. Users shall be required to change the temporary password upon their first successful login. Account recovery and forgot-password requests follow an identical channel where the system validates the submitted ID number as the username, generates a time-sensitive verification code, and utilizes the existing SMS infrastructure to deliver the reset code to the verified contact number on file to authorize secure credential modifications.

3.5 Institutional Logging Records Reference Module

The Institutional Logging Records Reference Module treats Gate Entry and Gate Exit logging at Gate 1 and Gate 2 and Library Entry and Library Exit logging at the Library as existing school infrastructure. ISSRMS does not create, edit, delete, mark, or recompute source logs or replace the school's logging systems. It provides only authorized, read-only access to permitted institutional logging records.

When a Student/Parent or authorized staff member requests logging history, the backend verifies the account and record-level permissions, retrieves the approved records through the school-approved read-only integration mechanism, and displays the permitted log type, location, date, and timestamp. Approved log types are Gate Entry, Gate Exit, Library Entry, and Library Exit. Approved locations are Gate 1, Gate 2, and Library.

3.6 Financial Services Module

The Financial Services Module manages canteen-only e-wallet transactions by providing cash top-ups, purchase processing, approved-credit tracking, authorized manual credit settlement, certified adjustment or reversal transactions, and immutable double-entry journal recording. It integrates with the Student/Parent Portal, the Canteen Management Module, and the School Operations desktop application. It does not calculate, store, consolidate, or transfer library fines or official school debts.

Each student is assigned a student E-Wallet linked securely to their system account, allowing funds to be used for supported canteen transactions. The financial data zone also maintains controlled system-owned wallets for Cash Collection, Canteen Revenue, Credit Receivable, and Adjustment Reserve. Every finalized financial transaction has one immutable transaction header and at least two balanced ledger-entry lines. The E-Wallet supports confirmed over-the-counter cash top-ups processed by the Registrar or authorized Canteen

Personnel. Canteen Personnel are limited to cash-only top-up processing and may not manage credit limits, settlements, adjustments, or reversals.

To accommodate institutional policies, the module supports pre-approved credit limits for authorized students, explicitly tracking any consumed credit as outstanding canteen credit to maintain clear financial liability.

The module validates every canteen financial transaction before creating its immutable transaction header and balanced ledger-entry lines, and before updating credit usage, sales records, and inventory movements. Library fines are calculated, adjusted, waived, and settled only within the Library Management Module and are never deducted from the E-Wallet. The school's official debts, billing records, and Statement of Account remain in the separate institutional financial process; ISSRMS does not generate, modify, query, consolidate, or integrate with them.

To ensure absolute database consistency and security, all internal financial operations are executed within database transactions governed by strict ACID (Atomicity, Consistency, Isolation, Durability) properties, preventing partial updates or data corruption in the event of hardware or network processing failures. The module uses an immutable Financial_Transactions header and Financial_Ledger_Entries lines. For every finalized transaction, the sum of debit entries must equal the sum of credit entries; a transaction with fewer than two balanced lines is rejected.

Students and parents may view the authorized student's active wallet balance, canteen credit used, and transaction history through the Student/Parent Portal. Confirmed cash-only top-ups may be processed by the Registrar or authorized Canteen Personnel. Credit-limit management, manual settlement, adjustments, reversals, and complete financial-record review remain restricted to the Registrar. The Registrar may inspect the itemized canteen transactions that contributed to outstanding canteen credit, including purchased items, quantities, purchase total, cash-funded portion, credit-funded portion, operator, date and time, sale reference, and financial transaction reference.

To prevent financial fraud and maintain audit integrity, the system prohibits direct editing or deletion of finalized financial transactions. A canteen or wallet error is corrected only through a certified adjustment or reversal that links to the original record and leaves a permanent audit trail. Library fines may be adjusted, waived, or cleared only within the Library Management Module. At every grading or payment period, the Registrar may execute the manual Settle-to-Zero action for canteen credit_used only after verifying, through an official receipt, payment reference, or other school-approved record, that the outstanding canteen credit has been paid. The action records the actor, timestamp, amount, reference, and resulting wallet state.

The Registrar may view unpaid library fine amounts and statuses for informational purposes only. The Administrator has no library-management write permission. Assessment, adjustment, reduction, waiver, and settlement are restricted to authorized Library Personnel.

The following examples show how one canteen financial event is represented by two equal accounting entries and recorded together by the backend.

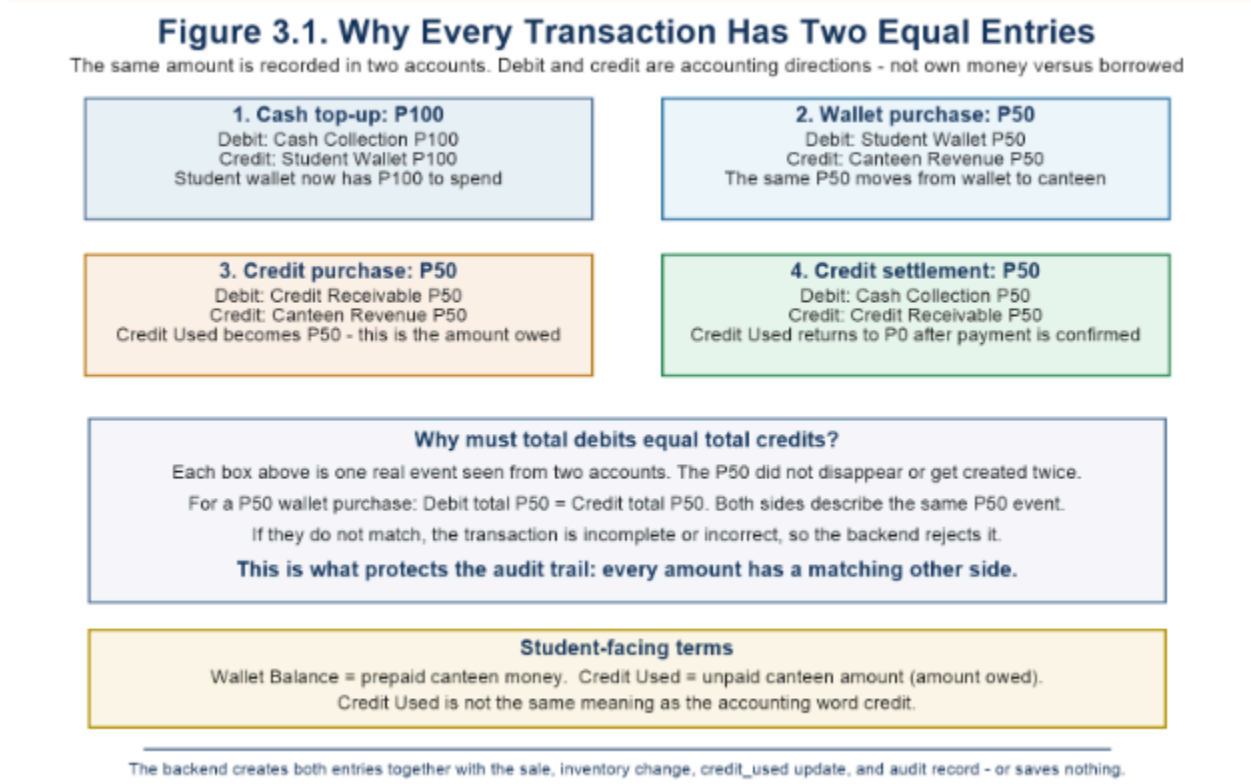


Figure 3.1. Double-Entry Financial Example

The Financial Services Module provides secure backend services for payment validation, balance verification, and transaction processing. Other system modules communicate with this centralized service to process financial transactions, ensuring that payment validation logic remains secure, isolated, and standardized across the entire ecosystem.

3.7 Mathematical Modeling of the Financial Module

Financial behavior is defined by explicit rules that the backend applies inside ACID database transactions. These rules cover only the canteen e-wallet and its financial ledger.

Table 3.7. Financial E-Wallet Mathematical Model

Formula	Meaning	Validation / Business Rule
$B = T_{total} - P_{total}$	Active spendable cash; T_{total} is confirmed top-ups and positive adjustments; P_{total} is cash-funded canteen purchases after valid reversals.	Derived from immutable ledger records.
$C_{avail} = C_{limit} - C_{used}$	Remaining approved credit; C_{limit} is the credit ceiling and C_{used} is unsettled canteen credit.	Store credit_limit and credit_used separately.
$P \leq B + C_{avail}$	P is the total canteen purchase amount.	Approve only when the purchase does not exceed cash plus available credit.
$Cash_{Used} = \min(P, B)$	Portion of the purchase paid from active cash.	Consume active cash first.
$Credit_{Added} = P - Cash_{Used}$	Uncovered portion of the purchase.	Only the uncovered portion increases credit_used.
$C_{used(new)} = C_{used(old)} + Credit_{Added}$	Credit balance after the purchase.	Apply after authorization, balance, limit, and ledger checks pass.
$C_{used(new)} = 0$	Canteen credit after verified manual settlement.	Registrar authorization and approved payment evidence are required; create a new auditable

		transaction.

3.8 Library Management Module

The Library Management Module digitizes the school's library operations by providing a centralized platform for catalog management, circulation, and book inventory management. The module is accessible through the School Operations desktop application for Library Personnel and through the Student/Parent Portal for students and parents.

The module maintains a centralized library catalog containing bibliographic information, book availability, categories, circulation status, and inventory records. Library Personnel can register new books, update catalog information, process borrowing and return transactions, monitor circulation activities, and manage the library inventory through the desktop application. Students and parents may browse the library catalog, search for available resources, monitor active borrowings, and view borrowing history through the Student/Parent Portal.

Borrowing and return transactions are associated with the student's account via RFID identity verification or through manual student ID number entry by authorized library personnel, ensuring continuous operations if a physical card is lost, damaged, or unavailable. The backend validates each transaction before updating circulation records, book availability, and inventory status, ensuring that all library records remain synchronized and consistent. All library activities are recorded within the operational database to support auditing and historical record keeping.

Library fines and penalties are generated and managed within the Library Management Module. They remain operationally isolated from the canteen e-wallet, Financial_Transactions, and Financial_Ledger_Entries.

The Registrar may view unpaid library fine amounts and statuses for informational purposes only. Authorized Library Personnel may assess, adjust, reduce, waive, or settle library fines.

3.9 Canteen Management Module

The Canteen Management Module provides an RFID-integrated, cashier-assisted Point-of-Sale (POS) system for managing canteen sales, product inventory, and transaction records. The module is accessible through the School Operations desktop application and is designed to streamline daily canteen operations while supporting the ISSRMS electronic wallet.

Students are identified through RFID identity verification or via manual student ID number entry by the cashier, allowing the system to retrieve the corresponding student profile and account status before processing purchases. This manual input capability ensures uninterrupted checkout service in the event of lost, damaged, or unavailable physical RFID cards. Cashiers complete transactions through a touch-based POS interface by selecting products from the available inventory. The backend validates product availability and the selected payment method before authorizing the transaction. Authorized Canteen Personnel may also process confirmed over-the-counter cash-only wallet top-ups. The operator identifies and confirms the student account through RFID or manual ID, records the cash amount received, confirms the transaction, and receives a unique transaction reference. Canteen Personnel may not change credit_limit, settle credit_used, create adjustments or reversals, or edit finalized financial records. GCash and other external electronic-loading methods are outside the current scope.

The module maintains its own product inventory, enabling authorized personnel to manage product information, monitor stock levels, update product availability, and record inventory movements resulting from sales or inventory adjustments. Inventory records are automatically synchronized with sales transactions to ensure accurate stock monitoring.

Payment processing is handled by the Financial Services Module, which derives the active cash balance from balanced immutable ledger-entry lines, validates approved credit, and verifies the supported payment method before authorizing purchases. Upon successful payment authorization, the backend records the sale, updates product inventory, and stores the transaction history within the

operational database. Database transactions are utilized to ensure that sales, inventory updates, and transaction records remain consistent in the event of processing failures.

3.10 Notifications Module

The Notifications Module is responsible for managing internal application reminders and system status updates. It generates non-disruptive notifications displayed within the web-based Student/Parent Portal and the School Operations desktop application upon user login. The module focuses primarily on core system events, including library reminders for upcoming book return deadlines or accrued fines, and financial alerts for low electronic wallet balances or cash top-up confirmations.

Because institutional logging belongs to separate existing school infrastructure, source-system logging notifications remain outside this application. ISSRMS does not process or transmit source logging alerts; it only displays authorized logging records retrieved through the school-approved read-only integration mechanism. Relational mapping in the operational database connects application alerts, such as library due-date reminders and wallet notifications, to the appropriate user account. Notification logs are retained for administrative audit and review.

3.11 Home and Announcements Module

The Home and Announcements Module serves as the primary landing page of the proposed system immediately following user authentication. It provides users with centralized access to critical announcements, institutional news, upcoming campus events, and administrative updates across the system interfaces. The module aggregates communication streams from two distinct sources to provide a comprehensive information feed. First, it utilizes an internal administrative channel where authorized school administrators and staff can publish announcements directly through the system interface. To manage this

flow of information, the backend applies Role-Based Access Control rules to filter and disseminate these internal notices to users based on their designated roles and permissions. Second, the module may incorporate an external campus feed that displays public announcements and posts from the school's official Facebook Page through a school-approved and technically supported embedding method. If implemented, the client applications shall use the school's internet connection to load the external content separately from the system's local transactions and operational database. The integration shall be subject to school approval, platform availability, network access, and applicable Facebook embedding requirements.

3.12 Database Architecture

The system implements a two-zone data architecture that preserves strict isolation. Zone 1 contains the existing institutional identity and logging sources and is accessed only through one school-approved read-only integration mechanism. Depending on the existing architecture and access approved by the school, the selected mechanism may use authenticated read-only API endpoints, including an approved adapter API, or a dedicated database connection restricted to approved SELECT operations. Only one method shall be selected and implemented after school-approved technical validation; the two methods are not simultaneous requirements. Zone 2 is the separate ISSRMS operational MySQL database and is accessed through a distinct least-privilege read/write connection.

The first component consists of the existing institutional identity and logging sources, which is treated as a strictly read-only data source by the system backend. It serves as the authoritative registry for student and staff identities, containing primary fields such as the official ID number and general profile demographics. The proposed platform does not modify, alter, or inject new tables into this pre-existing infrastructure.

The second component is a newly established, centralized MySQL database built specifically for the proposed platform. It maintains all application-specific

operational records, including securely hashed user credentials, student and system-owned wallet configuration, immutable financial transaction headers and ledger-entry lines, canteen product inventories, notification logs, and library circulation records.

Because a cross-database foreign key is intentionally absent, the backend validates the Zone 1 identity before creating or changing any Zone 2 record that requires an account reference. For a library borrowing request or canteen purchase, it verifies through the approved Zone 1 integration mechanism that the supplied ID number exists and is authorized before it performs the Zone 2 operation. If verification fails, the request is rejected. Zone 2 constraints, transactions, audit records, and periodic reconciliation provide operational integrity without modifying the institutional database; this logical reference does not claim database-engine enforcement across the two zones.

3.13 Security and Access Control

To ensure comprehensive data security, system integrity, and user privacy, the proposed system implements strict authentication mechanisms and Role-Based Access Control. Access to all system functionalities is explicitly gated by the backend API based on the verified role assigned to each user account. This authorization framework ensures that the interface and available features adapt dynamically to the specific permissions of the logged-in account, preventing unauthorized users from accessing restricted administrative modules or operational records.

Because access to the existing institutional identity and logging sources is strictly read-only and does not manage application credentials, the proposed platform handles authentication independently. All user accounts maintain credentials securely isolated within the new operational MySQL database. Clear-text passwords are never stored by the system. The backend enforces cryptographic hashing protocols, utilizing robust algorithms such as bcrypt or Argon2, to convert and store credential values. This one-way cryptographic conversion

ensures that user passwords remain protected against unauthorized extraction even in the event of direct database exposure.

External Student/Parent communication shall use HTTPS through the school-approved perimeter firewall and secure reverse proxy or web server before permitted requests reach the internal Node.js backend. Internal School Operations traffic shall use the secured School LAN. Every protected request shall require a valid, unexpired, and role-authorized session or token. The Node.js backend, Zone 1 integration, Zone 2 database, RFID interfaces, backup services, and internal administration services shall not be directly exposed to the Internet. Authorization shall be enforced by the backend even when a client interface hides an action. Data-processing and database credentials shall remain separated from client applications, and the Electron renderer shall receive only the validated RFID events and responses required for the active workflow.

Financial records receive additional protection. Every top-up, purchase, settlement, adjustment, and reversal is validated server-side and executed through ACID database transactions. A canteen sale, inventory deduction, credit update, Financial_Transactions header, and its Financial_Ledger_Entries lines either commit together or roll back together. Each finalized transaction has at least two immutable lines whose debit total equals its credit total; direct edit or deletion is prohibited. Corrections create linked certified adjustment or reversal transactions. Least-privilege roles restrict financial actions, while audit records capture the actor, timestamp, action, affected account or wallet, amount, reference, result, and reason when applicable. These controls preserve traceability and reduce the risk of unauthorized or partial financial changes.

In simple terms, this hashing algorithm acts as a one-way mathematical function that takes a user password and scrambles it into a random, fixed-length string of characters. Unlike standard encryption, which can be unlocked and reversed with a digital key, a hash is designed to be mathematically irreversible. When a user attempts to log in, the system applies the exact same mathematical scrambling

process to the newly entered password and checks if the resulting string matches the one saved in the database.

3.14 Development Technologies

The proposed system utilizes a localized, decoupled software stack engineered for consistent interface rendering, secure hardware interfacing, and isolated database operations on the school premises.

React serves as the core frontend library for building component-driven user interfaces, utilizing Tailwind CSS for utility-first styling. This frontend configuration is deployed across two targets. For the web-based Student/Parent Portal, the React production build is hosted through a school-approved web server or reverse proxy and delivered to authorized browsers using HTTPS. For the desktop client, the same React and Tailwind CSS interface is delivered as the School Operations desktop application. Electron serves as its wrapper and exposes only the controlled local runtime and hardware communication needed for the supported USB RFID readers.

Node.js functions as the primary server-side runtime environment to execute backend business logic, process incoming RFID serial data, and enforce role-based access control boundaries. The backend service layer runs on the school's secured on-premises server. Internal School Operations traffic uses the local network, while approved external Student/Parent requests reach the backend only through the protected web layer.

The data layer is managed via a two-zone data configuration using a school-approved read-only integration mechanism for Zone 1 and a separate database connection for the Zone 2 operational MySQL database

to maintain strict isolation. The backend retrieves authorized identity and logging records through the selected school-approved Zone 1 integration mechanism to verify student and staff ID records. Simultaneously, a separate connection targets a newly established local MySQL database dedicated exclusively to storing application-specific data, preventing any integration with or modification of institutional financial ledgers.

This chapter presented the technical background of the proposed Integrated School Services and Record Management System (ISSRMS). It discussed the centralized client-server architecture, RFID-based identity verification, and the major functional modules, including Institutional Logging Records reference, financial services, library management, canteen management, notifications, and home and announcements. It also described how the modules interact through a secure, application-enforced two-zone data architecture that preserves the read-only boundary of the existing institutional identity and logging records. The security, transaction-integrity, and technology concepts in this chapter establish the foundation for the system design, implementation, and evaluation presented in the succeeding chapters.

CHAPTER 4
DESIGN AND METHODOLOGY

This chapter presents the blueprint and methodology for developing the Integrated School Services and Record Management System (ISSRMS). It follows the prescribed capstone format by describing the conceptual framework, analysis and design artifacts, development model and approach, software tools, project-management plan, and verification, validation, and testing activities. Every design decision in this chapter is limited to the functions defined in Chapter III: RFID-assisted identity verification; read-only access to authorized institutional logging records; canteen e-wallet and inventory transactions; library circulation, inventory, and separately managed fines; notifications; announcements; and role-controlled web and desktop access.

The school's official debts, billing records, and Statement of Account are outside ISSRMS. No process, data flow, entity, interface, or equation in this chapter may imply that ISSRMS generates, updates, consolidates, or integrates with those institutional financial records. Library fines remain within the Library Management Module, while the Financial Services Module handles only the canteen e-wallet.

4.1 Conceptual Framework

The conceptual framework of the Integrated School Services and Record Management System (ISSRMS) follows an Input-Process-Output (IPO) model with a feedback loop to represent the iterative nature of system development and evaluation. The framework illustrates how the system specifications serve as the basis for identifying the required inputs and system processes, how these processes produce the intended system outputs, and how evaluation and validation provide feedback for further refinement. This framework ensures that the development of ISSRMS remains aligned with the identified requirements, operational needs, and limitations of Saint Joseph School of Mactan.

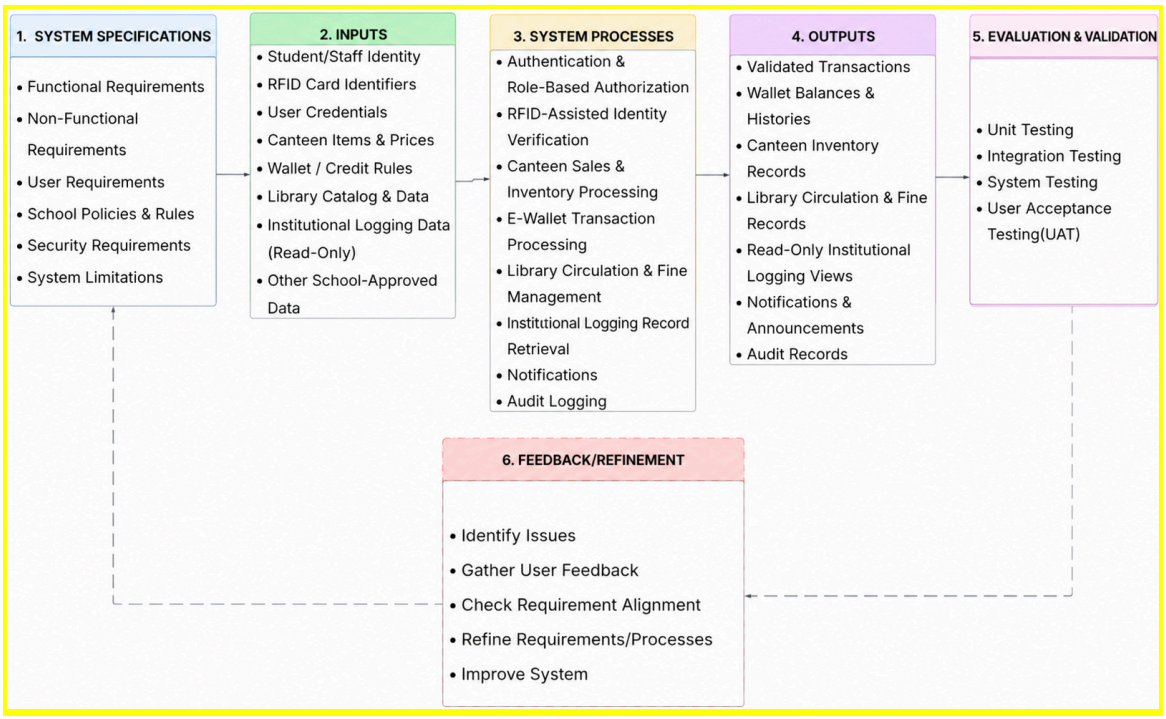


Figure 4.1.1. Conceptual Framework of the Integrated School Services and Record Management System

The System Specifications define the requirements and limitations of ISSRMS. These include functional and non-functional requirements, user requirements, school policies, security requirements, and system limitations. These specifications serve as the basis for the system's inputs, processes, and expected outputs.

The Inputs are the authorized data required by ISSRMS, including user identities, RFID identifiers, credentials, canteen data, wallet and credit rules, library data, and institutional logging data. Institutional identity and logging data are accessed through a read-only integration, while ISSRMS-generated records are stored in its own operational database.

The System Processes transform the inputs into the required system services and records. These include authentication, role-based authorization, RFID verification, canteen transactions, e-wallet processing, inventory management, library circulation and fines, logging record retrieval, notifications, and audit logging.

The Outputs are the records and services produced by ISSRMS. These include transaction records, wallet balances and histories, inventory records, library circulation and fine records, authorized logging views, notifications, announcements, and audit records.

The Evaluation and Validation stage determines whether ISSRMS meets its defined requirements and functions correctly. Unit testing, integration testing, system testing, and User Acceptance Testing (UAT) are conducted to identify errors and assess whether the system meets user requirements.

The Feedback and Refinement stage uses testing results and user feedback to identify issues and improve the system. Necessary changes are applied to the requirements or system processes, after which the system undergoes another cycle of testing and evaluation.

Overall, the framework shows that ISSRMS is developed from defined specifications, processes authorized inputs, produces the expected outputs, and undergoes continuous evaluation and refinement. This ensures that the system remains aligned with its requirements and the operational needs of its intended users.

4.2 Analysis and Design

The proponents will use Object-Oriented Analysis and Design (OOAD) and Unified Modeling Language (UML) to define the system's functional boundaries, actors, processes, data structures, components, and interfaces.

4.2.1 Functional Boundary and Actor Analysis

The principal actors are Student/Parent users, the Registrar, Canteen Personnel, Library Personnel, and the Administrator. Institutional identity and logging sources are external read-only sources rather than ISSRMS actors with write behavior. The Registrar operates the Financial Services Module and may process cash top-ups, manage approved credit limits, perform verified manual credit settlement, create certified adjustments or reversals, review detailed canteen-credit activity, and view unpaid library-fine information and related circulation activity for reference only. Authorized Canteen Personnel may process canteen sales, manage permitted canteen inventory, and perform confirmed cash-only wallet top-ups but may not change credit limits, settle outstanding credit, or perform adjustments or reversals. Library Personnel manage the catalog, circulation, inventory, and library-fine assessment, adjustment, reduction, waiver, and settlement actions. The Administrator manages only Zone 2 accounts and roles, approved non-financial settings, announcements, authorized read-only Institutional Logging Records, and read-only system-wide Audit Logs; it does not execute financial, canteen, or library operational transactions. All Administrator actions remain authenticated, server-authorized, validated, and audited.

4.2.2 Use Case Model

The UML Use Case Diagram shall show the services available to each actor and the role-based boundary around protected operations. The use case specification tables shall document the preconditions, trigger, main success flow, alternative flows, exceptions, postconditions, security rules, and audit result for each approved use case. Figure 4.2.5 shall identify the Administrator as a limited platform-administration actor connected only to account and role management, approved settings, announcements, read-only Institutional Logging Records, and read-only system-wide Audit Logs. Financial, canteen, and library operational use cases shall remain connected to their respective authorized personnel.

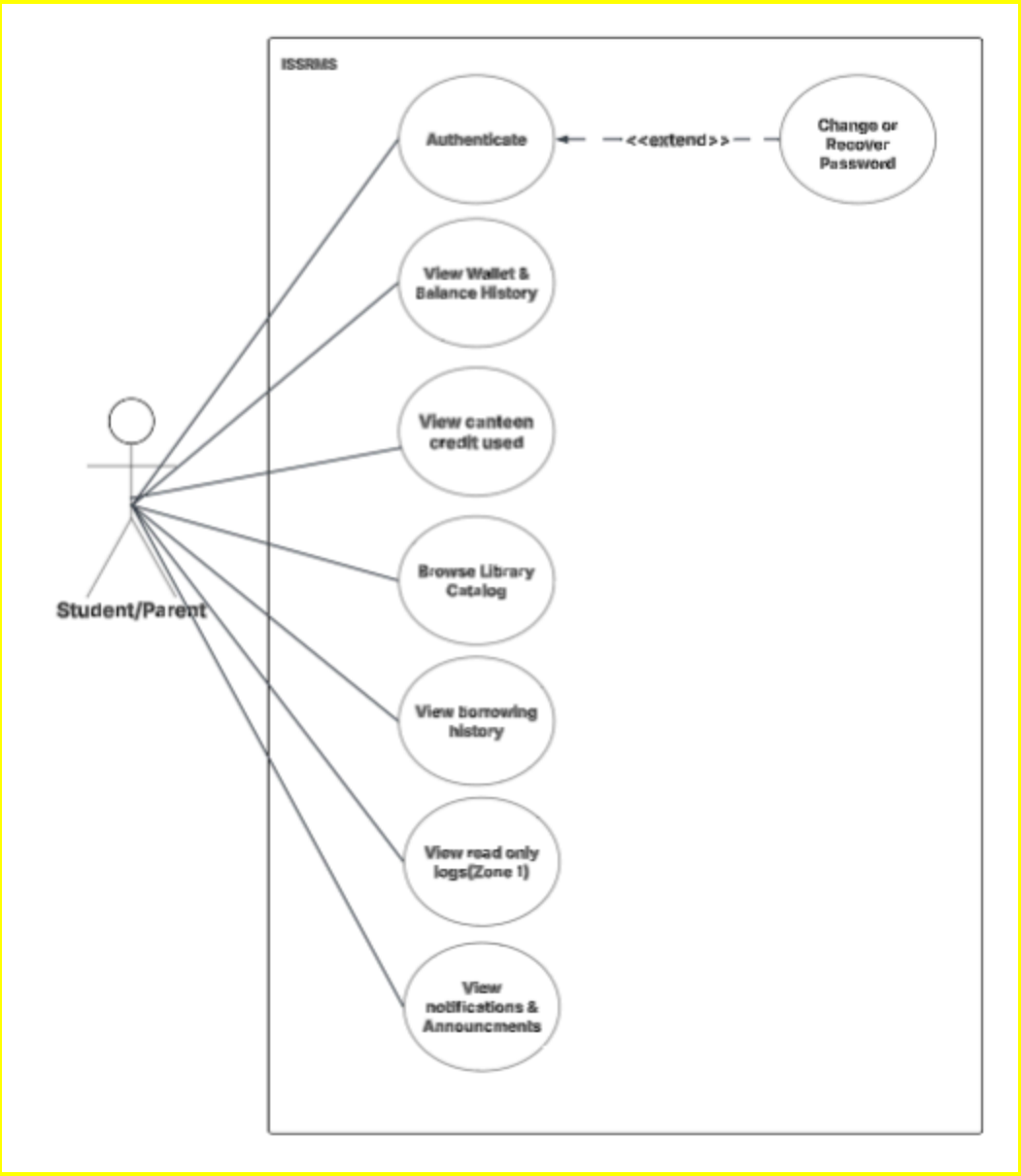


Figure 4.2.1 UML Use Case Diagram of ISSRMS

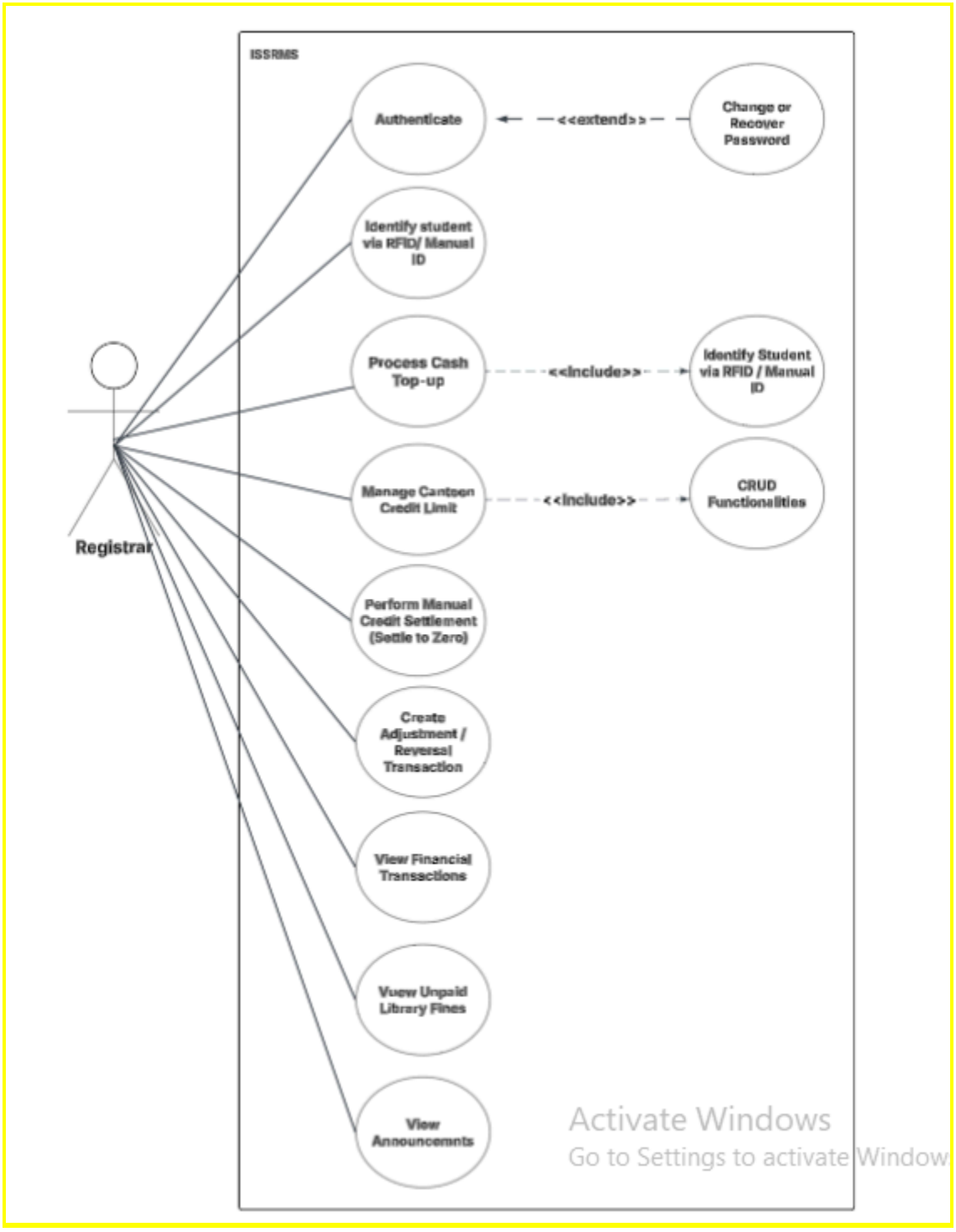


Figure 4.2.2 UML Use Case Diagram of ISSRMS



Figure 4.2.3 UML Use Case Diagram of ISSRMS

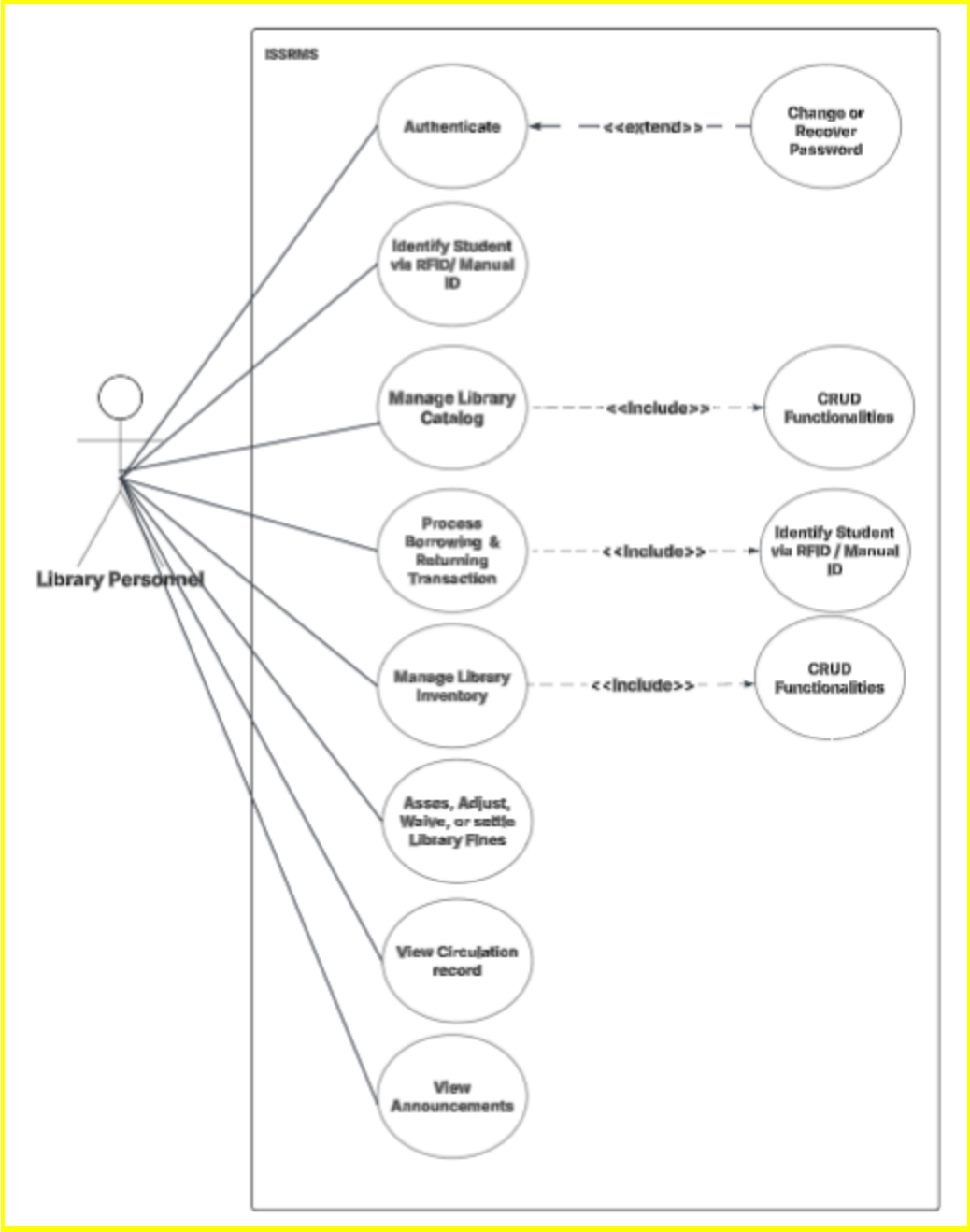


Figure 4.2.4 UML Use Case Diagram of ISSRMS

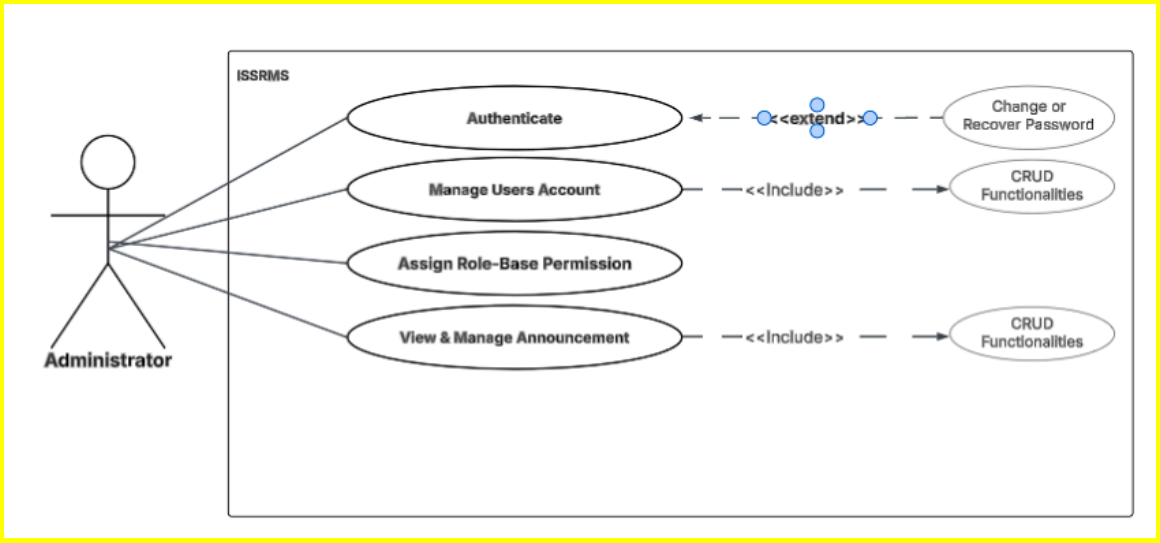


Figure 4.2.5 UML Use Case Diagram

UC-01 — Authenticate User

Field	Description
Use Case ID	UC-01
Use Case Name	Authenticate User
Primary Actor(s)	Student/Parent Portal User, Registrar, Canteen Personnel, Library Personnel, Administrator
Supporting Actor(s)	ISSRMS authentication service

Field	Description
Goal / Description	Authenticate an active ISSRMS account and display only the modules permitted by its assigned role.
Preconditions	An ISSRMS account exists in Zone 2, is active, and has assigned role permissions.
Trigger	The user submits an ID number and password.
Main Success Flow — Step 1 (Actor Action)	User opens the Student/Parent Portal or School Operations desktop application.
Step 2 — Actor Action	User enters the ID number and password.
Step 3 — System Response	The backend validates the account status and verifies the password hash.
Step 4 — System Response	The backend creates a valid session and determines the permitted role.
Step 5 — System Response	The system displays the role-authorized dashboard.
Alternative Flows	If a temporary password is used, the system requires a password change before protected functions are available.

Field	Description
Exceptions	Invalid credentials, an inactive account, an expired or revoked session, or an unauthorized request is rejected with a safe error message.
Postconditions	An authenticated session exists and only role-authorized functions and records are available.
Security and Authorization Rules	Credentials are validated by the backend; clear-text passwords are never stored; every protected endpoint rechecks authentication and authorization.
Audit Result	Record the supplied identifier, timestamp, action, result, and available request reference without recording the password.
Related SRS Requirements	PF-01; UI-01; SI-01; CI-02; REQ-4.1-1 to REQ-4.1-5; REQ-5.3-1, REQ-5.3-2, REQ-5.3-8

UC-02 — Top Up Canteen E-Wallet

Field	Description
Use Case ID	UC-02
Use Case Name	Top Up Canteen E-Wallet

Field	Description
Primary Actor(s)	Registrar; authorized Canteen Personnel
Supporting Actor(s)	Student; Financial Services Module
Goal / Description	Record a confirmed cash top-up for a student's canteen e-wallet.
Preconditions	The top-up operator is authenticated and authorized; the student has an active ISSRMS account and Student wallet; and the cash payment has been received.
Trigger	The Registrar or authorized Canteen Personnel selects the cash top-up function.
Main Success Flow — Step 1 (Actor Action)	Authorized top-up operator searches for and confirms the student account.
Step 2 — System Response	System displays the affected wallet and current derived cash balance.
Step 3 — Actor Action	Authorized top-up operator enters and confirms the authorized top-up amount.
Step 4 — System Response	Backend validates the amount and permissions.
Step 5 — System Response	Within one ACID transaction, the system creates an immutable

Field	Description
	Financial_Transactions header, balanced Financial_Ledger_Entries, and the audit event.
Step 6 — System Response	System displays the transaction reference and newly derived cash balance.
Alternative Flows	Authorized manual ID entry may be used to locate the student account.
Exceptions	Unknown or inactive account, invalid amount, duplicate submission, authorization failure, or posting failure causes rejection or complete rollback.
Postconditions	The finalized top-up is immutable, debit and credit totals are equal, and the derived cash balance reflects the confirmed top-up.
Security and Authorization Rules	Only the Registrar or authorized Canteen Personnel may process a confirmed over-the-counter cash top-up. Canteen Personnel may not change credit_limit, settle credit_used, create adjustments or reversals, or edit finalized financial records. Confirmation and server-side validation are required.

Field	Description
Audit Result	Record actor, student/wallet reference, amount, timestamp, transaction type, result, and unique transaction reference.
Related SRS Requirements	PF-04; UI-02; REQ-4.3-1, REQ-4.3-2, REQ-4.3-7 to REQ-4.3-10; REQ-5.2-1 to REQ-5.2-4; REQ-5.3-4

UC-03 — Purchase Canteen Items

Field	Description
Use Case ID	UC-03
Use Case Name	Purchase Canteen Items
Primary Actor(s)	Student; Canteen Personnel
Supporting Actor(s)	Financial Services Module
Goal / Description	Complete a cashier-assisted canteen purchase using available wallet cash and approved credit.

Field	Description
Preconditions	The operator is authorized; the student account is active; selected items are active and available; the purchase satisfies $P \leq B + C_{avail}$.
Trigger	The student presents an RFID card or the cashier begins authorized manual-ID lookup.
Main Success Flow — Step 1 (System Response)	System validates the RFID identifier or manually entered ID.
Step 2 — Actor Action	Cashier selects available items and quantities.
Step 3 — System Response	System calculates the total P using active prices.
Step 4 — System Response	Backend derives B and C_{avail} and checks $P \leq B + C_{avail}$.
Step 5 — System Response	Cash is consumed first; only the uncovered amount increases credit_used.
Step 6 — System Response	Within one ACID transaction, the system posts the sale, sale items, stock deduction, credit update, immutable financial header, balanced ledger entries, and audit event.

Field	Description
Step 7 — System Response	System displays the successful transaction reference.
Alternative Flows	If the RFID card is unavailable or unreadable, authorized Canteen Personnel may enter the student ID manually and undergo the same validation.
Exceptions	Unmatched or inactive account, unavailable item, invalid quantity, insufficient B + C_avail, reader/network failure, or database error causes rejection or complete rollback.
Postconditions	The sale and financial records are finalized together; stock and credit values are consistent; no partial transaction remains.
Security and Authorization Rules	The backend revalidates identity, operator permission, active prices, stock, balance, credit limit, and transaction integrity.
Audit Result	Record operator, student/wallet, sale and financial references, items, amount, cash-used portion, credit-added portion, timestamp, and result.

Field	Description
Related SRS Requirements	PF-02; PF-04; PF-05; REQ-4.2-1 to REQ-4.2-3; REQ-4.3-1 to REQ-4.3-10; REQ-4.4-1 to REQ-4.4-5; ACC-02

UC-04 — Borrow Library Material

Field	Description
Use Case ID	UC-04
Use Case Name	Borrow Library Material
Primary Actor(s)	Student; Library Personnel
Supporting Actor(s)	Library Management Module
Goal / Description	Create an authorized library circulation record for an available material.
Preconditions	Authorized Library Personnel is authenticated; the student account is active; the material is available. Any additional borrowing restriction must come from an approved school library policy.
Trigger	The student requests to borrow a material.
Main Success Flow — Step 1 (Actor Action)	Student presents the RFID card or ID number.

Field	Description
Step 2 — System Response	System validates the identifier and account status.
Step 3 — Actor Action	Authorized Library Personnel selects the material.
Step 4 — System Response	System validates availability and approved borrowing rules.
Step 5 — System Response	System records the borrowing date, due date, and circulation status.
Step 6 — System Response	System changes the material status to Borrowed and displays confirmation.
Alternative Flows	If the RFID card is unavailable or unreadable, authorized Library Personnel may enter the student ID manually under the same validation rules.
Exceptions	Unmatched or inactive account, unavailable material, unauthorized operator, invalid circulation data, or database failure causes rejection without a partial record.
Postconditions	An auditable circulation record exists, the due date is assigned, and material availability is synchronized.

Field	Description
Security and Authorization Rules	Only authorized Library Personnel may create circulation records; portal users have read-only access to their permitted library records.
Audit Result	Record actor, student account, material, borrow date, due date, result, and circulation reference.
Related SRS Requirements	PF-02; PF-06; REQ-4.2-2, REQ-4.2-3; REQ-4.5-1, REQ-4.5-2; REQ-5.3-2 to REQ-5.3-4

UC-05 — Return Library Material and Assess Fine

Field	Description
Use Case ID	UC-05
Use Case Name	Return Library Material and Assess Fine
Primary Actor(s)	Library Personnel
Supporting Actor(s)	Student; Library Management Module;
Goal / Description	Complete a return and determine any fine according to the approved school library policy.
Preconditions	An active circulation record exists for the returned material.

Field	Description
Trigger	Authorized Library Personnel receives a returned material.
Main Success Flow — Step 1 (Actor Action)	Authorized Library Personnel identifies the material and retrieves its active circulation record.
Step 2 — System Response	System compares the return date with the due date.
Step 3 — System Response	Any fine is calculated or assessed only under the approved library fine policy.
Step 4 — System Response	If there is no fine, the system completes the return and updates material availability.
Step 5 — System Response	If a fine is assessed, the system updates the related Library_Circulation fine amount and fine status and creates a library audit event.
Step 6 — System Response	The canteen E_Wallet, cash balance, credit_limit, credit_used, Financial_Transactions, and Financial_Ledger_Entries remain unchanged.

Field	Description
Step 7 — System Response	The system displays the return result, fine amount, fine status, and circulation reference.
Alternative Flows	Authorized Library Personnel may correct permitted return or fine information through the separate library fine-management workflow with a required audit reason.
Exceptions	Missing circulation record, invalid dates or amount, unauthorized operator, or database failure prevents completion and rolls back the return-and-fine transaction.
Postconditions	The circulation record, material availability, fine amount, fine status, and library audit record reflect the completed action. No wallet, financial transaction, or financial-ledger record is created or changed.
Security and Authorization Rules	Only authorized Library Personnel may complete returns or assess library fines. Library-fine actions remain inside the Library Management Module and shall not affect the canteen wallet or financial ledger.

Field	Description
Audit Result	Record the actor, circulation and material references, return date, fine amount and status, action, reason when applicable, timestamp, and result.
Related SRS Requirements	PF-06; UI-05; REQ-4.5-2 to REQ-4.5-5; CON-05; BR-02; BR-08; ACC-02; ACC-03

UC-06 — Manage Canteen Products and Inventory

Field	Description
Use Case ID	UC-06
Use Case Name	Manage Canteen Products and Inventory
Primary Actor(s)	Canteen Personnel
Supporting Actor(s)	Canteen Management Module
Goal / Description	Maintain authorized canteen products, active prices, availability, and stock quantities without altering historical sales.
Preconditions	The user is authenticated and has canteen inventory permission.
Trigger	The user opens Canteen Inventory.

Field	Description
Main Success Flow — Step 1 (System Response)	System displays the authorized product list and stock values.
Step 2 — Actor Action	User adds a product or selects an existing product.
Step 3 — Actor Action	User enters or updates permitted product information, price, availability, and stock quantity.
Step 4 — System Response	Backend validates required fields, values, permissions, and duplicate rules.
Step 5 — System Response	System saves the change and displays confirmation.
Alternative Flows	A product that should no longer be sold is deactivated rather than deleting historical sale references.
Exceptions	Invalid price or quantity, duplicate product, unauthorized change, or database failure causes rejection without a partial update.
Postconditions	The current inventory reflects the validated change and historical sale-item references remain intact.

Field	Description
Security and Authorization Rules	Only authorized Canteen Personnel may change product or inventory data; all inputs are validated server-side.
Audit Result	Record actor, item, action, before/after values where applicable, reason for adjustment, timestamp, and result.
Related SRS Requirements	PF-05; REQ-4.4-1 to REQ-4.4-5; REQ-5.2-2; REQ-5.3-2 to REQ-5.3-4

UC-07 — Manage ISSRMS User Accounts

Field	Description
Use Case ID	UC-07
Use Case Name	Manage ISSRMS User Accounts
Primary Actor(s)	Administrator
Supporting Actor(s)	Institutional identity source through read-only validation
Goal / Description	Provision and maintain Zone 2 authentication accounts, roles, and account status without modifying institutional identity data.
Preconditions	The Administrator is authenticated and authorized; the institutional ID can be

Field	Description
	validated through the approved read-only mechanism.
Trigger	The Administrator opens User Management.
Main Success Flow — Step 1 (Actor Action)	Administrator searches for or selects an institutional ID.
Step 2 — System Response	Backend validates the ID against the authorized read-only identity source.
Step 3 — Actor Action	Administrator creates or updates only Zone 2 account credentials, role permissions, and active status.
Step 4 — System Response	Backend validates uniqueness and role rules.
Step 5 — System Response	System saves the Zone 2 account change and confirms the result.
Alternative Flows	Authorized batch provisioning may create accounts from permitted active institutional IDs during deployment while storing only project-generated credentials and role data in Zone 2.
Exceptions	Unknown institutional ID, duplicate ISSRMS account, invalid role, missing required data, or authorization failure causes rejection.

Field	Description
Postconditions	The ISSRMS account reflects the approved Zone 2 change; no Zone 1 identity record is modified.
Security and Authorization Rules	Only Administrators may provision accounts or assign roles; the backend enforces least privilege and prevents writes to Zone 1.
Audit Result	Record administrator, affected account/ID, account action, role/status change, timestamp, and result.
Related SRS Requirements	PF-01; CON-01; CON-02; SI-02; SI-03; REQ-4.1-2, REQ-4.1-4, REQ-4.1-5; REQ-5.3-2, REQ-5.3-4

UC-08 — View Institutional Logging Records

Field	Description
Use Case ID	UC-08
Use Case Name	View Institutional Logging Records
Primary Actor(s)	Student/Parent Portal User; Administrator (read-only); explicitly authorized staff
Supporting Actor(s)	Institutional logging source through read-only integration

Field	Description
Goal / Description	Display permitted Gate Entry, Gate Exit, Library Entry, and Library Exit records without changing the institutional source system.
Preconditions	The user is authenticated and authorized to view the requested person and logging categories.
Trigger	The user opens Institutional Logging Records.
Main Success Flow — Step 1 (Actor Action)	User selects the permitted person, logging category, and date period.
Step 2 — System Response	Backend verifies identity and record-level permission.
Step 3 — System Response	Backend retrieves approved records through the school-approved read-only mechanism.
Step 4 — System Response	The system displays the student or permitted account, ID number, log type, location, date, and timestamp and labels the information as read-only. Gate records use Gate 1 or Gate 2 as the location, while Library Entry and Library Exit records use Library.

Field	Description
Alternative Flows	If no permitted records exist for the period, the system displays a no-records message without creating a source event.
Exceptions	Zone 1 unavailability, unauthorized request, invalid period, or integration timeout produces a safe error and does not corrupt Zone 2.
Postconditions	Permitted records are displayed; no institutional source record is created, edited, deleted, marked, or recomputed.
Security and Authorization Rules	Zone 1 credentials are least-privilege and read-only; the backend enforces role and record-level authorization.
Audit Result	Record the requesting account, requested scope, timestamp, result, and available request reference without copying unauthorized data.
Related SRS Requirements	PF-03; UI-04; CON-01; SI-02; SI-03; REQ-4.2-4, REQ-4.2-5; REQ-5.3-5; ACC-03

UC-09 — Change or Recover Password

Field	Description
Use Case ID	UC-09
Use Case Name	Change or Recover Password
Primary Actor(s)	ISSRMS User
Supporting Actor(s)	Administrator or school-approved secure communication service when authorized
Goal / Description	Allow an authorized user to replace a temporary or forgotten password without exposing credentials.
Preconditions	An ISSRMS account exists; the user can complete the approved identity-verification procedure.
Trigger	The user is required to replace a temporary password or requests password recovery.
Main Success Flow — Step 1 (Actor Action)	User begins password change or recovery.
Step 2 — System Response	Backend validates the account and approved verification evidence.
Step 3 — System Response	If an approved communication service is available, the system sends a time-limited verification code through that service.

Field	Description
Step 4 — Actor Action	User submits the valid code and a compliant new password.
Step 5 — System Response	Backend stores only the new password hash, revokes affected sessions when required, and confirms completion.
Alternative Flows	If school-approved SMS access is unavailable, an approved administrator-assisted recovery procedure may be used according to school policy.
Exceptions	Unknown account, invalid or expired code, failed identity verification, weak password, excessive attempts, or service unavailability causes rejection.
Postconditions	The new password hash is stored, temporary credentials are no longer valid, and affected sessions are handled according to policy.
Security and Authorization Rules	Codes are time-limited and rate-limited; clear-text passwords and codes are not written to logs; recovery does not reveal whether unauthorized records exist.

Field	Description
Audit Result	Record account reference, recovery method, timestamp, session action, and result without recording the password or verification code.
Related SRS Requirements	PF-01; REQ-4.1-1, REQ-4.1-3 to REQ-4.1-5; REQ-5.3-1, REQ-5.3-2, REQ-5.3-7, REQ-5.3-8

UC-10 — Manage Internal Announcements

Field	Description
Use Case ID	UC-10
Use Case Name	Manage Internal Announcements
Primary Actor(s)	Administrator
Supporting Actor(s)	Home and Announcements Module
Goal / Description	Create, edit, publish, or deactivate internal ISSRMS announcements
Preconditions	The Administrator is authenticated and has announcement-management permission.
Trigger	The Administrator opens Announcements Management.

Field	Description
Main Success Flow — Step 1 (System Response)	System displays existing internal announcements.
Step 2 — Actor Action	Administrator creates or selects an announcement.
Step 3 — Actor Action	Administrator enters or updates the title, content, type, and permitted audience.
Step 4 — System Response	Backend validates the content and authorization.
Step 5 — Actor Action	Administrator confirms publication or the selected management action.
Step 6 — System Response	System stores the change and displays confirmation.
Alternative Flows	An announcement may be saved as Draft, published as Active, or changed to Inactive. A previously published announcement is deactivated rather than permanently deleted so its history and audit record remain available.
Exceptions	Missing required fields, invalid audience, unauthorized action, or database failure causes rejection.
Postconditions	The announcement has the approved Draft, Active, or Inactive status and is

Field	Description
	visible only to its permitted audience when Active.
Security and Authorization Rules	Only authorized Administrators may manage announcements; content and audience rules are enforced by the backend.
Audit Result	Record actor, announcement reference, action, audience, timestamp, and result.
Related SRS Requirements	PF-07; REQ-4.8-2; REQ-4.7-1; REQ-5.3-2 to REQ-5.3-4

UC-11 — View Wallet and Canteen Transactions

Field	Description
Use Case ID	UC-11
Use Case Name	View Wallet and Canteen Transactions
Primary Actor(s)	Student/Parent Portal User
Supporting Actor(s)	Financial Services Module
Goal / Description	Display the user’s permitted canteen wallet and purchase information.

Field	Description
Preconditions	The portal user is authenticated and authorized for the selected student record.
Trigger	The user opens Wallet and Canteen Transactions.
Main Success Flow — Step 1 (Actor Action)	User opens Wallet and Canteen Transactions.
Step 2 — System Response	Backend verifies account and record-level permission.
Step 3 — System Response	System derives the active cash balance B from valid ledger entries.
Step 4 — System Response	System retrieves credit_used, permitted wallet transactions, and canteen purchase history.
Step 5 — System Response	System displays cash balance and canteen credit used as separate values with transaction references and timestamps.
Alternative Flows	If there is no transaction history, the system displays the current permitted values and a no-history message.
Exceptions	Unauthorized record request, data-retrieval failure, or temporary

Field	Description
	server unavailability produces a safe error.
Postconditions	The permitted financial information is displayed without allowing a protected financial action.
Security and Authorization Rules	Portal access is read-only and restricted to permitted records; protected top-up, settlement, credit-limit, adjustment, and reversal controls are unavailable.
Audit Result	Record the requesting account, record scope, timestamp, and result according to the approved access-audit policy.
Related SRS Requirements	PF-07; REQ-4.6-1 to REQ-4.6-3; DATA-01 to DATA-04; REQ-5.3-2

UC-12 — View Notifications

Field	Description
Use Case ID	UC-12
Use Case Name	View Notifications
Primary Actor(s)	Student/Parent Portal User; authorized staff

Field	Description
Supporting Actor(s)	Notifications Module
Goal / Description	Display permitted internal notifications for approved library, wallet, and system events.
Preconditions	The user is authenticated and has access to the associated account.
Trigger	The user opens Notifications.
Main Success Flow — Step 1 (Actor Action)	User opens Notifications.
Step 2 — System Response	Backend verifies the user and notification scope.
Step 3 — System Response	System retrieves permitted notifications.
Step 4 — System Response	System displays each notification’s type, message, and timestamp and records an opened notification as read.
Alternative Flows	If no notifications are available, the system displays a no-notifications message.
Exceptions	Unauthorized access or retrieval failure produces a safe error and does not expose another user’s notification.

Field	Description
Postconditions	Permitted notifications are displayed and opened notifications have the correct read status.
Security and Authorization Rules	Notification access is account- and role-restricted; institutional source logging alerts are not generated or transmitted by ISSRMS.
Audit Result	Record notification delivery/read status and relevant system events according to retention policy.
Related SRS Requirements	PF-07; REQ-4.8-1; REQ-4.6-1, REQ-4.6-2; REQ-5.3-2

UC-13 — Manage Canteen Credit Limit

Field	Description
Use Case ID	UC-13
Use Case Name	Manage Canteen Credit Limit
Primary Actor(s)	Registrar
Supporting Actor(s)	Financial Services Module
Goal / Description	Set or change a student’s approved canteen credit ceiling.

Field	Description
Preconditions	The Registrar is authenticated and authorized; the student wallet exists; the requested limit is supported by approved school policy.
Trigger	The Registrar selects Credit Limit Management.
Main Success Flow — Step 1 (Actor Action)	The authorized financial actor searches for and confirms the student wallet.
Step 2 — System Response	System displays credit_limit and credit_used separately.
Step 3 — Actor Action	The authorized financial actor enters the approved new credit limit and supporting reason/reference when required.
Step 4 — System Response	Backend validates permission, bounds, and policy rules.
Step 5 — System Response	System records the change and displays confirmation.
Alternative Flows	If policy requires additional approval, the change remains pending until the authorized approval is recorded.

Field	Description
Exceptions	Invalid limit, unauthorized request, missing required evidence, or database failure causes rejection.
Postconditions	The approved credit_limit is stored separately from credit_used; existing finalized financial transactions remain unchanged.
Security and Authorization Rules	Only the Registrar may change the limit; server-side bounds and confirmation are required.
Audit Result	Record actor, wallet, previous and new credit limit, reason/reference, timestamp, approval when applicable, and result.
Related SRS Requirements	PF-04; UI-02; REQ-4.3-6; REQ-4.7-1; REQ-5.2-2; REQ-5.3-4

UC-14 — Settle Canteen Credit

Field	Description
Use Case ID	UC-14
Use Case Name	Settle Canteen Credit

Field	Description
Primary Actor(s)	Registrar
Supporting Actor(s)	Financial Services Module
Goal / Description	Settle a student's canteen credit_used after separate payment has been verified.
Preconditions	The Registrar is authorized; credit_used is greater than zero; an official receipt, payment reference, or other school-approved payment record has been verified.
Trigger	The Registrar selects Manual Credit Settlement.
Main Success Flow — Step 1 (Actor Action)	The authorized financial actor searches for and confirms the student wallet.
Step 2 — System Response	System displays credit_used and requests the verified payment reference and reason.
Step 3 — Actor Action	The authorized financial actor confirms the settlement.
Step 4 — System Response	Backend revalidates authority and evidence.
Step 5 — System Response	Within one ACID transaction, the system posts an immutable

Field	Description
	Credit_Settlement transaction with balanced ledger entries, sets credit_used to zero, and records the audit event.
Step 6 — System Response	System displays the settlement reference and resulting credit_used.
Alternative Flows	Settlement is performed according to school policy at each grading or payment period after valid payment confirmation.
Exceptions	Missing or invalid payment evidence, zero credit_used, unauthorized request, duplicate settlement, or posting failure causes rejection or complete rollback.
Postconditions	credit_used is zero only after a finalized auditable settlement; prior financial records remain immutable.
Security and Authorization Rules	Only the Registrar may execute Settle-to-Zero; payment verification, reason/reference, confirmation, and backend authorization are mandatory.
Audit Result	Record actor, wallet, settled amount, payment reference, reason, timestamp, balanced transaction

Field	Description
	reference, previous/resulting credit_used, and result.
Related SRS Requirements	PF-04; UI-03; REQ-4.3-1, REQ-4.3-7 to REQ-4.3-12; BR-07; ACC-02

UC-15 — Record Financial Adjustment or Reversal

Field	Description
Use Case ID	UC-15
Use Case Name	Record Financial Adjustment or Reversal
Primary Actor(s)	Registrar
Supporting Actor(s)	Financial Services Module
Goal / Description	Correct a canteen-wallet error without editing or deleting the original finalized transaction.
Preconditions	The Registrar is authorized; the original transaction exists and is finalized; a certified reason and related reference are available.
Trigger	The Registrar selects Adjustment or Reversal for a financial record.

Field	Description
Main Success Flow — Step 1 (Actor Action)	The authorized financial actor searches for and reviews the original transaction.
Step 2 — System Response	System displays the immutable original header and ledger entries.
Step 3 — Actor Action	The authorized financial actor selects adjustment or reversal and enters the certified reason and required values.
Step 4 — System Response	Backend validates authority, relationship, amounts, and debit-credit equality.
Step 5 — System Response	Within one ACID transaction, the system creates a new linked immutable transaction and balanced ledger entries.
Step 6 — System Response	System displays the corrective transaction reference while preserving the original record.
Alternative Flows	A partial adjustment may be used only when authorized by policy; otherwise a full reversal is recorded.
Exceptions	Original transaction not found, invalid amount, missing reason, duplicate reversal, unauthorized action, or

Field	Description
	posting failure causes rejection or rollback.
Postconditions	A linked corrective record exists; the original remains unchanged; all finalized ledger entries remain balanced.
Security and Authorization Rules	Direct update or deletion of finalized financial records is prohibited; reason, confirmation, and server-side validation are required.
Audit Result	Record actor, original and corrective transaction references, action type, amount, reason, timestamp, and result.
Related SRS Requirements	PF-04; CON-03; CON-04; UI-03; REQ-4.3-7 to REQ-4.3-10; BR-03; BR-04; ACC-02

UC-16 — Manage Library Catalog and Inventory

Field	Description
Use Case ID	UC-16
Use Case Name	Manage Library Catalog and Inventory
Primary Actor(s)	Library Personnel

Field	Description
Supporting Actor(s)	Library Management Module
Goal / Description	Maintain permitted library catalog, availability, and inventory information.
Preconditions	Authorized Library Personnel is authenticated and authorized.
Trigger	The user opens Library Catalog and Inventory Management.
Main Success Flow — Step 1 (System Response)	System displays the catalog and inventory records permitted to the role.
Step 2 — Actor Action	User creates or selects a library material.
Step 3 — Actor Action	User enters or updates bibliographic, category, inventory, and availability information.
Step 4 — System Response	Backend validates required fields, identifiers, permissions, and circulation dependencies.
Step 5 — System Response	System saves the change and confirms the result.
Alternative Flows	A material may be deactivated or marked unavailable instead of deleting a record needed by circulation history.

Field	Description
Exceptions	Duplicate material identifier, invalid data, active-circulation conflict, unauthorized action, or database failure causes rejection.
Postconditions	Catalog and inventory records reflect the approved change without breaking circulation history.
Security and Authorization Rules	Only authorized Library Personnel may change catalog or inventory records; portal users have read-only access.
Audit Result	Record actor, material, action, before/after values when applicable, timestamp, and result.
Related SRS Requirements	PF-06; REQ-4.5-1; REQ-4.7-1; REQ-5.2-2; REQ-5.3-2 to REQ-5.3-4

UC-17 — Manage Library Fines

Field	Description
Use Case ID	UC-17
Use Case Name	Manage Library Fines
Primary Actor(s)	Library Personnel

Field	Description
Supporting Actor(s)	Library Management Module
Goal / Description	Assess, adjust, reduce, waive, or settle a library fine within the Library Management Module without affecting the canteen E_Wallet, credit_used, Financial_Transactions, or Financial_Ledger_Entries.
Preconditions	The related circulation record exists, and the Library Personnel account is authenticated and authorized for the selected fine-management action.
Trigger	Authorized Library Personnel selects Fine Management for a circulation record.
Main Success Flow — Step 1 (System Response)	The system displays the circulation record, material information, fine_amount, fine_status, and fine-action history.
Step 2 — Actor Action	The authorized fine-management actor enters the required amount, status, reason, and library-approved reference when applicable.
Step 3 — System Response	The backend validates the actor's Library Personnel role, circulation

Field	Description
	record, fine amount, status, and required reason.
Step 4 — System Response	Backend validates library policy, role permission, and posting data.
Step 5 — System Response	The system updates the library fine record and creates an auditable library action record.
Step 6 — System Response	The canteen E_Wallet, active cash, credit_limit, credit_used, Financial_Transactions, and Financial_Ledger_Entries remain unchanged.
Step 7 — System Response	The system displays confirmation and the circulation or library-fine reference.
Alternative Flows	The Registrar may open the corresponding read-only view through UC-18 but may not perform a fine-management action. Only authorized Library Personnel may perform fine-management actions through this use case.
Exceptions	Invalid amount or status, missing required reason, unauthorized actor, missing circulation record, duplicate submission, or database failure

Field	Description
	causes rejection without a partial update.
Postconditions	The library fine amount, status, circulation record, and audit event reflect the authorized action. No canteen-wallet or financial-ledger value is changed.
Security and Authorization Rules	Only authorized Library Personnel may assess, adjust, reduce, waive, or settle a library fine. Registrar access is informational only.
Audit Result	Record the actor, circulation reference, previous and resulting fine amount and status, action, reason or library-approved reference, timestamp, and result.
Related SRS Requirements	PF-06; UI-05; CON-05; REQ-4.5-3 to REQ-4.5-6; BR-08; BR-11; ACC-03.

UC-18 — View Library Records

Field	Description
Use Case ID	UC-18
Use Case Name	View Library Records

Field	Description
Primary Actor(s)	Student/Parent Portal User; Registrar for permitted fine information
Supporting Actor(s)	Library Management Module
Goal / Description	Display permitted catalog, circulation, borrowing-history, and library-fine information.
Preconditions	The user is authenticated and authorized for the requested records.
Trigger	The user opens Library Records.
Main Success Flow — Step 1 (System Response)	Backend verifies role and record-level permission.
Step 2 — System Response	System retrieves the permitted catalog, active borrowing, borrowing history, and fine information.
Step 3 — System Response	System displays the permitted records.
Step 4 — Actor Action	For Registrar access, the system displays the unpaid fine amount and status together with permitted related circulation information, including the material title or identifier, circulation reference, borrow date, due date, return date when applicable, overdue duration, and latest Library Personnel

Field	Description
	action. The interface displays “Read Only — Managed by Library Personnel.” Fine-management controls are available only through UC-17 to authorized Library Personnel.
Alternative Flows	If no permitted records exist, the system displays a no-records message.
Exceptions	Unauthorized request or retrieval failure produces a safe error without exposing another user’s records.
Postconditions	Permitted library information is displayed without providing unauthorized catalog, circulation, or fine-management controls.
Security and Authorization Rules	Students and parents may view only the linked student’s permitted records. Registrar access to unpaid fines and related circulation activity is informational only, and assessment, adjustment, reduction, waiver, payment, and settlement controls shall not be displayed to the Registrar. Library catalog, circulation, inventory, and fine-management controls remain

Field	Description
	restricted to authorized Library Personnel through UC-16 and UC-17.
Audit Result	Record requesting account, record scope, timestamp, and result according to access-audit policy.
Related SRS Requirements	PF-07; REQ-4.5-6; REQ-4.6-1 to REQ-4.6-3; REQ-4.7-5; BR-11

UC-19 — View Announcements

Field	Description
Use Case ID	UC-19
Use Case Name	View Announcements
Primary Actor(s)	Student/Parent Portal User; authorized staff
Supporting Actor(s)	Home and Announcements Module
Goal / Description	Display internal announcements permitted for the authenticated role or audience.
Preconditions	The user is authenticated and the announcement is active and permitted for the user’s role.

Field	Description
Trigger	The user opens the Home or Announcements page.
Main Success Flow — Step 1 (Actor Action)	User opens the Home or Announcements page.
Step 2 — System Response	Backend verifies the user's role and audience permissions.
Step 3 — System Response	System retrieves active permitted announcements.
Step 4 — System Response	System displays announcement title, content, type, and timestamp.
Alternative Flows	If no internal announcements are available, the system displays a no-announcements message. Any optional external public feed is loaded separately only when school approval and platform support exist.
Exceptions	Unauthorized content request, data-retrieval failure, or optional external-feed unavailability does not affect local ISSRMS transactions.
Postconditions	Permitted announcements are displayed in read-only form.
Security and Authorization Rules	Role/audience filters are enforced by the backend; viewing does not grant

Field	Description
	announcement-management permission.
Audit Result	Record relevant access or delivery events according to the approved audit and retention policy.
Related SRS Requirements	PF-07; REQ-4.6-1, REQ-4.6-2; REQ-4.8-2; REQ-5.3-2

UC-20 — Review Audit Events

Field	Description
Use Case ID	UC-20
Use Case Name	Review Audit Events
Primary Actor(s)	Administrator
Supporting Actor(s)	ISSRMS audit service
Goal / Description	Review permitted operational, security, and financial audit events without altering finalized source records.
Preconditions	The Administrator is authenticated and has audit-log oversight permission.

Field	Description
Trigger	The Administrator opens Audit Events.
Main Success Flow — Step 1 (System Response)	System displays permitted filter options.
Step 2 — Actor Action	Administrator selects the event type, actor, target/reference, result, or period.
Step 3 — System Response	Backend validates audit-log permission and retrieves matching records.
Step 4 — System Response	System displays actor/reference, timestamp, action, result, and permitted contextual details.
Alternative Flows	No other role may access the system-wide Audit Logs interface. Operational personnel may view ordinary transaction or circulation references within their authorized module screens, but those references do not provide access to the Administrator Audit Logs interface.
Exceptions	Unauthorized request, invalid filter, or retrieval failure produces a safe error and does not change audit data.

Field	Description
Postconditions	Permitted audit events are displayed read-only; finalized financial and institutional source records remain unchanged.
Security and Authorization Rules	Only authenticated Administrator accounts may access the system-wide Audit Logs interface. Audit events are read-only and shall not provide edit or delete controls. Credentials, tokens, connection details, and unauthorized record contents shall not be displayed.
Audit Result	The audit-log access itself records the requesting actor, query scope, timestamp, and result.
Related SRS Requirements	PF-08; REQ-4.1-5; REQ-5.3-4; REQ-5.3-7; ACC-02; ACC-04

Table 4.2.2 *Use Case Specifications*

4.2.3 Activity Diagrams

Activity diagrams shall document the sequence, decision points, authorization checks, exception handling, rollback behavior, and audit events for core workflows. The following figures are required. Existing images may be retained only after their labels and flows pass the scope checks stated below.

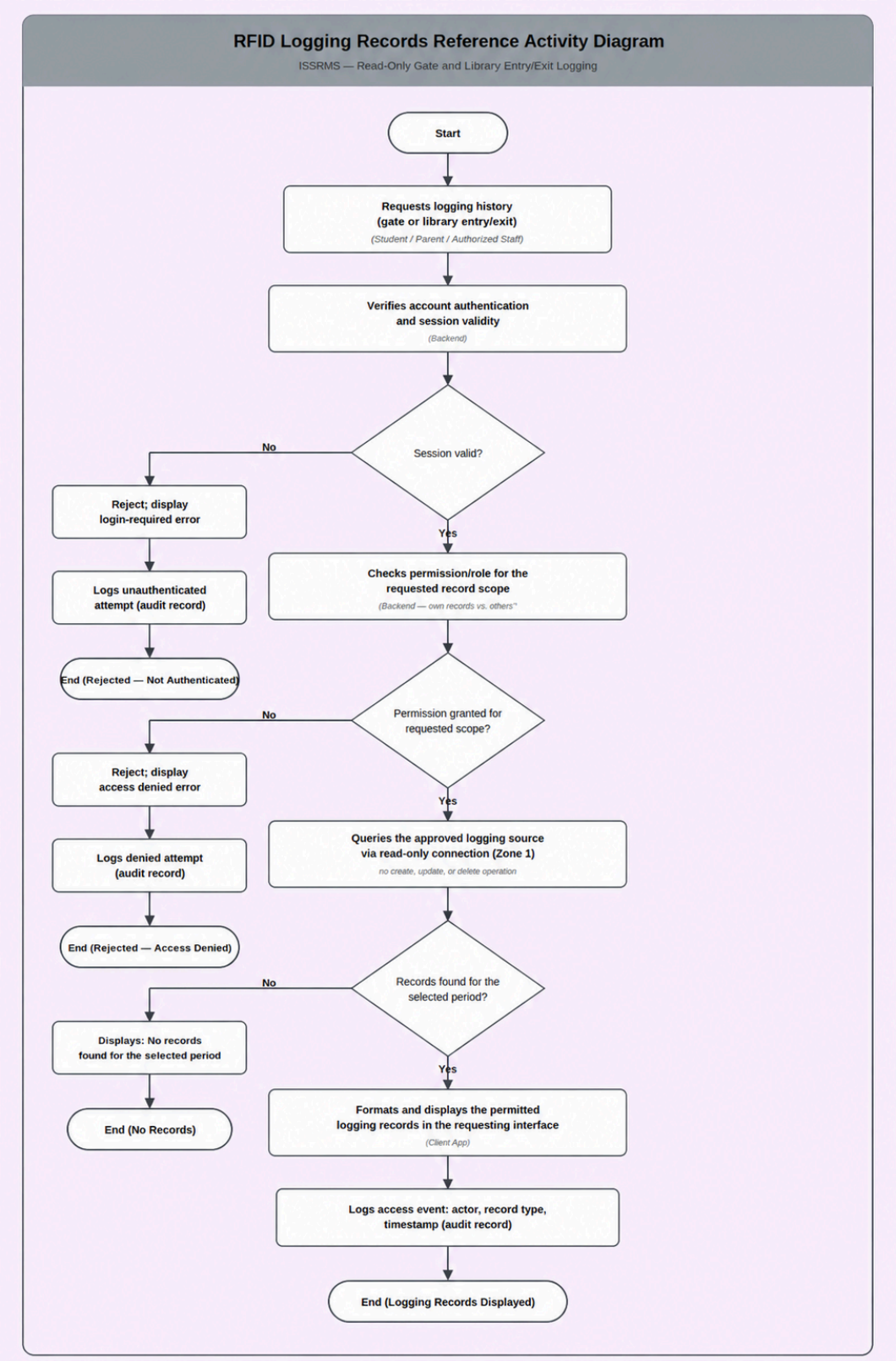


Figure 4.3. Institutional Logging Records Reference Activity Diagram

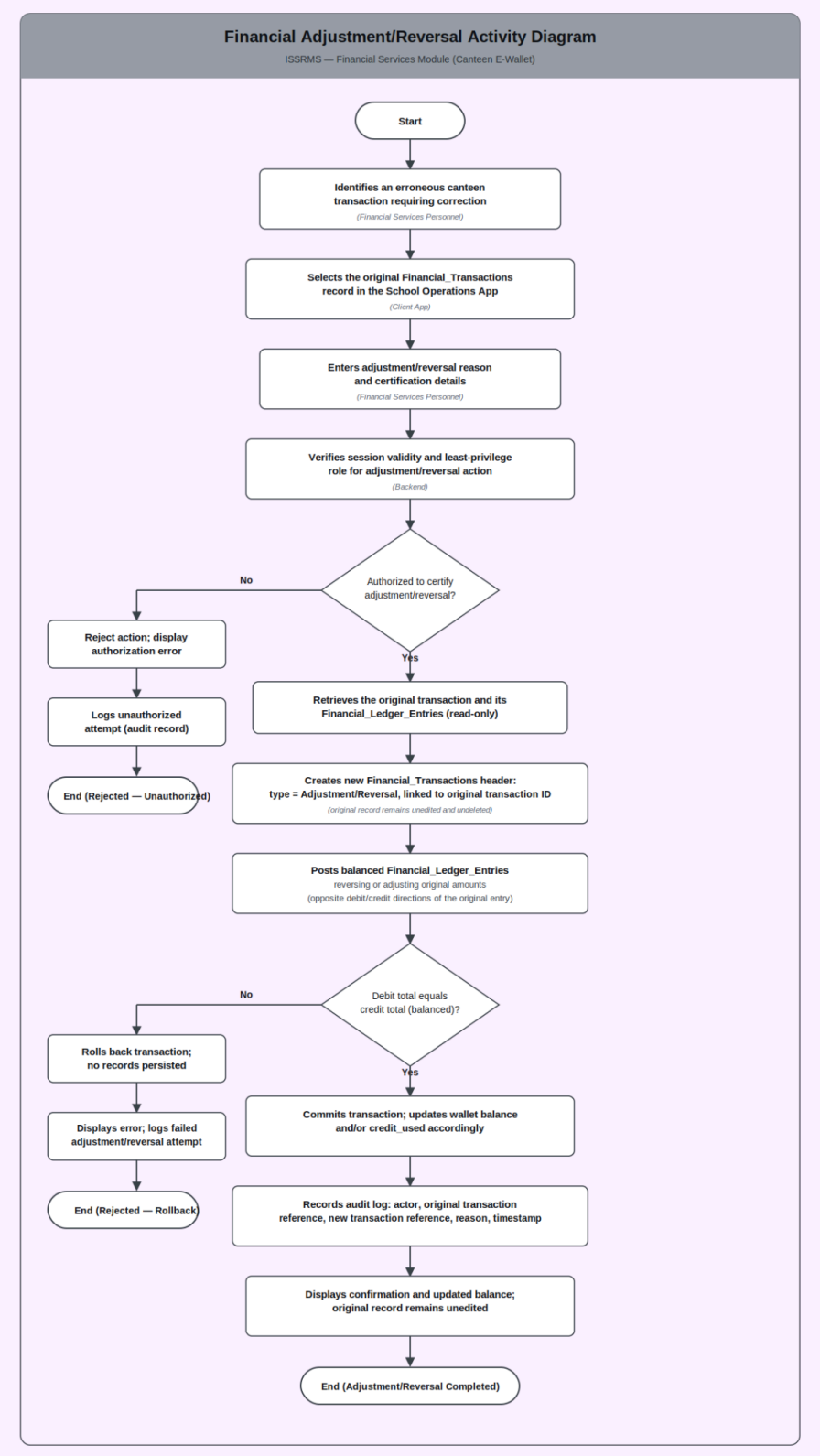


Figure 4.4. Financial Services Activity Diagram

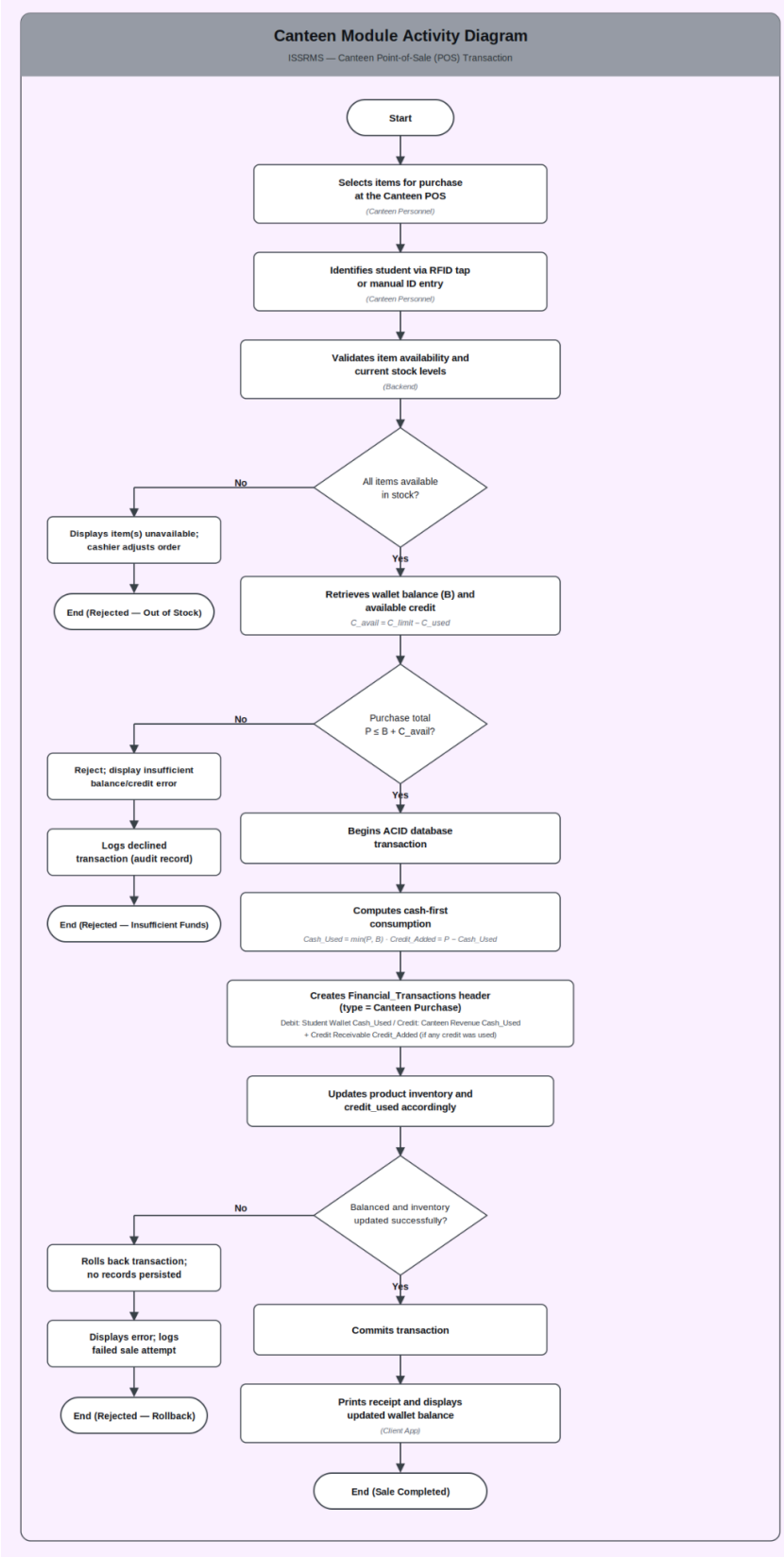


Figure 4.5. Canteen Module Activity Diagram

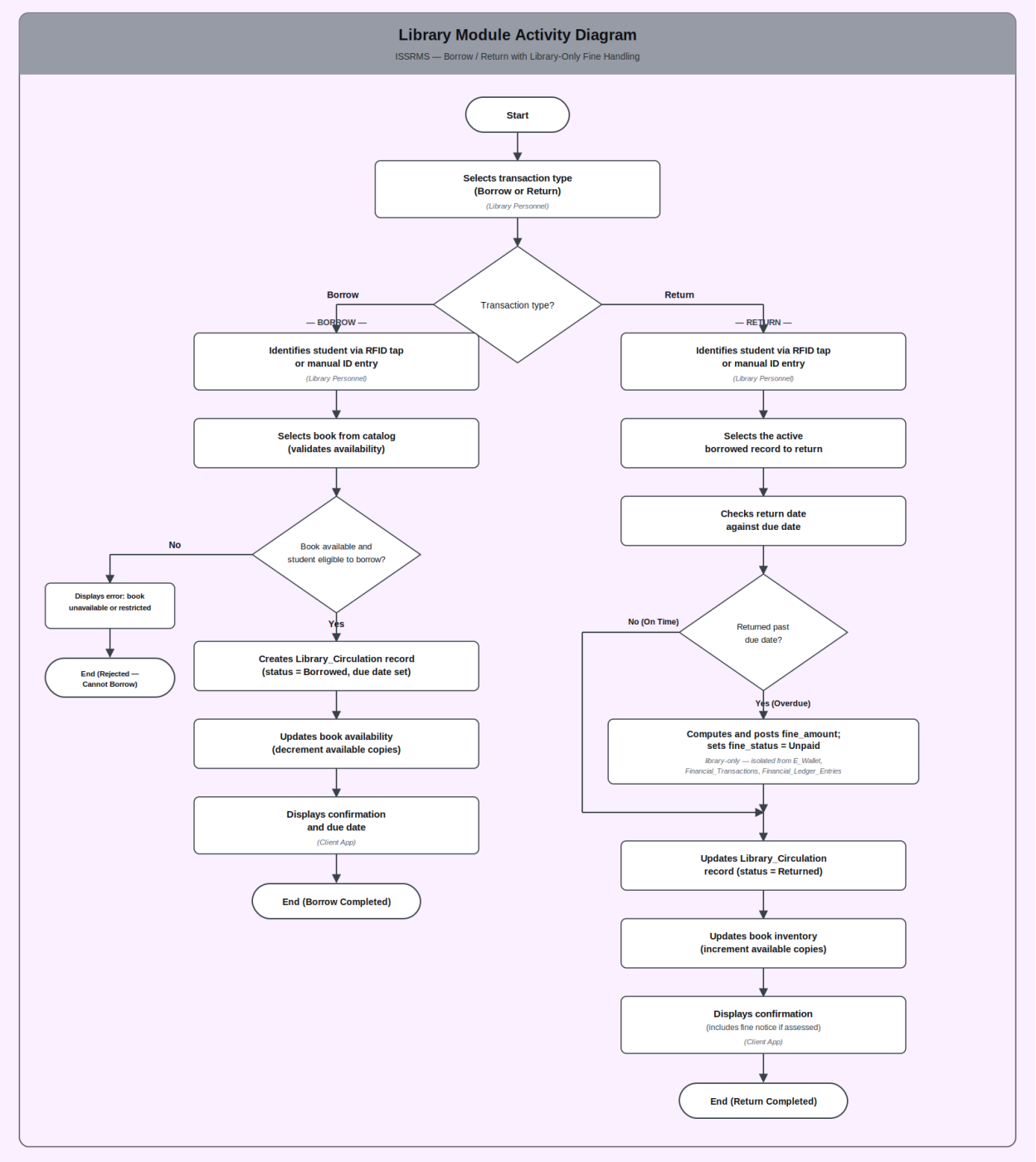


Figure 4.6. Library Module Activity Diagram

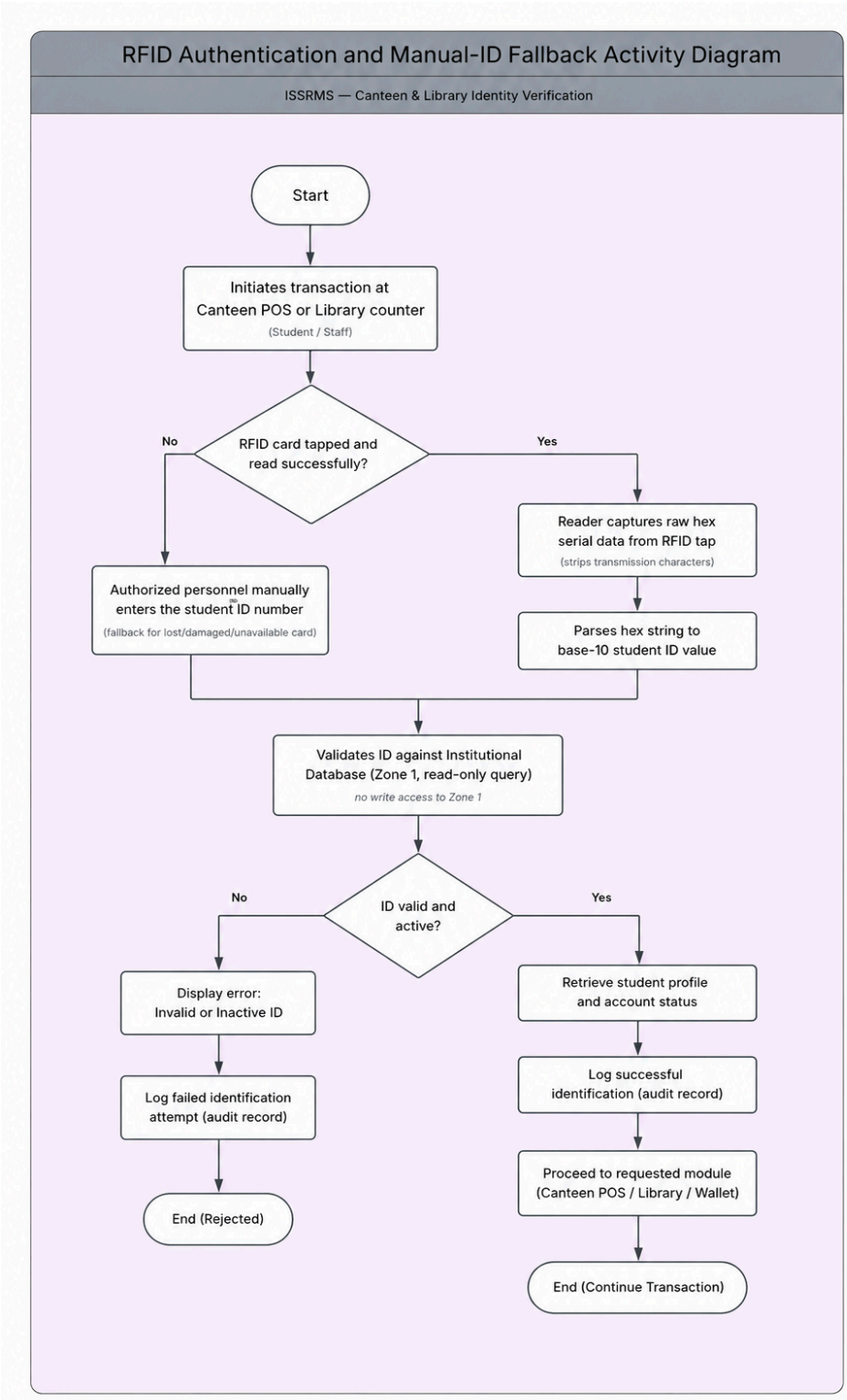


Figure 4.7. Authentication and Role-Based Access Activity Diagram

4.2.4 Business Process Model and Notation

Business Process Model and Notation (BPMN) diagrams shall illustrate the operational workflows of the ISSRMS by defining the sequence of business activities, participant responsibilities, system validations, decision gateways, exception handling, and transaction completion for core institutional services. The following BPMN diagrams model the primary processes for canteen purchases, financial adjustment or reversal, manual credit settlement (settle-to-zero), and library material borrowing.

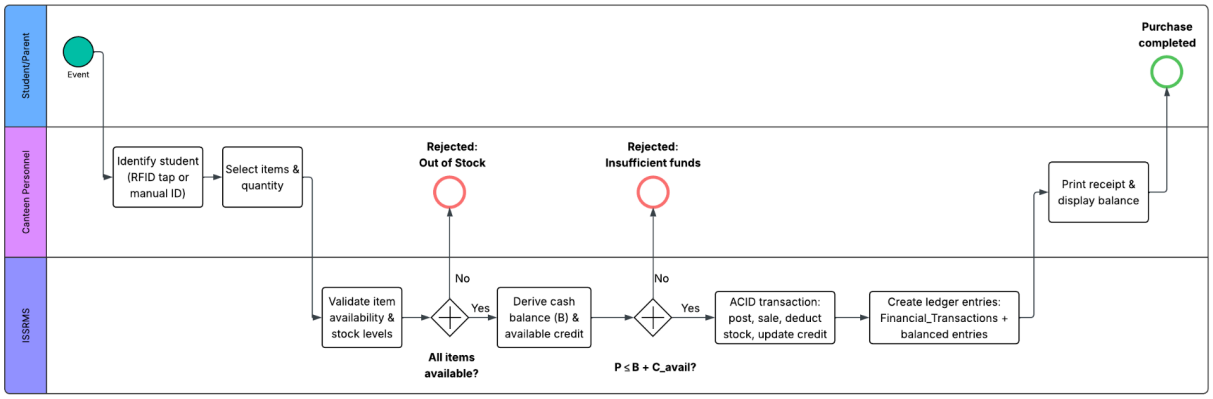


Figure 4.8. Canteen Purchase BPMN

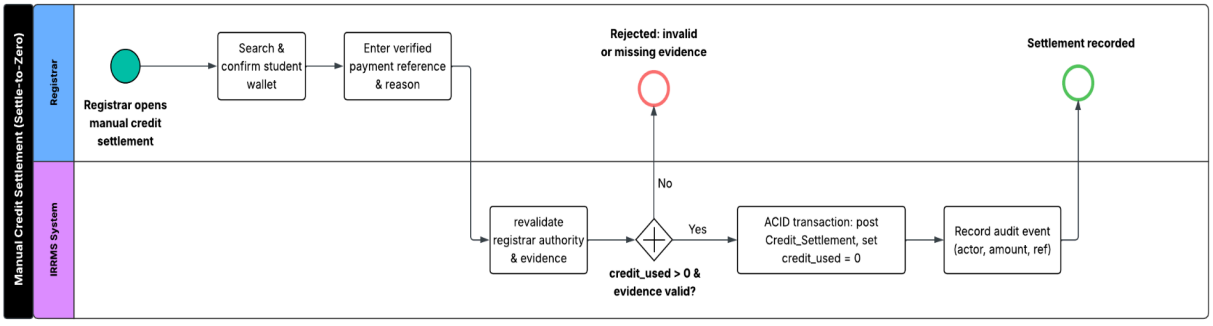


Figure 4.9. Manual Credit Settlement (Settle-to-Zero) BPMN

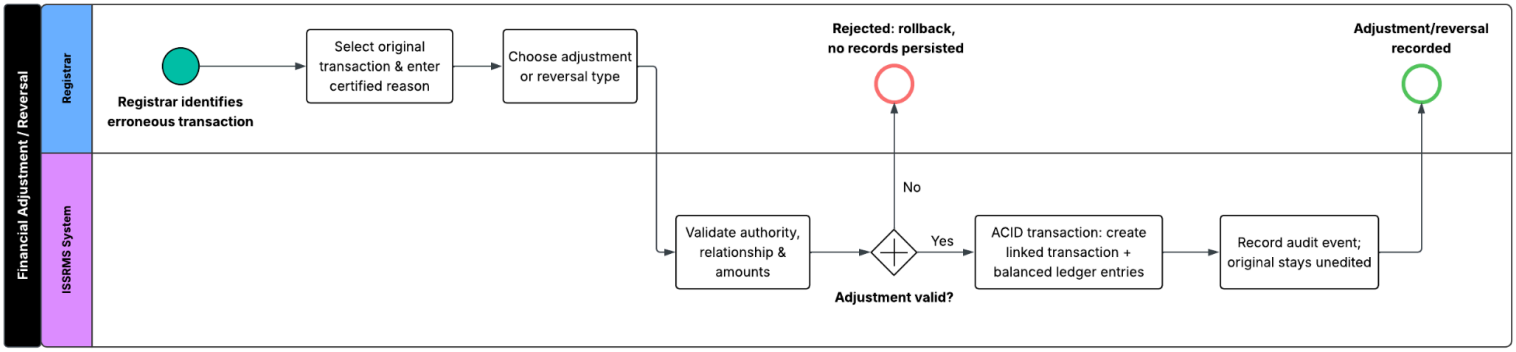


Figure 4.10. Financial Adjustment Reversal BPMN

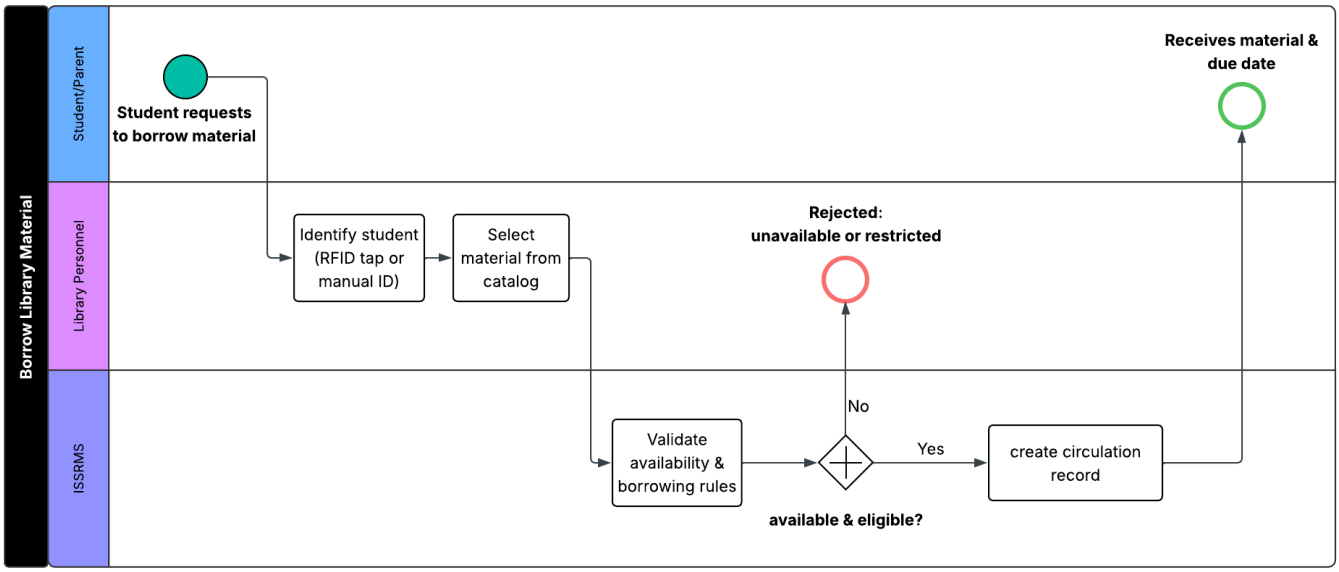


Figure 4.11. Borrow Library Material BPMN

4.2.5 Data Flow Diagrams

The data flow diagrams shall distinguish external entities, ISSRMS processes, Zone 1 read-only sources, and Zone 2 operational data stores. Arrow directions must make the separation explicit: ISSRMS may read permitted institutional identity and logging records, but it may not write to those sources. No flow shall connect ISSRMS to the school's official billing or Statement of Account process.

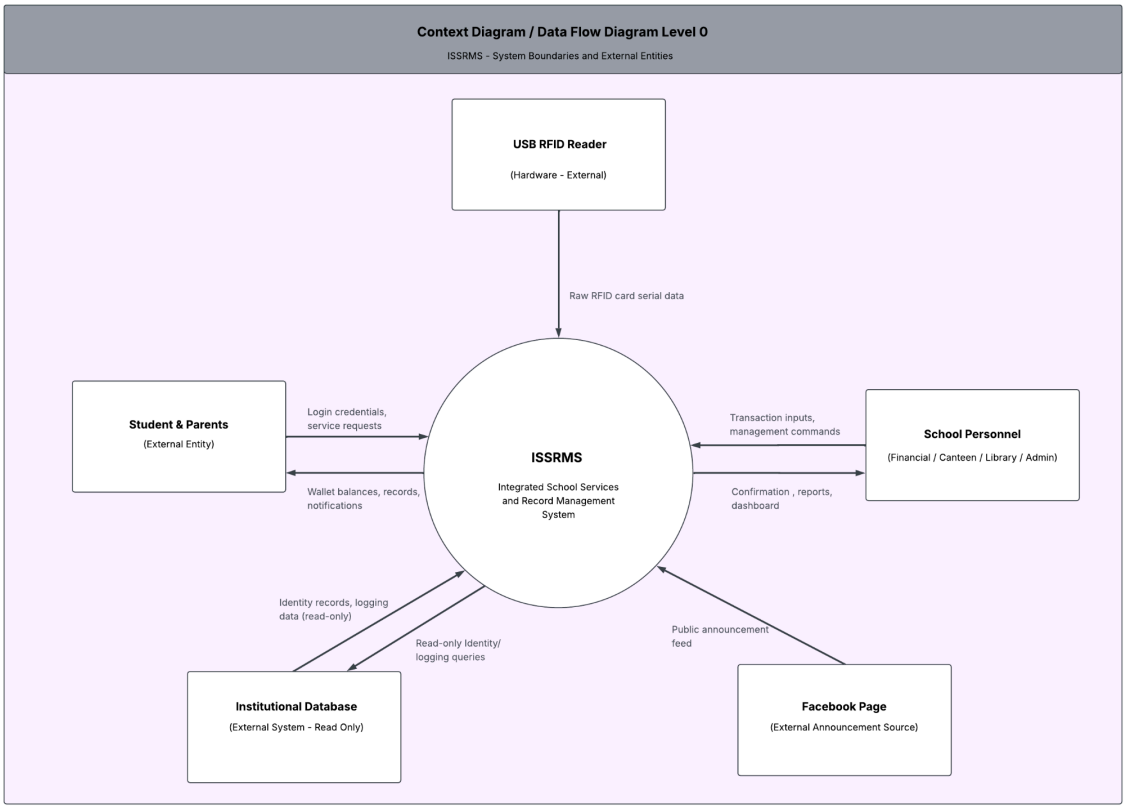


Figure 4.12. Context Diagram / Data Flow Diagram Level 0

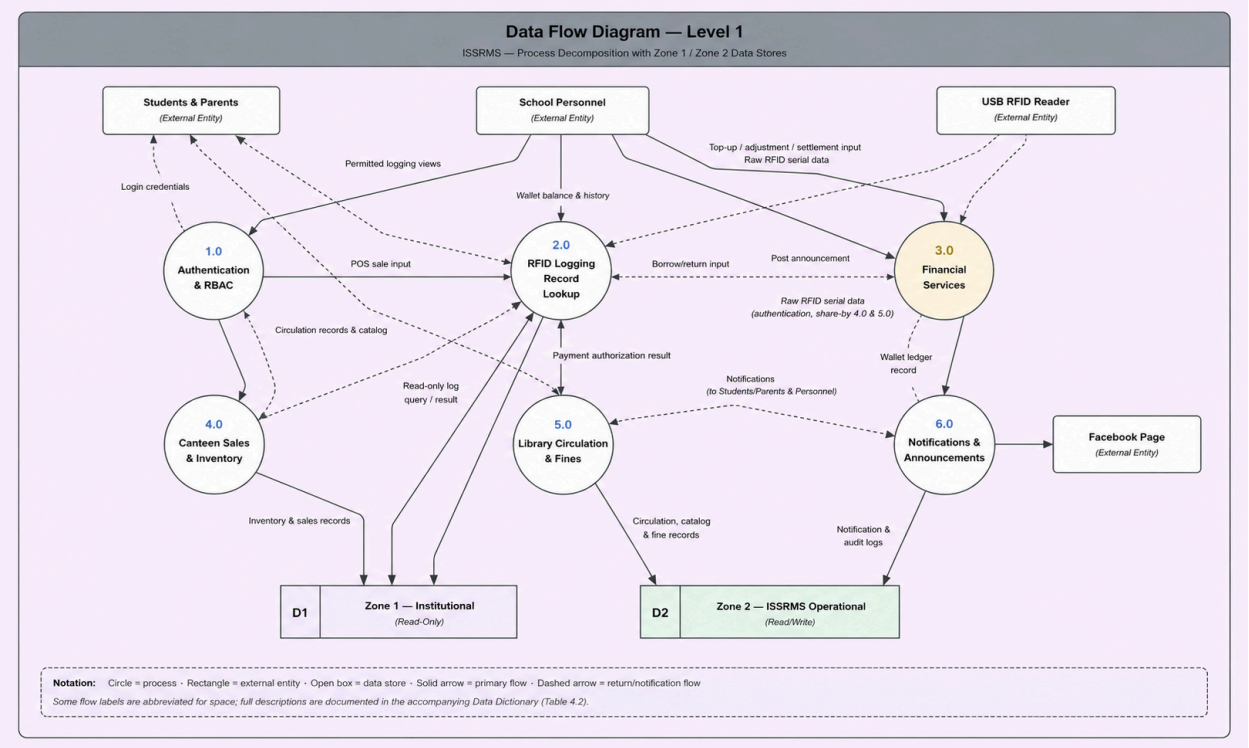


Figure 4.13. Data Flow Diagram Level 1

4.2.6 Data Design and Entity-Relationship Diagram

The data design uses two isolated data zones. Institutional identity and logging data remain read-only in Zone 1. System_Accounts, E_Wallet, Financial_Transactions, Financial_Ledger_Entries, canteen, library, notification, announcement, and audit records belong to Zone 2. The backend validates the institutional ID before creating or using an operational account. Each finalized financial transaction has at least two Financial_Ledger_Entries whose debit and credit totals are equal; neither entity has a library_circulation_id relationship or a Library_Fine transaction type. Library_Circulation stores fine_amount and fine_status independently. Figure 4.14 presents only the Zone 2 operational entities and their relationships; Zone 1 remains an external read-only integration source documented in the system architecture and interface descriptions. A certified reversal is recorded as a new immutable Financial_Transactions record rather than modifying or deleting the original transaction. The optional reversal_of_transaction_id foreign key links the reversal transaction to the

original Financial_Transactions record. Authorization, validation, balanced ledger posting, and duplicate-reversal prevention are enforced by the backend.

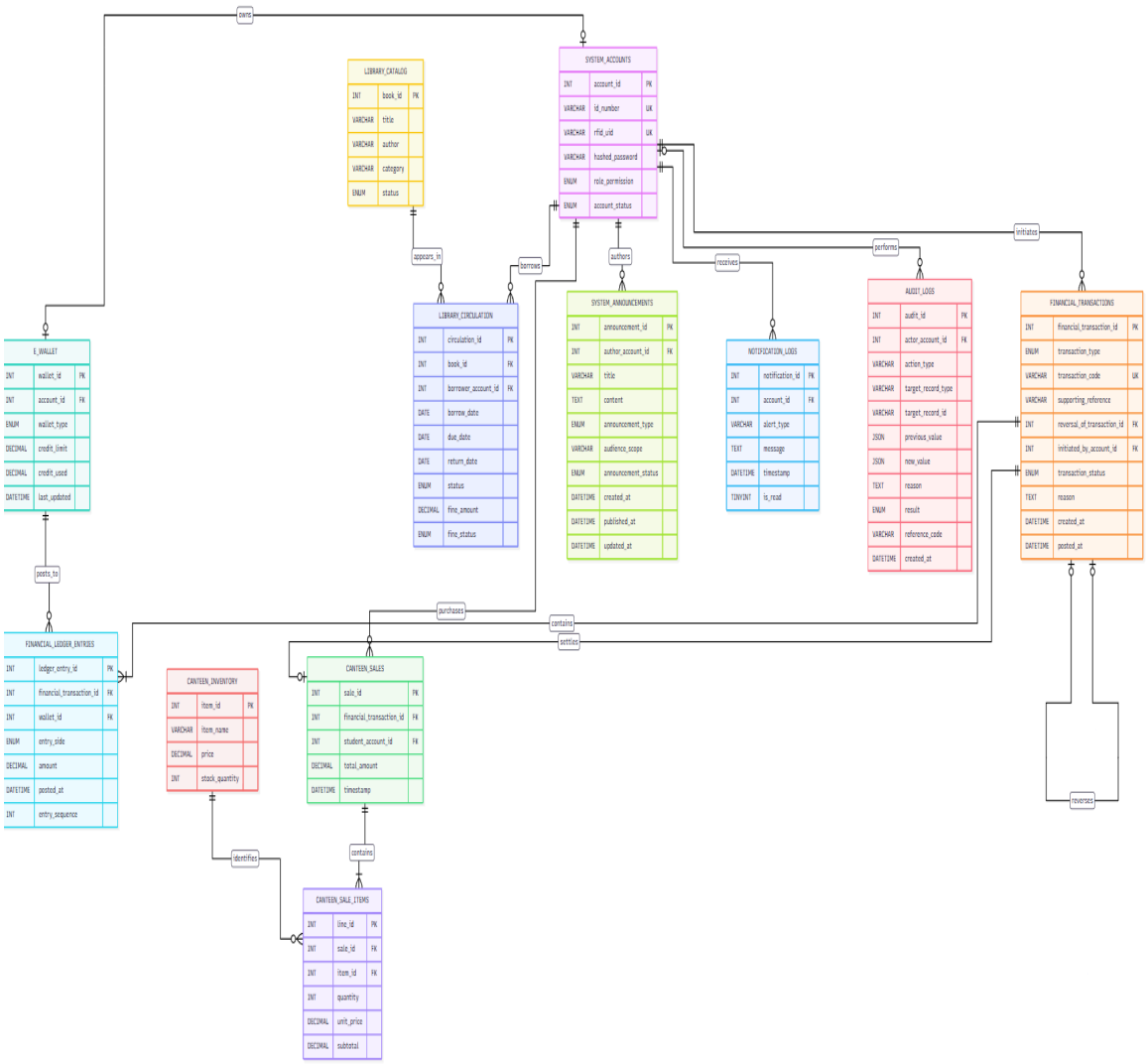


Figure 4.14. Entity-Relationship Diagram of ISSRMS

4.2.7 Data Dictionary

The data dictionary defines each table, attribute, data type, key, constraint, allowed value, default, and operational purpose. It shall be kept synchronized with the implemented schema and the updated ERD. Financial and security-sensitive fields require explicit constraints and audit behavior. Zone 1 structures are documented only as external read-only integration contracts and are not included as ISSRMS entities in Figure 4.14.

Institutional Database (Read-Only)

Table: Institutional_Registry
Description: The school's pre-existing database containing official student and staff identities.

Field Name	Data Type	Length	Key	Nullable	Description
id_number	VARCHAR	50	PK	No	The official ID number assigned by the institution.
first_name	VARCHAR	100		No	The legal first name of the individual.
last_name	VARCHAR	100		No	The legal last name of the individual.
demographics	VARCHAR	255		Yes	Grade level, section, or department designation.

Table/View: Institutional_Logging_Records

Description: Read-only representation of the school’s existing Gate Entry and Gate Exit records from Gate 1 and Gate 2 and Library Entry and Library Exit records from the Library. Confirm the approved physical table or view mapping before implementation.

Field Name	Data Type	Length	Key	Nullable	Description
log_id	INT	11	PK	No	Unique identifier for the institutional logging event.
id_number	VARCHAR	50	FK	No	Links to the Institutional_Registry.
log_type	ENUM	-		No	'Gate_Entry', 'Gate_Exit', 'Library_Entry', or 'Library_Exit'
log_location	VARCHAR	100		No	The gate, library point, or scanner where the ID was read.
timestamp	DATETIME	-		No	Exact date and time of the Gate Entry, Gate Exit, Library Entry, or Library Exit event.

ZONE 2: ISSRMS Operational Database (Read & Write)

Core System & Communications Module

Table: System_Accounts
Stores ISSRMS authentication, RFID, role, and account-status information. The id_number field uniquely identifies the institutional record associated with the account and serves as a logical foreign key to the institutional registry. The relationship is validated by the application rather than enforced through a cross-database foreign-key constraint.

Field Name	Data Type	Length	Key	Nullable	Description
account_id	INT	11	PK	No	Unique auto-incrementing identifier for the operational account.
id_number	VARCHAR	50	UK	No	Unique institutional identifier used as an application-level logical foreign key to INSTITUTIONAL_REGISTRY.id_number. The backend validates this

					reference, but no cross-database FOREIGN KEY constraint is created or enforced by MySQL.
rfid_uid	VARCHAR	100	UK	Yes	Hardware identifier from the physical RFID card scan.
hashed_password	VARCHAR	255		No	Securely hashed credential string (e.g., Argon2/Bcrypt).
role_permission	ENUM	-		No	Role constraint: 'Student_Parent', 'Registrar', 'Canteen_Personnel', 'Library_Personnel', or 'Administrator'.

account_status	ENUM	-		No	Account state: 'Pending', 'Active', 'Inactive', or 'Locked'.

4. Table: System_Announcements

Stores internal school bulletins published by authorized administrators.

Field Name	Data Type	Length	Key	Nullable	Description
announcement_id	INT	11	PK	No	Unique auto-incrementing identifier.
author_account_id	INT	11	FK	No	References the authorized account that authored the announcement.
title	VARCHAR	150		No	Headline or subject of the announcement.
content	TEXT	-		No	Full body text, capable of storing long paragraphs.

announcement_type	ENUM	-		No	Strict categorization ('Academic', 'Event', 'Admin').
audience_scope	VARCHAR	100		No	Role or audience permitted to view the announcement.
announcement_status	ENUM			No	'Draft', 'Active', or 'Inactive'; default is 'Draft'.
created_at	DATETIME			No	Date and time the record was created.
published_at	DATETIME			Yes	Date and time the announcement became Active.
updated_at	DATETIME			Yes	Date and time of the latest authorized update.

Table: Notification_Logs

Centralized asynchronous messaging system that alerts users to cross-module events.

Field Name	Data Type	Length	Key	Nullable	Description
notification_id	INT	11	PK	No	Unique auto-incrementing identifier.
account_id	INT	11	FK	No	Links to the user receiving the alert.
alert_type	VARCHAR	50		No	Category (e.g., 'Financial', 'Library', 'Security').
message	TEXT	-		No	The body of the notification alert.
timestamp	DATETIME	-		No	Date and time the alert was triggered.
is_read	TINYINT	1		No	Boolean flag indicating user acknowledgment (1 = Read, 0 = Unread).

Table: Audit_Logs

Description: Stores immutable, read-only audit events for authentication, account, financial, inventory, library-fine, announcement, and other protected system actions.

Field Name	Data Type	Length	Key	Nullable	Description
audit_id	INT	11	PK	No	Unique auto-incrementing identifier for the audit event.
actor_account_id	INT	11	FK	Yes	References the System_Accounts actor when known; may be null for unauthenticated or system-generated events.
action_type	VARCHAR	100		No	Controlled action name such as Login, Top_Up, Credit_Settlement, Fine_Adjustment, or Role_Update.
target_record_type	VARCHAR	100		No	Identifies the affected entity or protected resource.
target_record_id	VARCHAR	100		Yes	Identifier of the affected wallet, transaction, circulation, account, announcement, or other record.
previous_value	JSON	-		Yes	Approved before-state snapshot when the action changes protected data.

new_value	JSON	-		Yes	Approved after-state snapshot when the action changes protected data.
reason	TEXT	-		Yes	Required justification for adjustments, reversals, waivers, settlements, and other sensitive actions when applicable.
result	ENUM	-		No	'Success', 'Rejected', or 'Failed'.
reference_code	VARCHAR	100		Yes	Related transaction, circulation, announcement, or security reference when applicable.
created_at	DATETIME	-		No	Date and time the auditable event occurred.

Financial Management Module

Table: E_Wallet

Stores student wallets and controlled system-owned accounting wallets, approved credit limits, and current canteen credit liability. Active cash balance is derived from Financial_Ledger_Entries.

Field Name	Data Type	Length	Key	Nullable	Description

wallet_id	INT	11	PK	No	Unique auto-incrementing identifier.
account_id	INT	11	FK, UK	Yes	Links to the student owner; null only for a controlled system-owned counterpart wallet.
wallet_type	ENUM	-		No	Required wallet classification: 'Student', 'Cash_Collection', 'Canteen_Revenue', 'Credit_Receivable', or 'Adjustment_Reserve'. Active cash balance is derived from Financial_Ledger_Entries, not stored here.
credit_limit	DECIMAL	10,2		No	The absolute static overdraft

					threshold allowed by school policy.
credit_used	DECIMAL	10,2		No	Current outstanding canteen credit liability; default 0.00.
last_updated	DATETIME	-		No	Timestamp of the most recent credit-limit or credit-used update.

Table: Financial_Ledger_Entries

Immutable double-entry ledger lines. Each line posts one debit or credit amount to one student or controlled system-owned wallet and belongs to a Financial_Transactions header.

Field Name	Data Type	Length	Key	Nullable	Description
ledger_entry_id	INT	11	PK	No	Unique auto-incrementing identifier for the immutable ledger-entry line.

financial_transaction_id	INT	11	FK	No	Links this line to its immutable Financial_Transactions header.
wallet_id	INT	11	FK	No	Wallet affected by this ledger-entry line; student or controlled system-owned wallet.
entry_side	ENUM	-		No	Entry direction constrained to 'Debit' or 'Credit'. The transaction header defines the action type; no Library_Fine entry is allowed.
amount	DECIMAL	10,2		No	Monetary amount posted on this debit or credit entry line.
posted_at	DATETIME	-		No	Timestamp when this immutable

					ledger-entry line was posted.
entry_sequence	INT	-		No	Immutable line order within the financial transaction.

Table: Financial_Transactions

Immutable header record for one financial event. A finalized header must have at least two Financial_Ledger_Entries whose total debit amount equals total credit amount.

Field Name	Data Type	Length	Key	Nullable	Description
financial_transaction_id	INT	11	PK	No	Immutable header identifier for one financial event.
transaction_type	ENUM	-		No	Top_Up, Purchase, Adjustment, Reversal, or Credit_Settlement; no Library_Fine.

transaction_code	VARCHAR	64	UK	No	Unique system-generated reference used for audit and receipt review.
supporting_reference	VARCHAR	100		Yes	Optional official receipt or payment reference for manual settlement or review.
reversal_of_transaction_id	INT	11	FK	Yes	Self-reference to the original Financial_Transactions record corrected by a reversal.
initiated_by_account_id	INT	11	FK	No	Authorized System_Accounts actor who

					initiated the transaction .
transaction_status	ENUM	-		No	Posting state: 'Pending' or 'Posted'.
reason	TEXT	-		Yes	Required for an adjustment , reversal, or manual settlement when applicable.
created_at	DATETIME	-		No	Date and time the transaction header was created.
posted_at	DATETIME	-		Yes	Date and time the balanced transaction was finalized;

					null while pending.

Canteen Management Module

Table: Canteen_Inventory

Master list of items available for purchase and current stock levels.

Field Name	Data Type	Length	Key	Nullable	Description
item_id	INT	11	PK	No	Unique auto-incrementing identifier for the product.
item_name	VARCHAR	150		No	Description/Name of the food or beverage.
price	DECIMAL	10,2		No	The standard retail cost of the item.
stock_quantity	INT	11		No	Current physical count available for sale.

Table: Canteen_Sales

Header table recording the overall transaction of a point-of-sale checkout.

Field Name	Data Type	Length	Key	Nullable	Description
sale_id	INT	11	PK	No	Unique auto-incrementing identifier for the receipt.
financial_transaction_id	INT	11	FK, UK	No	References the Financial_Transactions header that settled the sale.
student_account_id	INT	11	FK	No	References the System_Accounts record of the student purchaser.
total_amount	DECIMAL	10,2		No	The final summed cost of all items in the transaction.
timestamp	DATETIME	-		No	Date and time the sale was finalized.

Table: Canteen_Sale_Items

Line-item detail table linking individual inventory products to a specific sale receipt.

Field Name	Data Type	Length	Key	Nullable	Description
line_id	INT	11	PK	No	Unique auto-incrementing identifier for the line item.
sale_id	INT	11	FK	No	Links to the parent transaction in Canteen_Sales.
item_id	INT	11	FK	No	Links to the purchased product in Canteen_Inventory.
quantity	INT	11		No	Amount of this specific item purchased.
unit_price	DECIMAL	10,2		No	Unit price recorded at the time of sale so later price changes do not alter the receipt.
subtotal	DECIMAL	10,2		No	Computed value (price × quantity) for this specific line.

Library Management Module

Table: Library_Catalog

The physical master list of all books stored in the library system.

Field Name	Data Type	Length	Key	Nullable	Description
book_id	INT	11	PK	No	Unique auto-incrementing identifier for the physical book.
title	VARCHAR	255		No	Title of the publication.
author	VARCHAR	150		No	Primary author of the publication.
category	VARCHAR	100		No	Subject, genre, or Dewey Decimal classification.
status	ENUM	-		No	Current physical state ('Available', 'Borrowed', 'Lost').

Table: Library_Circulation

Tracks library borrowing, return status, and operational fine information independently of the canteen e-wallet.

Field Name	Data Type	Length	Key	Nullable	Description
circulation_id	INT	11	PK	No	Unique auto-incrementing identifier for the borrow event.
book_id	INT	11	FK	No	Links to the specific book checked out.
borrower_account_id	INT	11	FK	No	References the System_Accounts record of the student borrower.
borrow_date	DATE	-		No	The day the book was issued.
due_date	DATE	-		No	The scheduled return date used by the library module when determining overdue penalties.
return_date	DATE	-		Yes	Actual return date; null until the material is returned.
status	ENUM	-		No	Circulation state ('Borrowed', 'Returned', or 'Lost').

fine_amount	DECIMAL	10,2		No	Operational overdue penalty amount maintained by the library module; default 0.00.
fine_status	ENUM	-		No	Fine state constrained to 'No_Fine', 'Unpaid', or 'Paid'.

4.2.8 System Architecture and Deployment Design

The deployment design separates external Student/Parent access, the protected web layer, and the internal application and data services. Student and Parent browsers access the portal through HTTPS, the school's perimeter firewall, and a reverse proxy or web server. Authorized personnel use the School Operations desktop application through the internal school network, with Electron providing controlled local access to supported USB RFID readers. The Node.js backend and both data zones remain inside the secured school environment, and clients do not connect directly to either data zone.

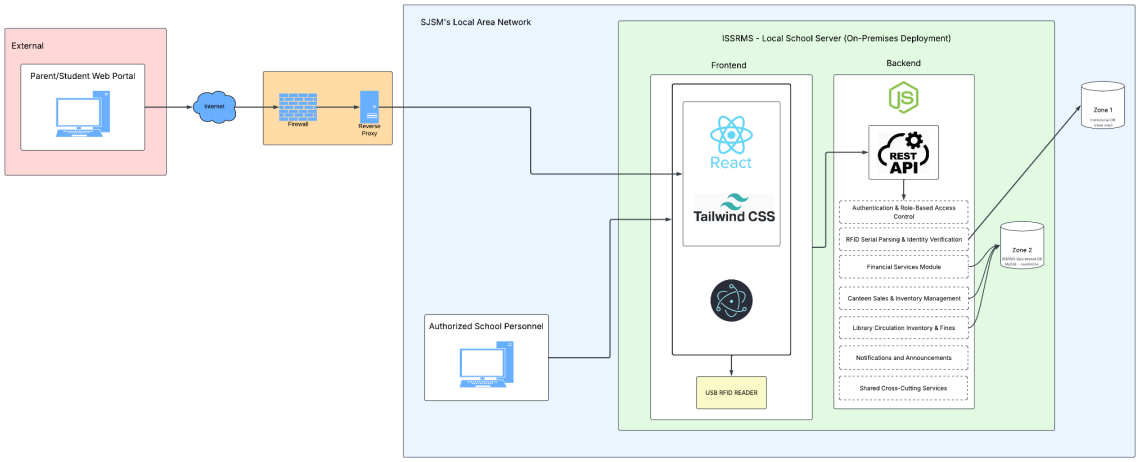


Figure 4.15. System Architecture and Network Topology

4.2.9 User Interface Design

The user interface design shall present role-specific navigation, clear transaction totals, confirmation prompts for protected financial actions, readable errors, and transaction references. Financial screens must distinguish top-up, purchase, credit-used, settlement, adjustment, and reversal actions. Library screens must keep fines within the library workflow. Logging screens must visibly state that their records are read-only. The Administrator interface shall provide Zone 2 account and role management, approved non-financial settings, announcement management, read-only Institutional Logging Records, and read-only system-wide Audit Logs. Financial, canteen, and library operational controls shall not be displayed to the Administrator.

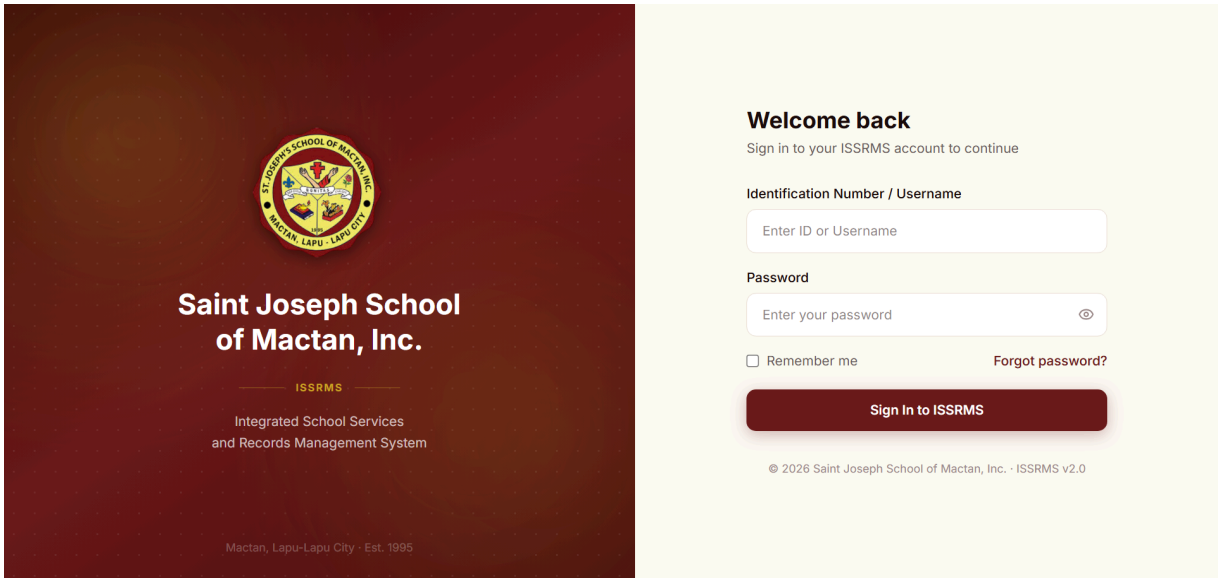


Figure 4.16.1. Login Page Wireframe

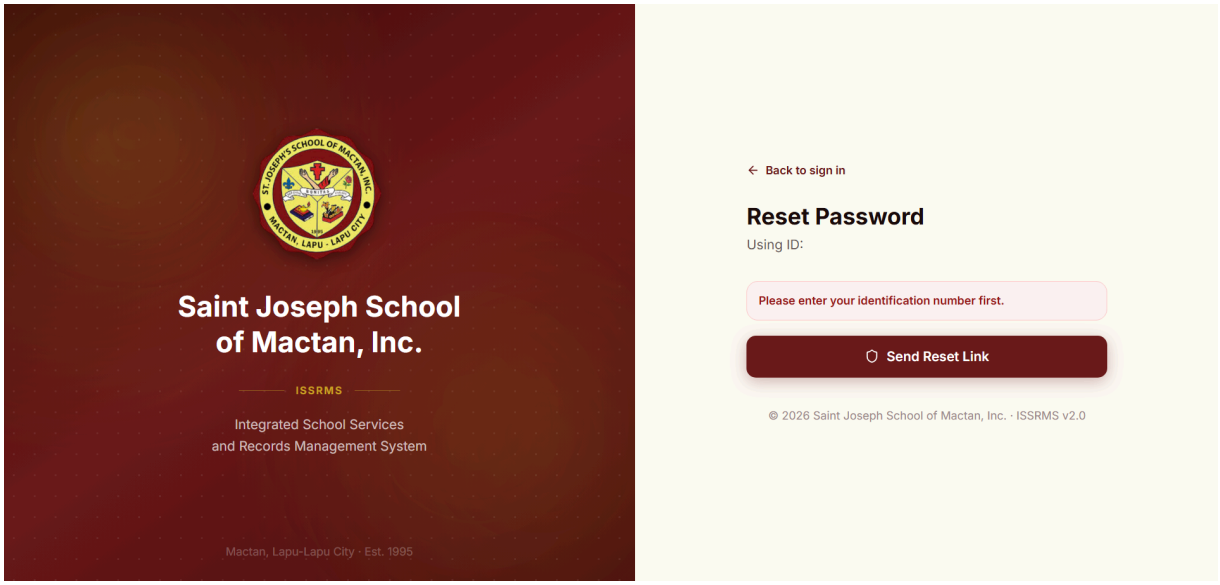


Figure 4.16.2. Forgot Password Page Wireframe

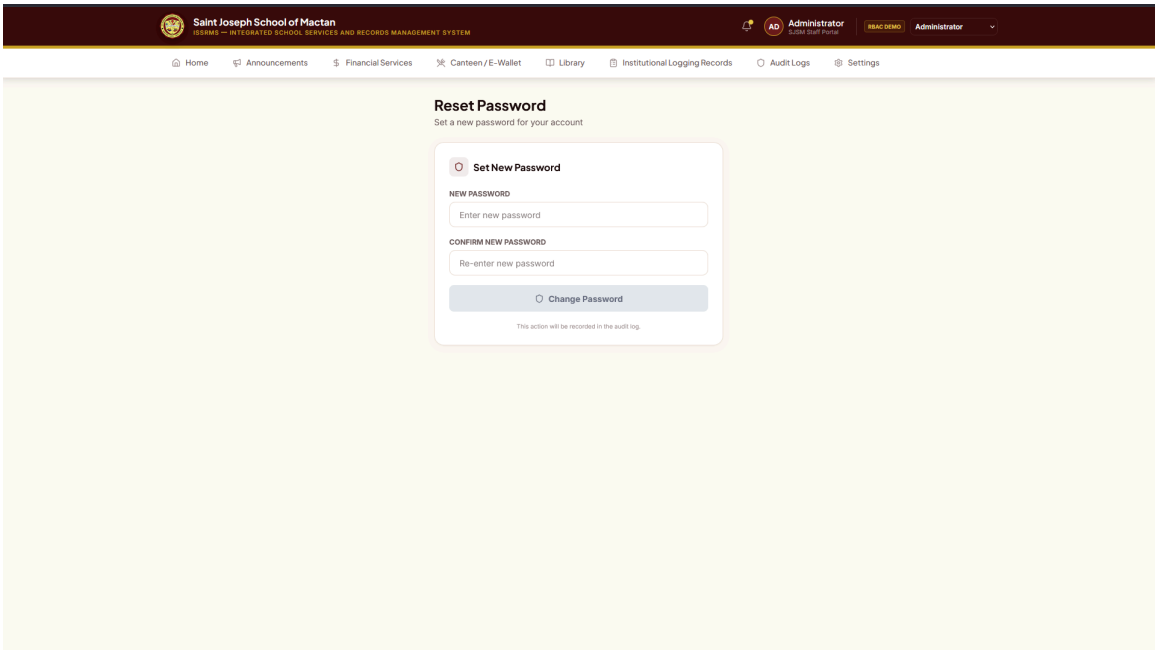


Figure 4.16.3. Reset Password Page Wireframe

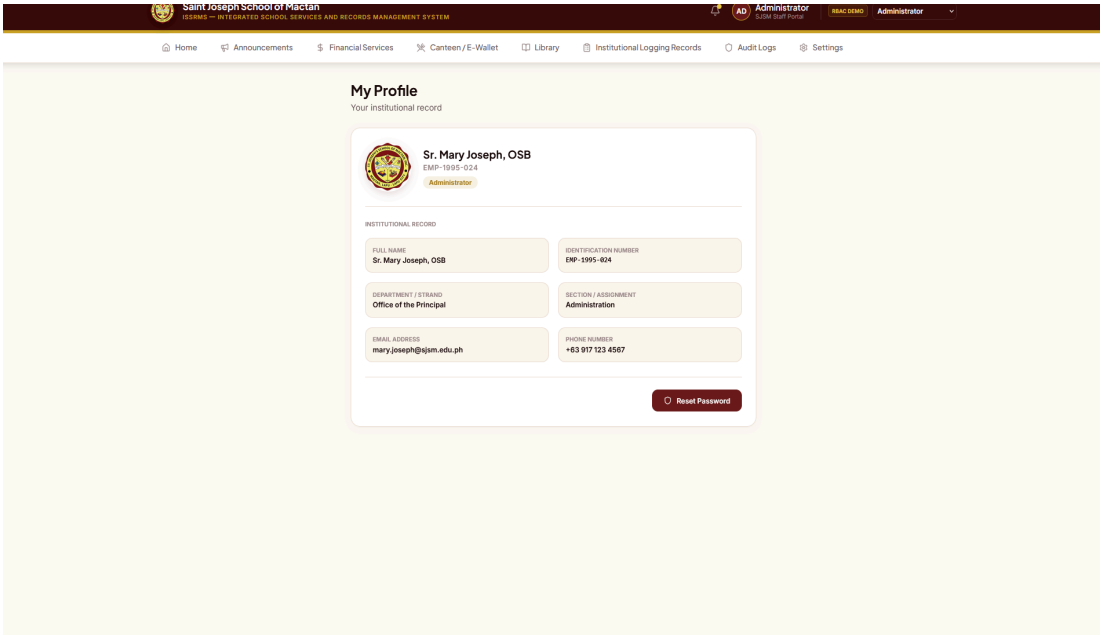


Figure 4.16.4. My Profile Page

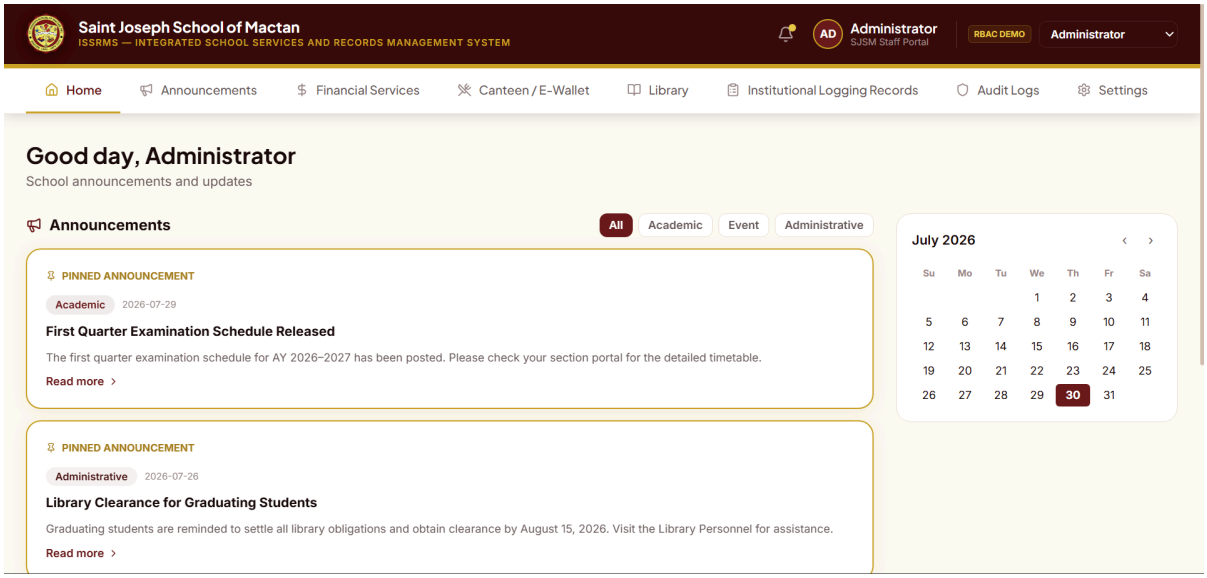


Figure 4.16.5. Administrator Home Page Wireframe

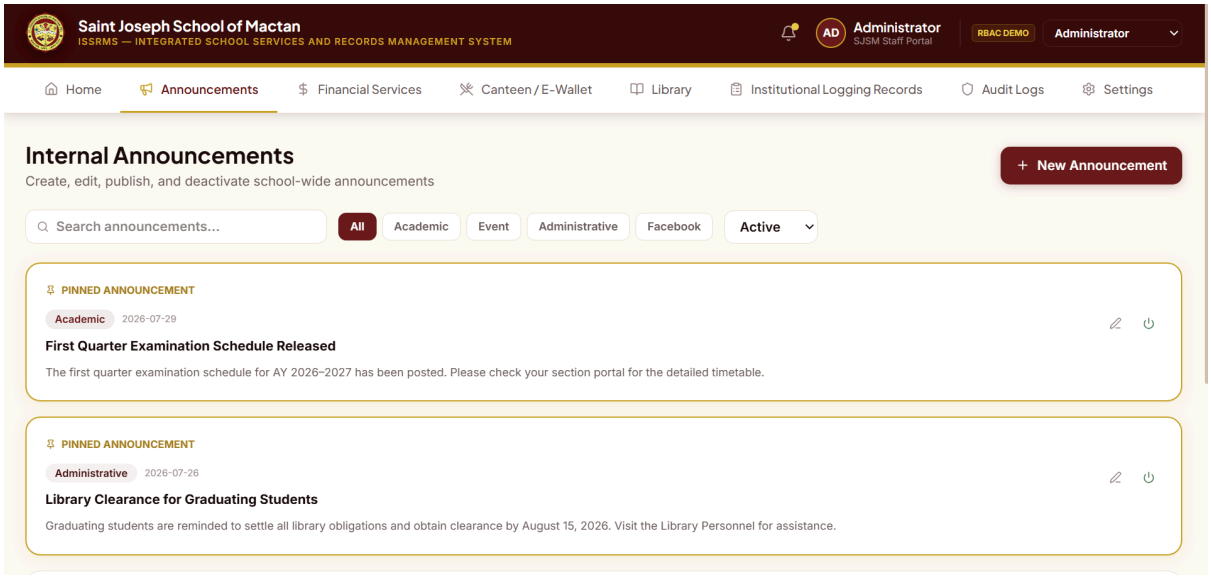


Figure 4.16.6. Administrator Announcements Wireframe

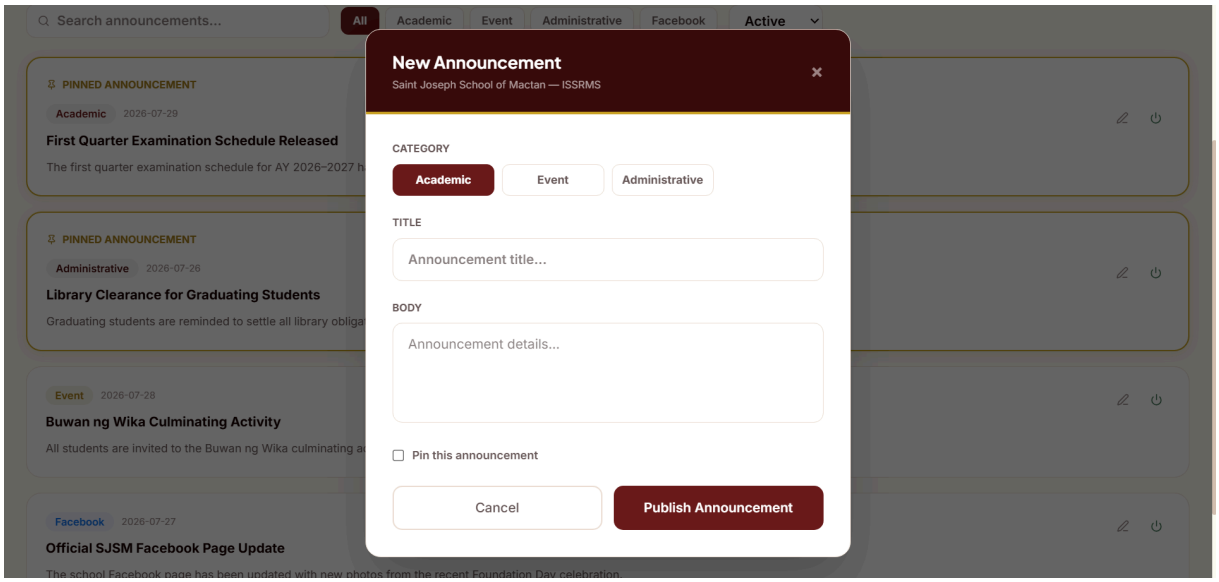


Figure 4.16.7. Administrator New Announcement Wireframe

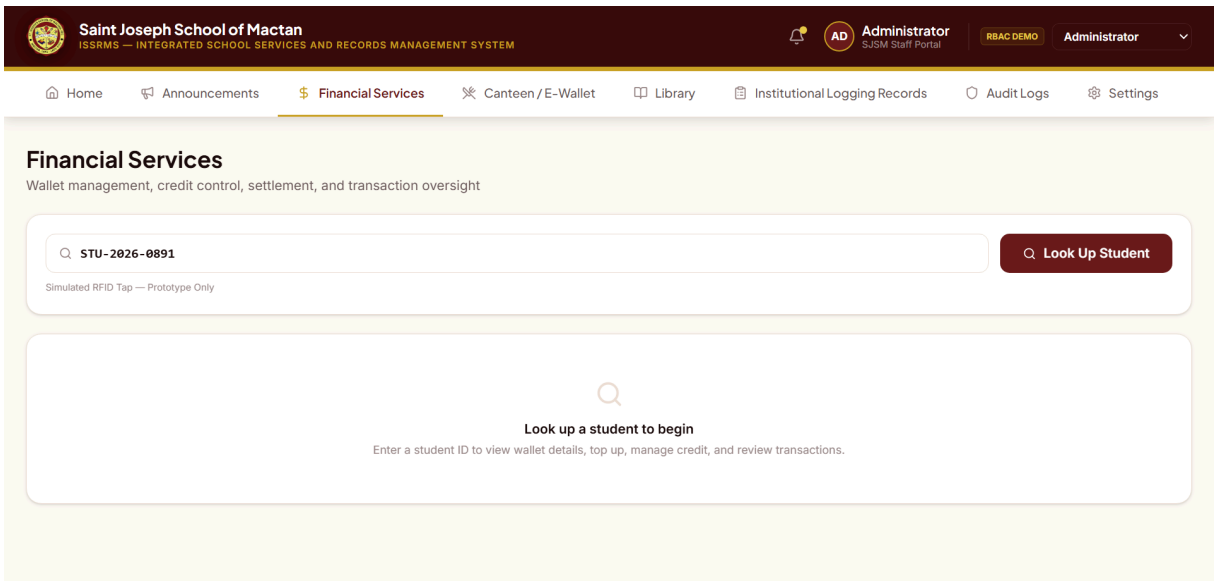


Figure 4.16.8. Registrar Financial Services Wireframe

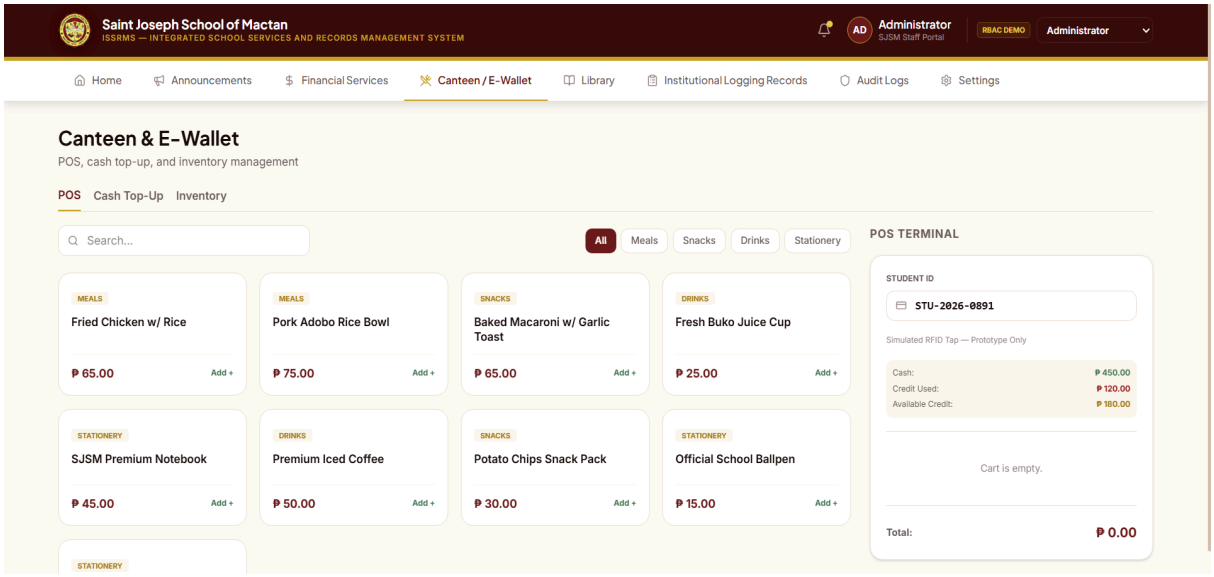


Figure 4.16.9. Canteen Personnel POS Wireframe

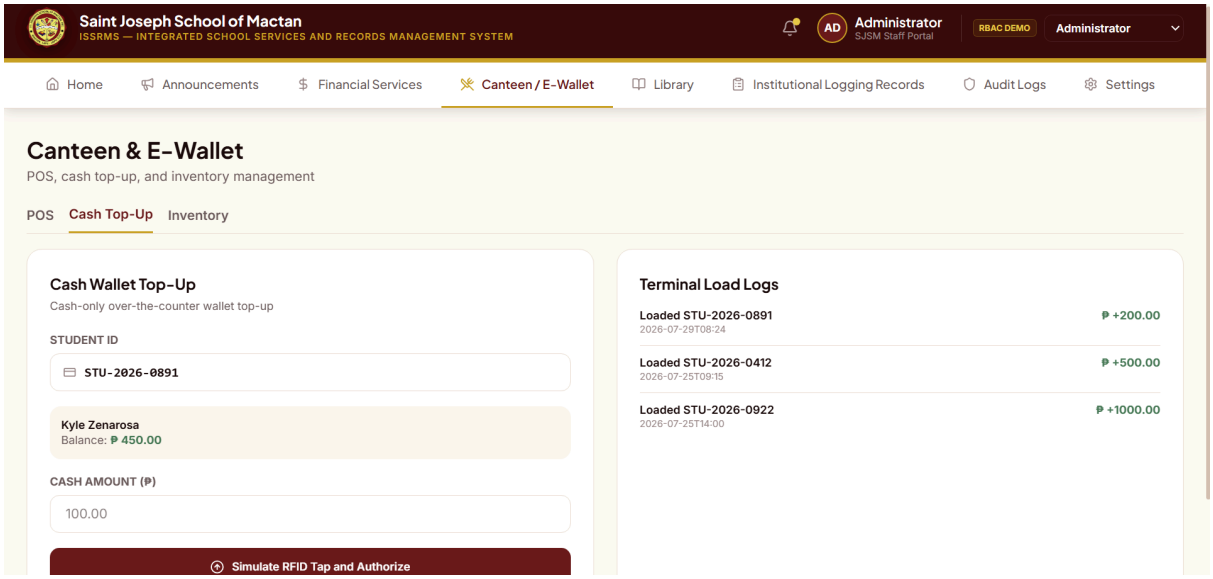


Figure 4.16.10. Canteen Personnel Cash Top-up Wireframe

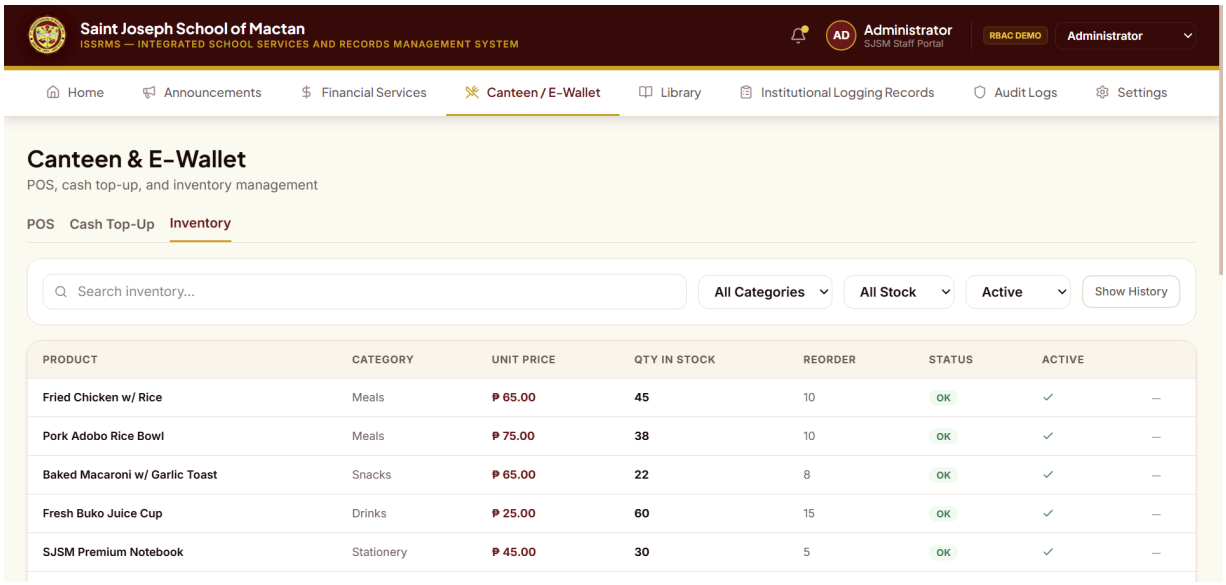


Figure 4.16.11. Canteen Personnel Canteen Inventory Management Wireframe

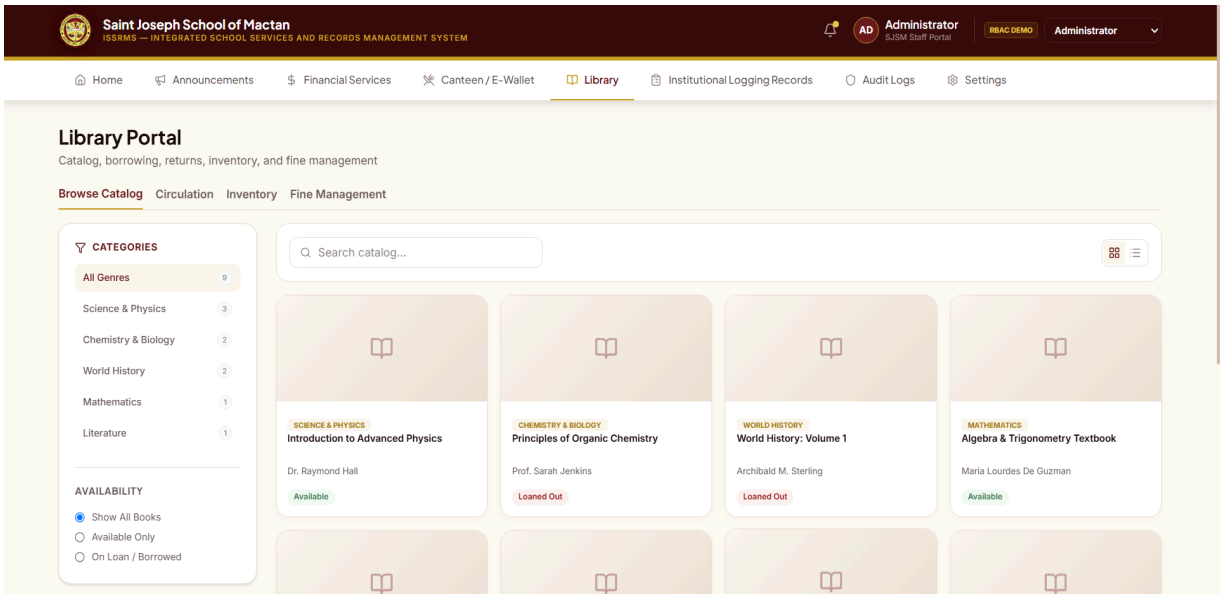


Figure 4.16.12. Library Personnel Library Catalog Wireframe

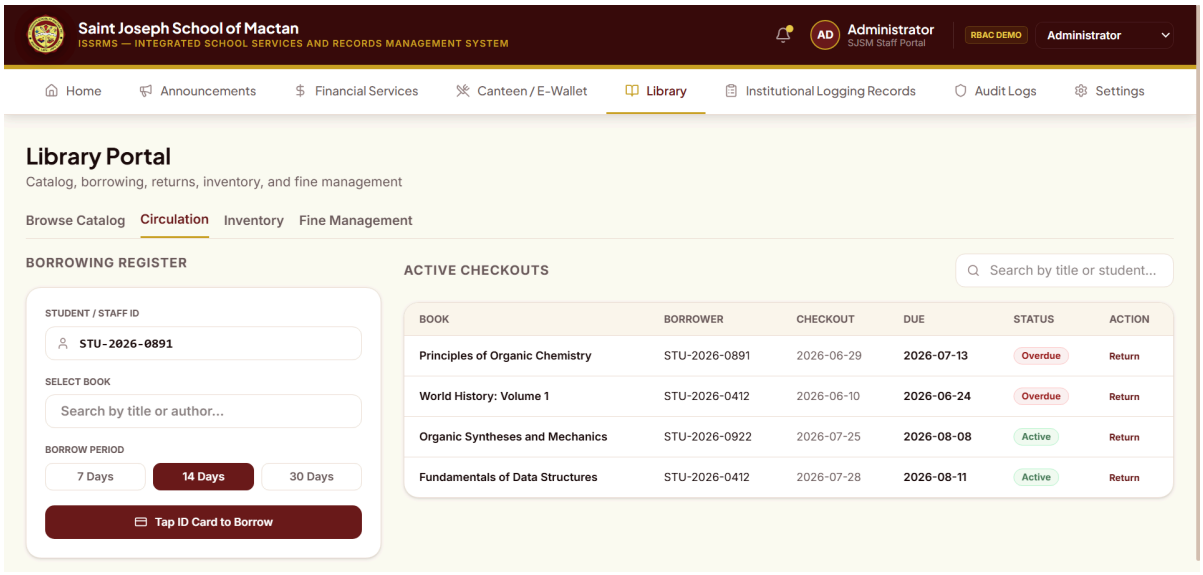


Figure 4.16.13. Library Personnel Library Circulation Wireframe

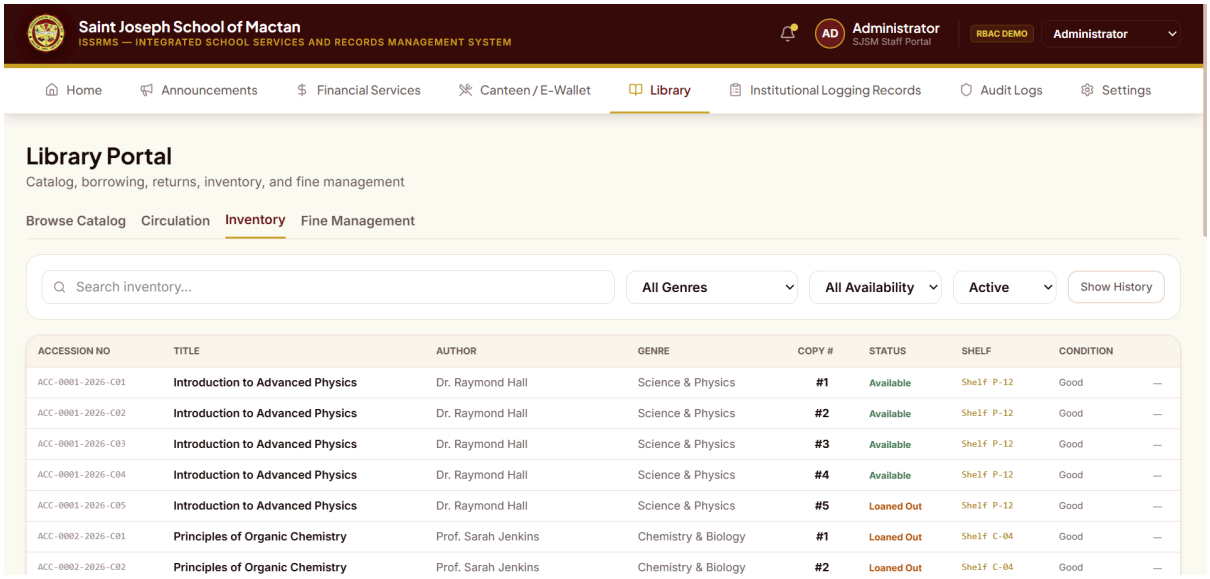


Figure 4.16.14. Library Personnel Library Inventory Wireframe

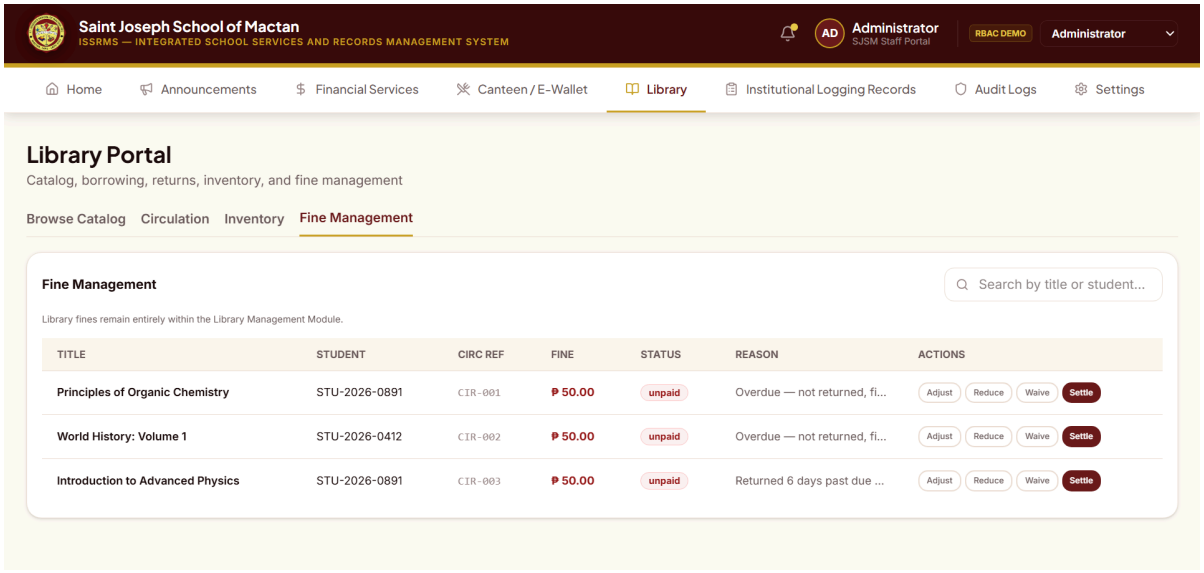


Figure 4.16.15. Library Personnel Fine Management Wireframe

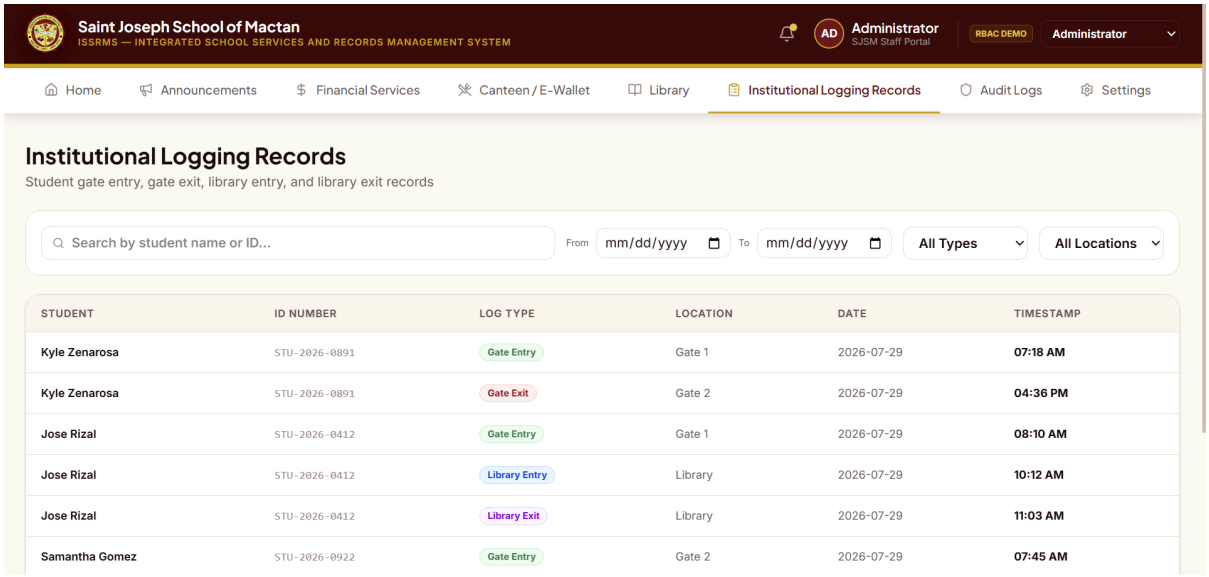


Figure 4.16.16. Administrator Institutional Logging Records Wireframe (Read-Only)

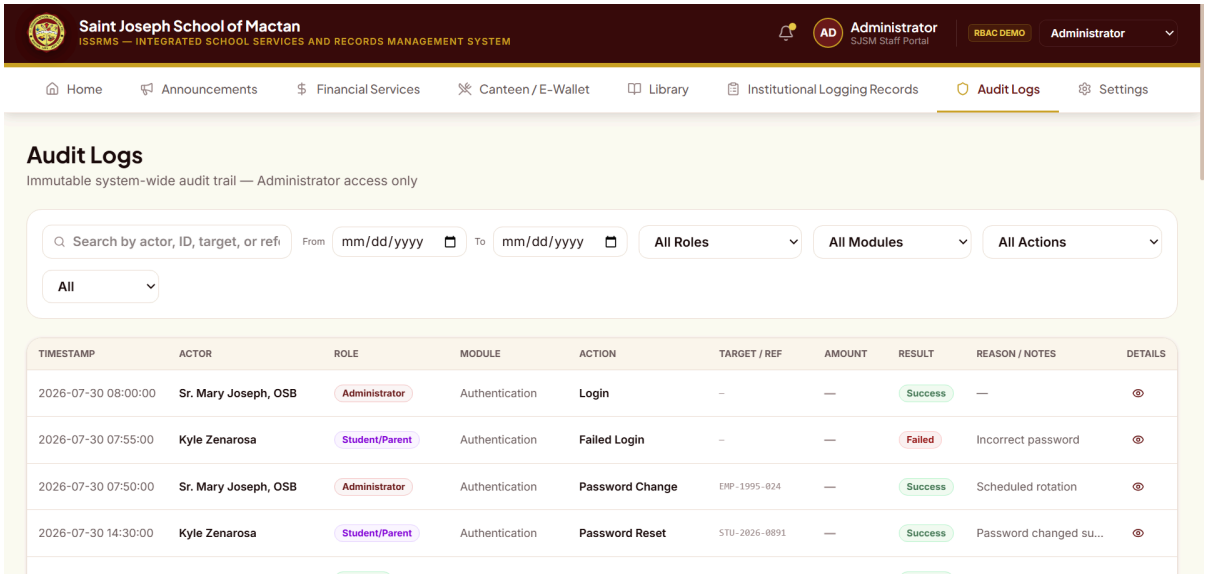


Figure 4.16.17. Administrator System-Wide Audit Logs Wireframe (Read-Only)

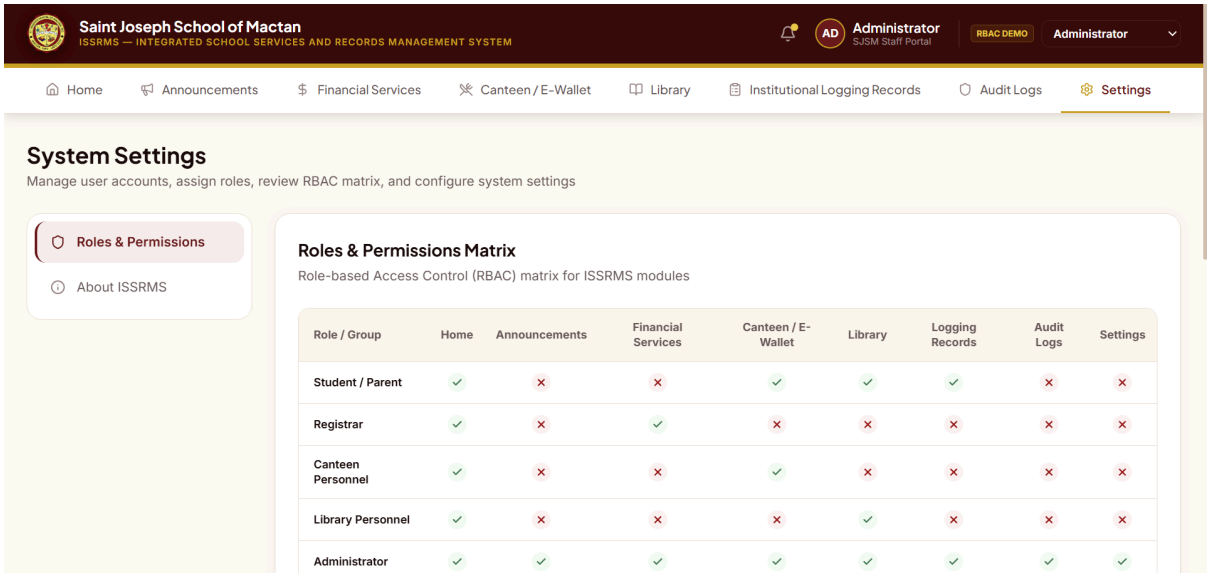


Figure 4.16.18. Administrator Approved Non-Financial Settings Wireframe

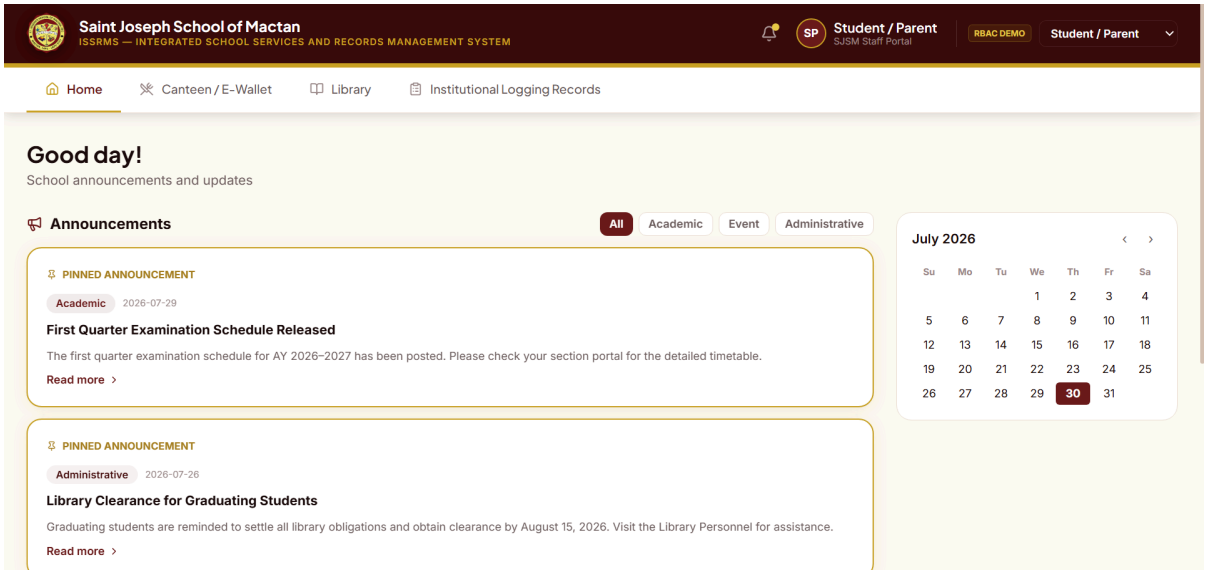


Figure 4.16.19. Student/Parent Home Page Wireframe

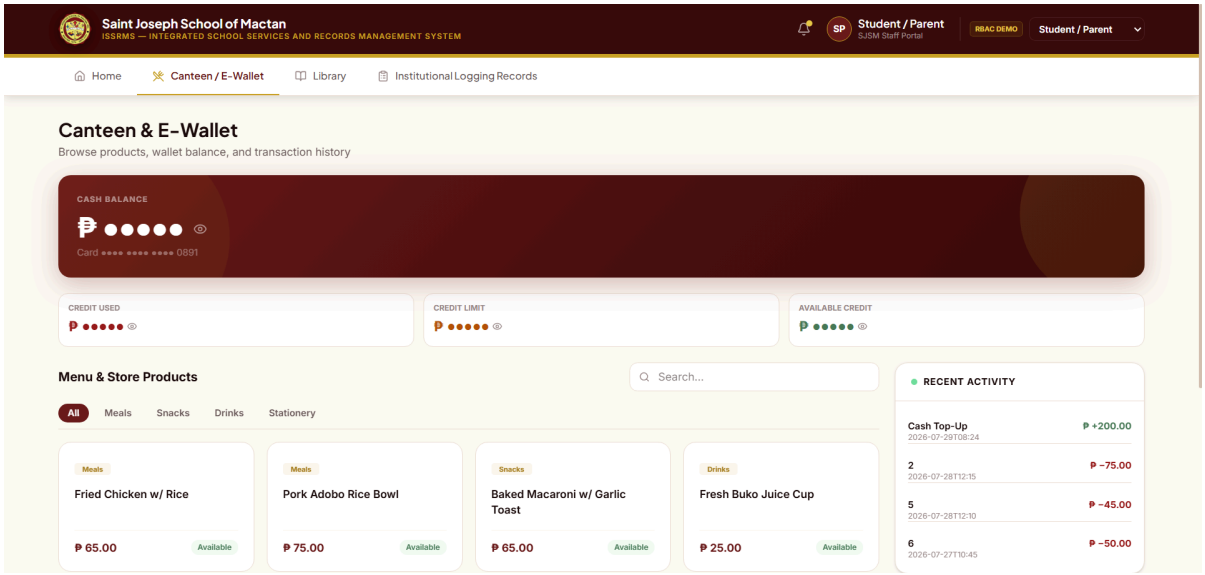


Figure 4.16.20. Student/Parent Canteen E-Wallet Wireframe

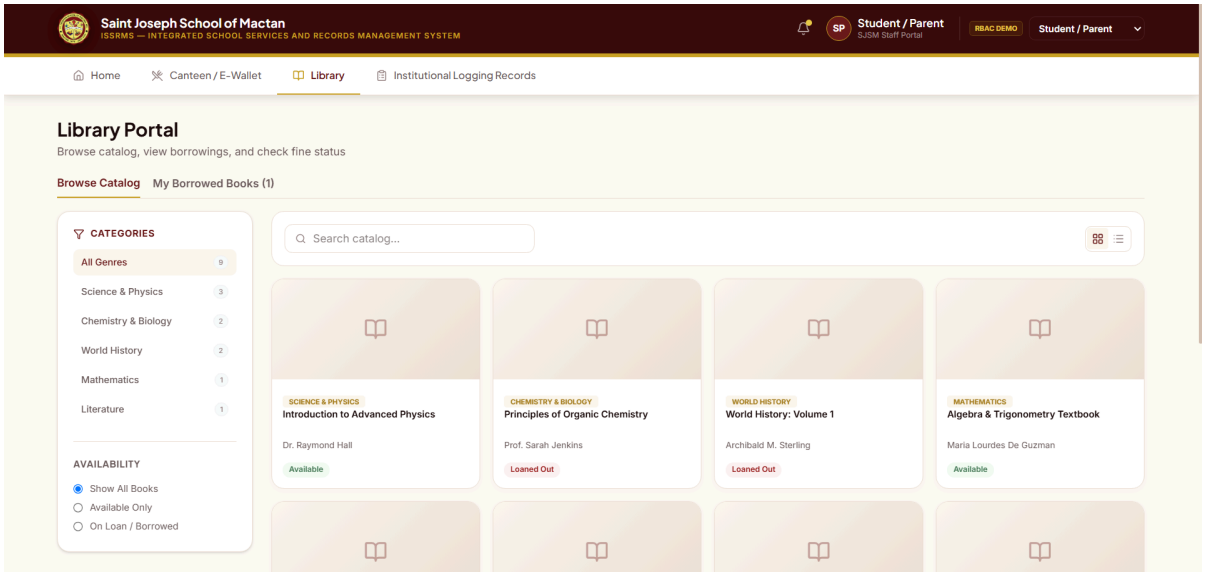


Figure 4.16.21. Student/Parent Library Catalog Wireframe

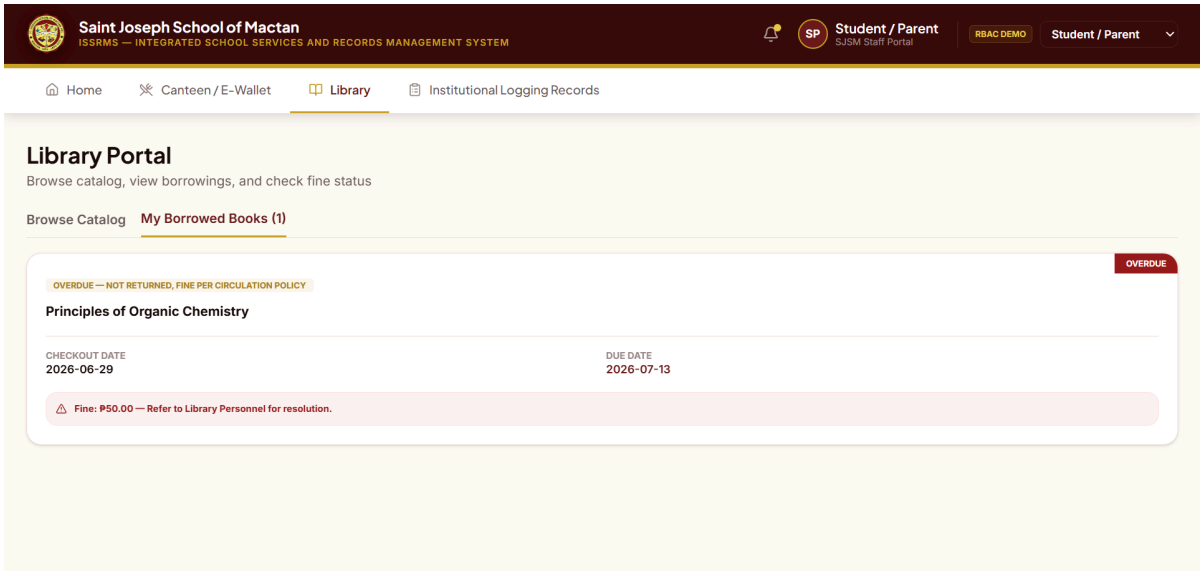


Figure 4.16.22. Student/Parent Library Borrowed Books Wireframe

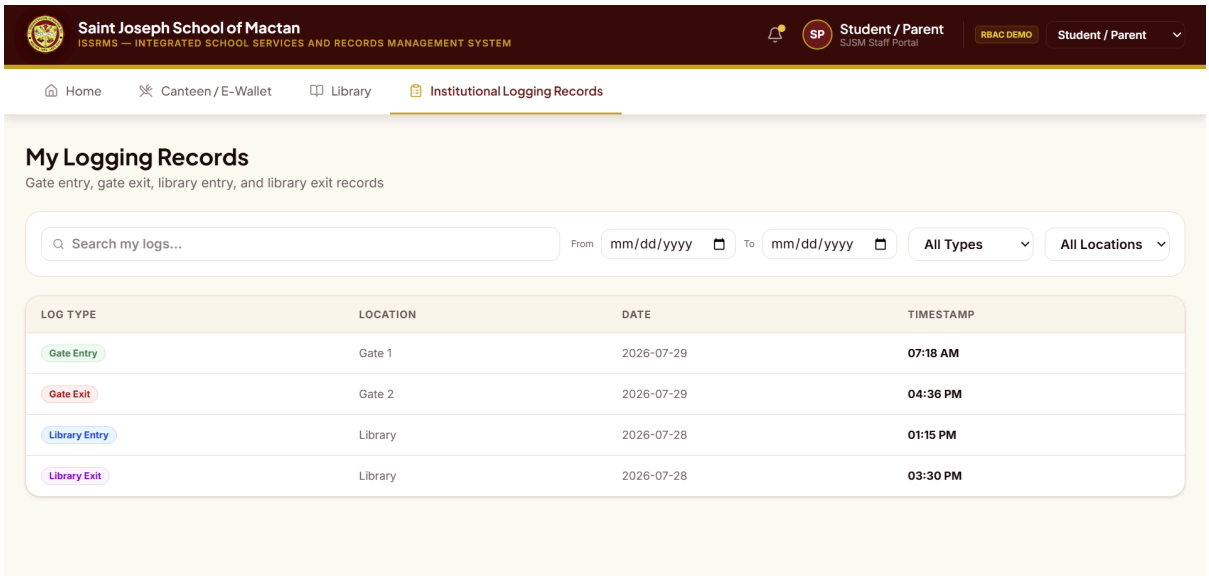


Figure 4.16.23. Student/Parent Institutional Logging Records Wireframe

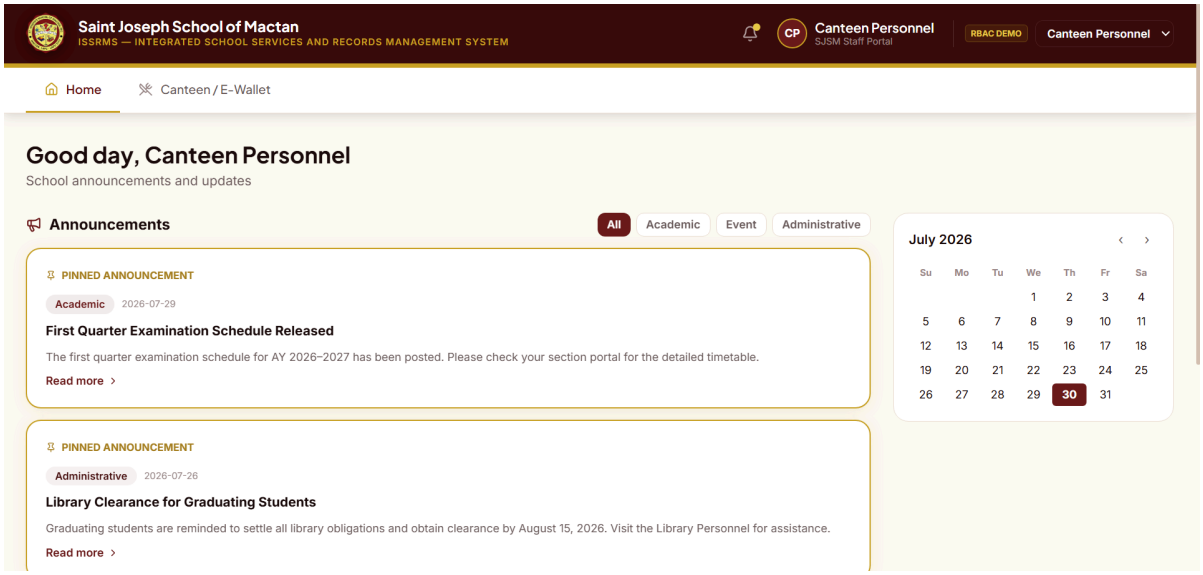


Figure 4.16.24. Canteen Personnel Home Page Wireframe

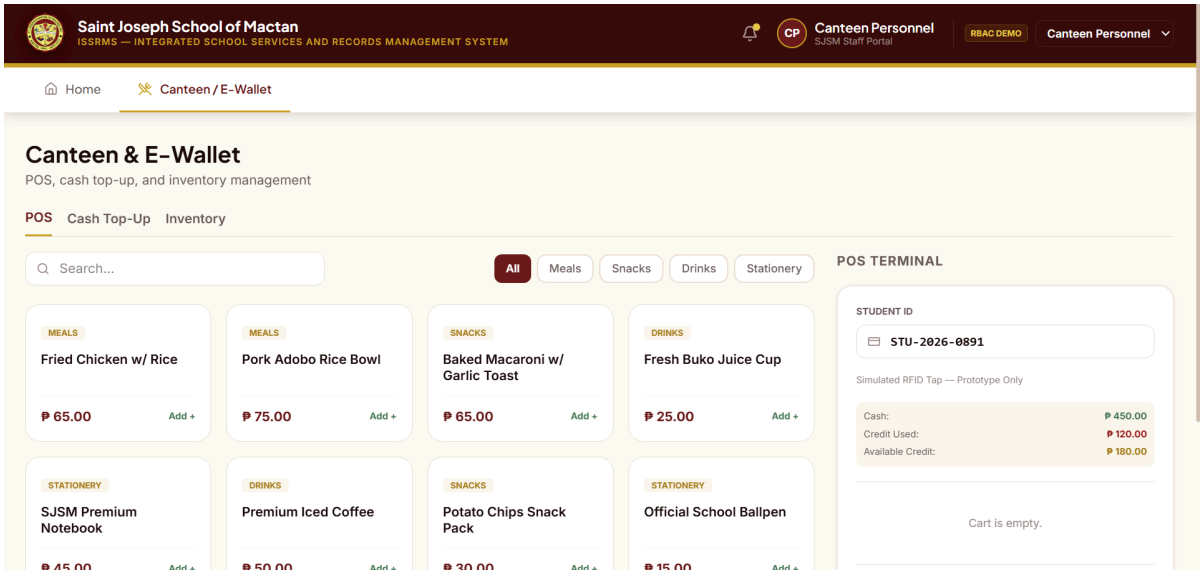


Figure 4.16.25. Canteen Personnel POS Wireframe

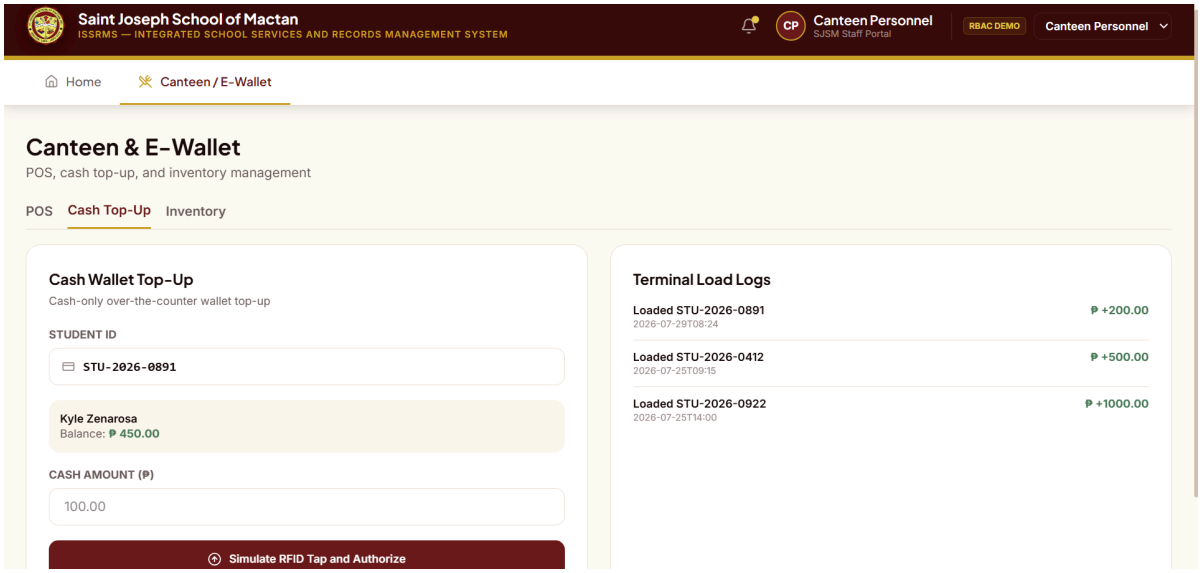


Figure 4.16.26. Canteen Personnel Cash Top-up Wireframe

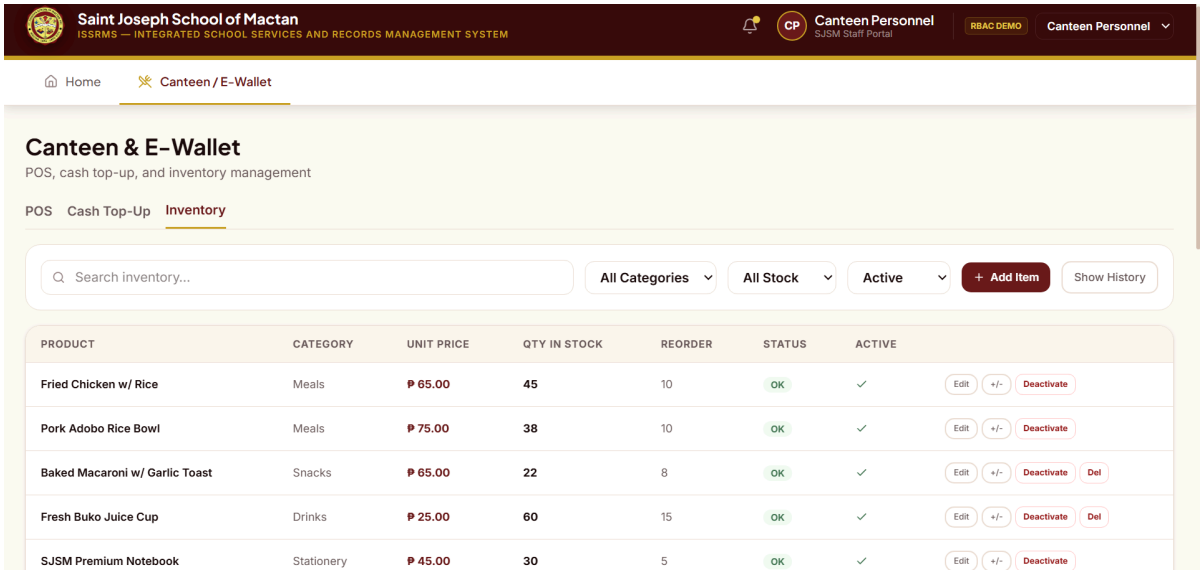


Figure 4.16.27. Canteen Personnel Canteen Inventory Management Wireframe

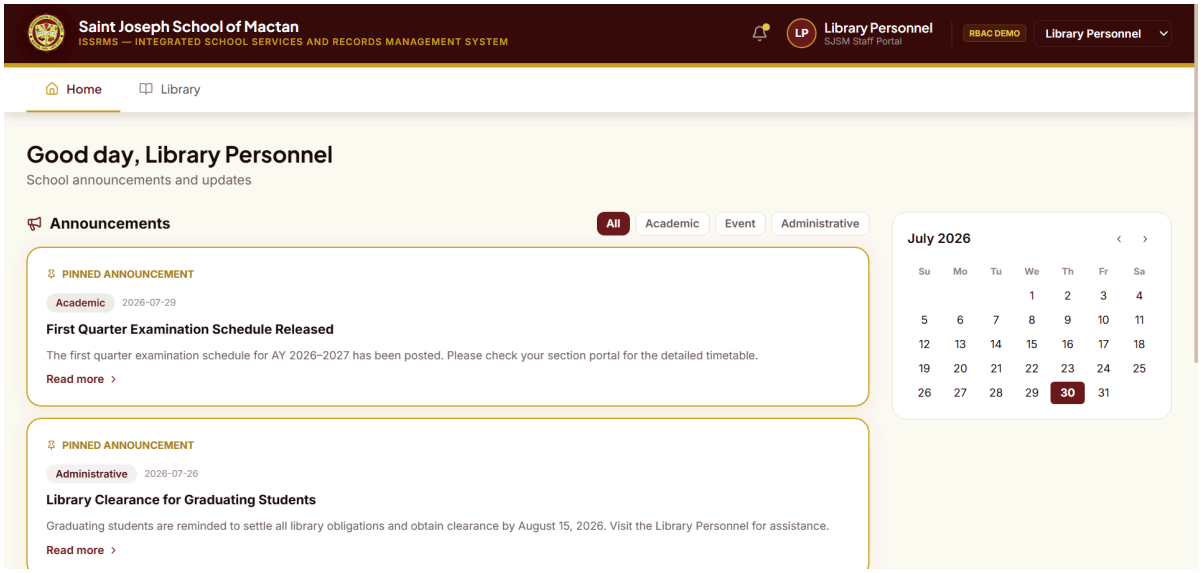


Figure 4.16.28. Library Personnel Home Page Wireframe

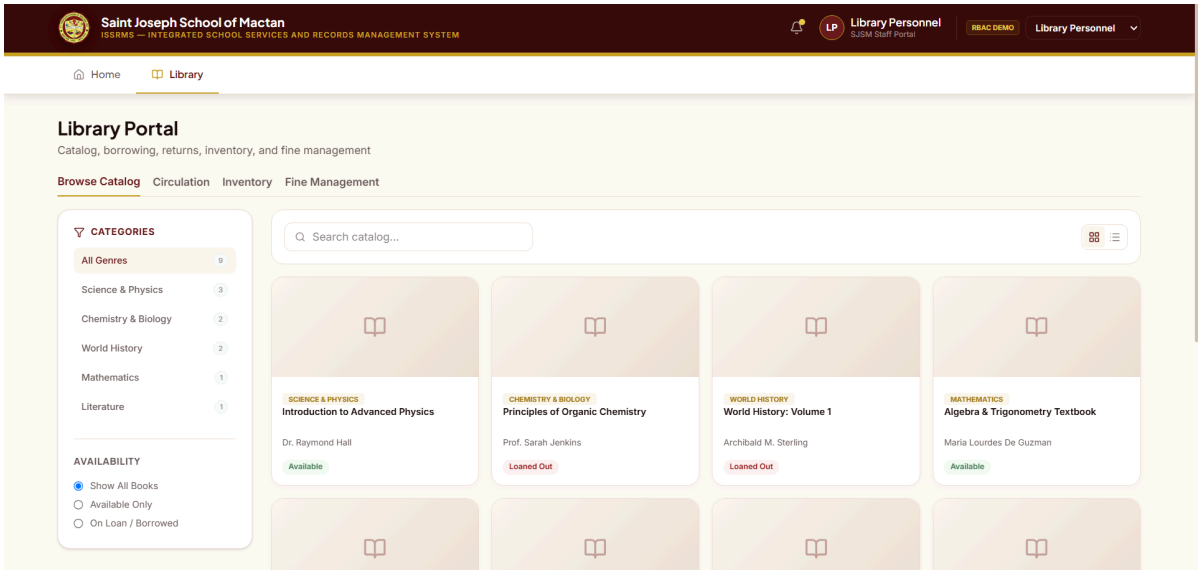


Figure 4.16.29. Library Personnel Library Catalog Wireframe

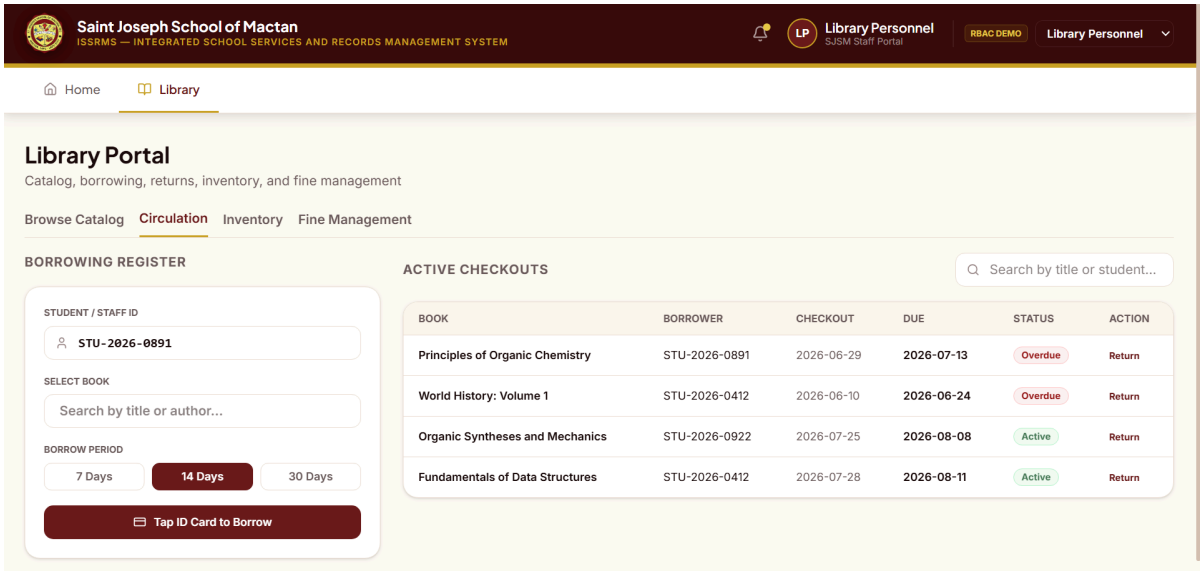


Figure 4.16.30. Library Personnel Library Circulation Wireframe

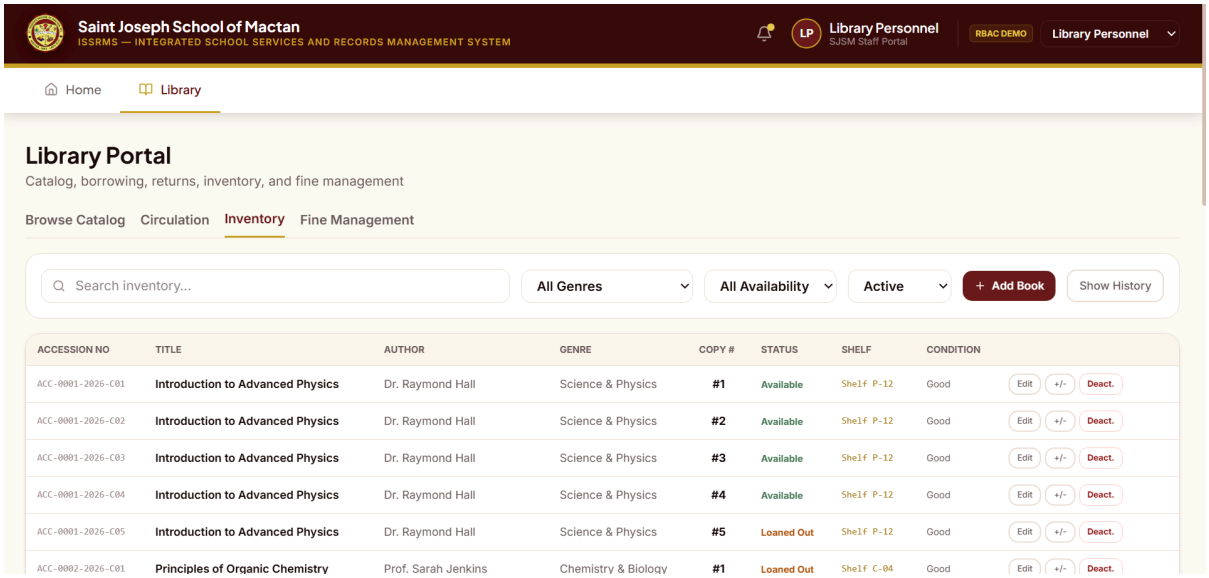


Figure 4.16.31. Library Personnel Library Inventory Wireframe

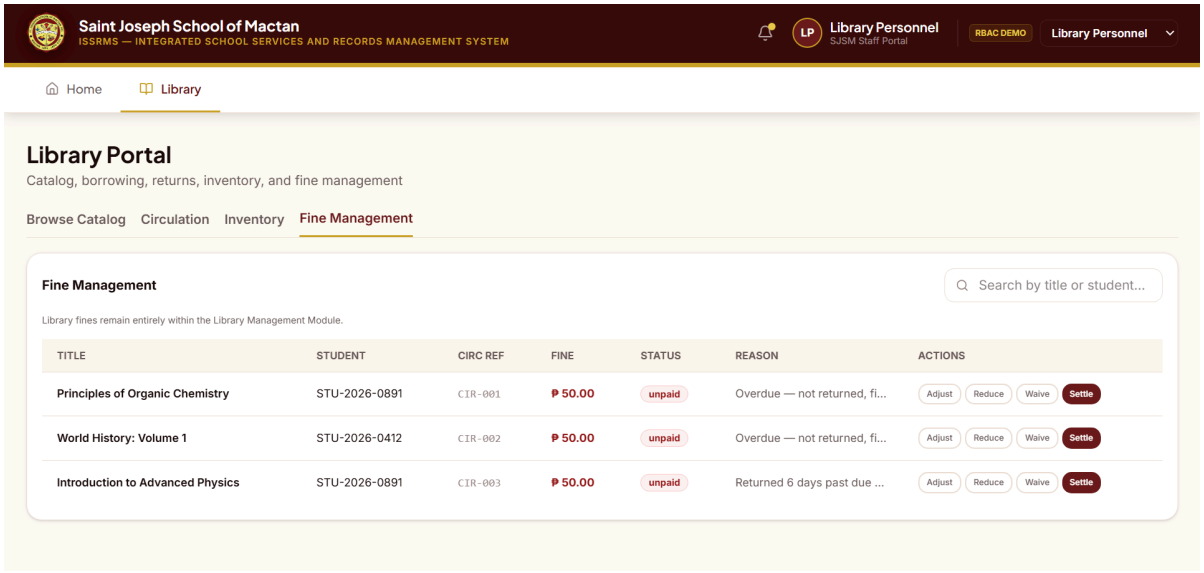


Figure 4.16.32. Library Personnel Fine Management Wireframe

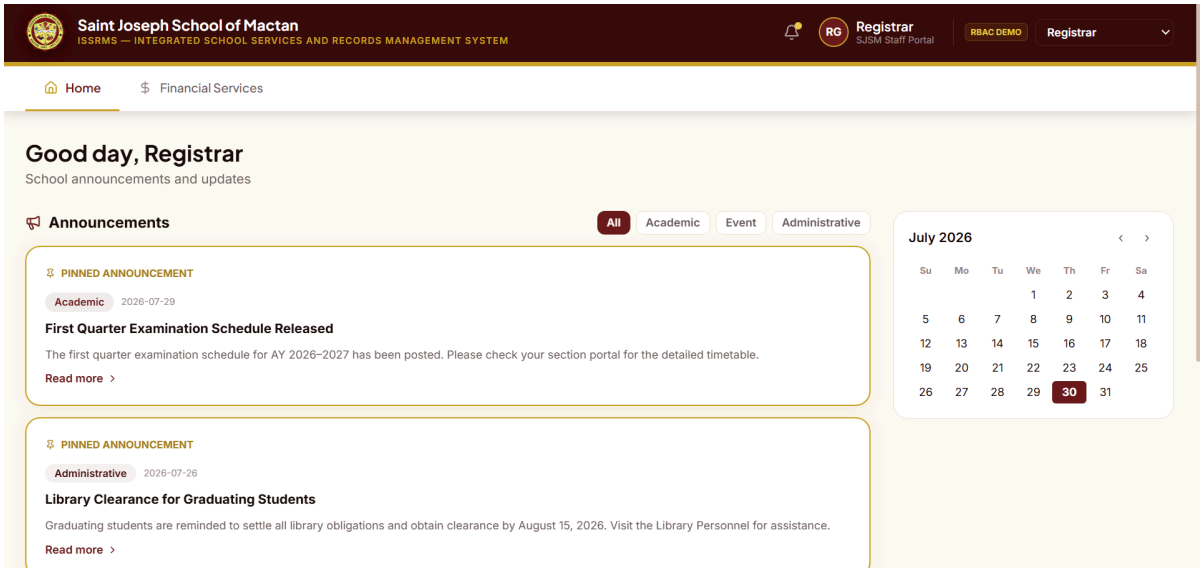


Figure 4.16.33. Registrar Home Page Wireframe

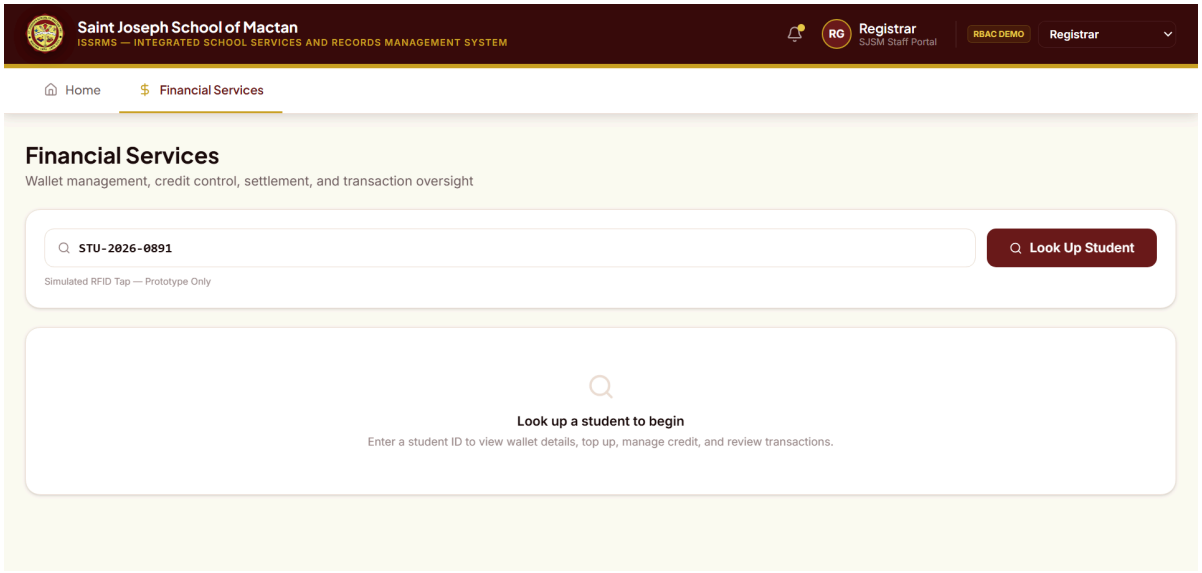


Figure 4.16.34. Registrar Financial Services Wireframe

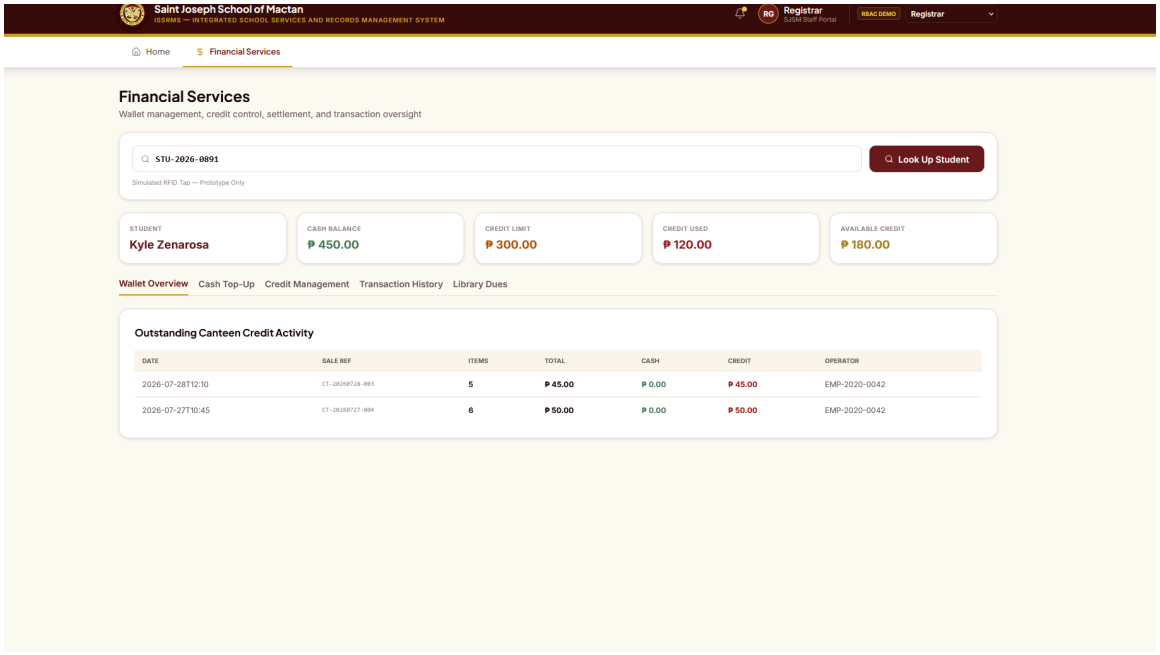


Figure 4.16.35. Registrar Financial Services Wallet Overview Wireframe

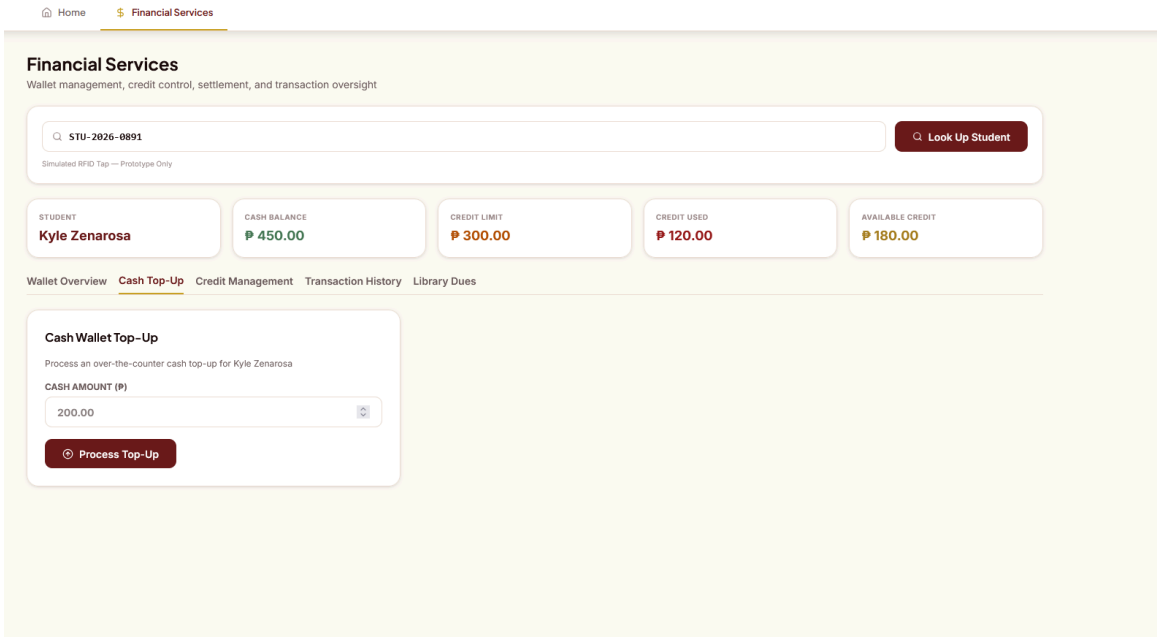


Figure 4.16.36. Registrar Financial Services Cash Top-Up Wireframe

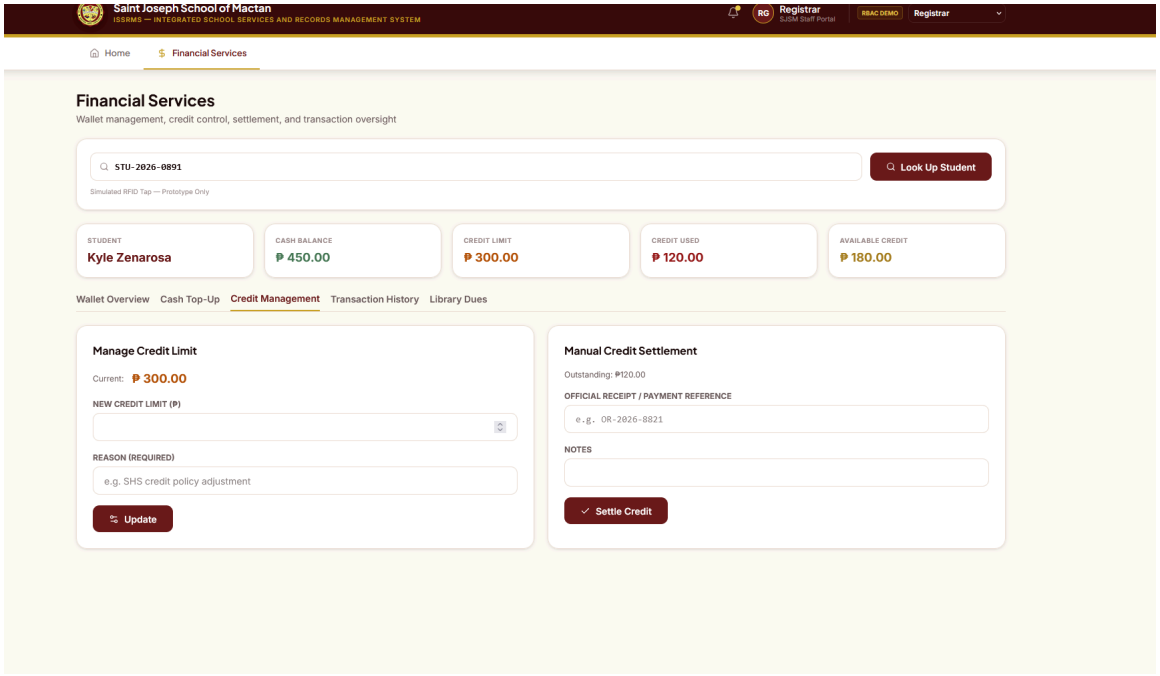


Figure 4.16.37. Registrar Financial Services Credit Management Wireframe

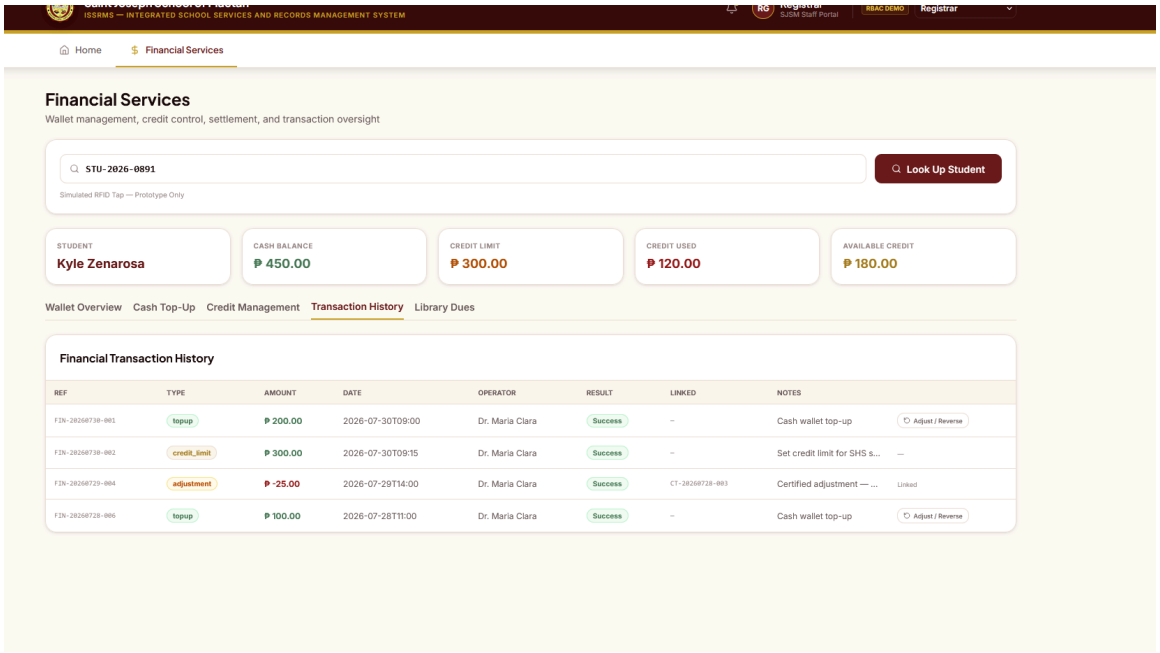


Figure 4.16.38. Registrar Financial Services Transaction History Wireframe

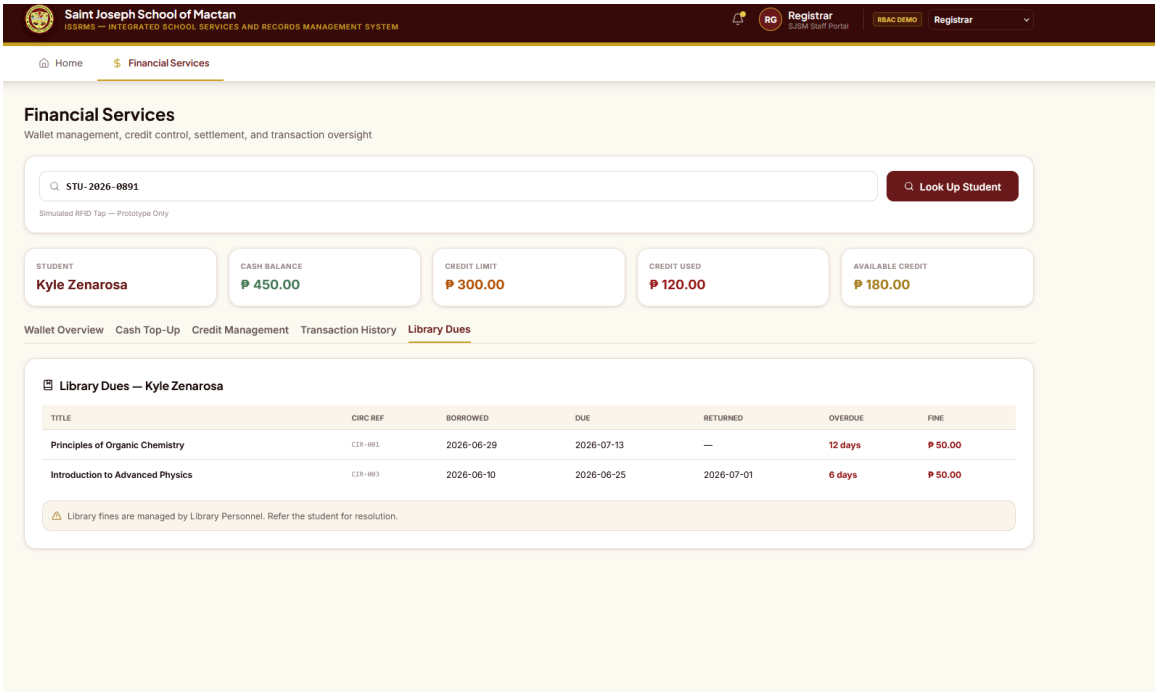


Figure 4.16.39. Registrar Library Fine and Circulation Reference Wireframe

4.2.10 Financial Transaction and Mathematical Design

The backend shall apply the following mathematical rules only to the canteen e-wallet. All related database changes are executed inside ACID transactions so each transaction is atomic, preserves constraints, is isolated from concurrent changes, and remains durable after commit.

$$B = T_{total} - P_{total}$$

B is the active spendable cash balance. T_total is the cumulative value of confirmed cash top-ups and positive financial adjustments. P_total is the cumulative value of canteen purchase debits after valid reversals. The immutable ledger provides the auditable source records for the calculation.

$$C_{avail} = C_{limit} - C_{used}$$

C_avail is the remaining approved credit, C_limit is the school-approved credit ceiling, and C_used is the unsettled canteen credit. credit_limit and credit_used are stored as separate values.

$$P \leq B + C_{avail}$$

A purchase P is approved only when it is within the combined active cash and available credit. The system consumes active cash before increasing credit_used.

$$Cash_Used = \min(P, B)$$

$$Credit_Added = P - Cash_Used$$

$$C_{used(new)} = C_{used(old)} + Credit_Added$$

The canteen sale, inventory deduction, credit update, Financial_Transactions header, and balanced Financial_Ledger_Entries lines commit together. If validation or posting fails, the transaction rolls back and none of those records is finalized.

$$C_{used(new)} = 0, \text{ only after authorized manual settlement}$$

At each grading or payment period, the Registrar may perform the Settle-to-Zero action only after verifying an official receipt, payment reference, or other school-approved payment record. The settlement creates a new auditable financial entry and never edits or deletes a finalized ledger record. These equations do not include library fines, official school debts, billing records, or the institutional Statement of Account.

4.2.11 Security and Transaction Controls

Financial security follows defense-in-depth controls: least-privilege RBAC, salted password hashing, validated sessions or tokens, server-side input and authorization checks, restricted integration credentials, service accounts, access tokens, and database credentials, ACID transactions, immutable Financial_Transactions headers, balanced Financial_Ledger_Entries lines, debit-credit validation, certified adjustment/reversal transactions, detailed audit logs, backup and recovery procedures, and safe error messages. Audit records shall identify the actor, timestamp, action, target wallet or record, amount, reference, result, and reason when applicable. Direct edit or deletion of a finalized financial transaction is prohibited.

4.3 Software Development Model

For the development of the proposed system, the Agile Software Development Model will be used. Agile is a flexible, iterative, and incremental approach to software development that has a primary focus on continuous collaboration and feedback with the client. Through this approach, adaptive planning can be applied, allowing for constant refinement of the website as issues are identified and changes are implemented throughout the development process. This approach allows for the development of the system in small increments called sprints, thus enabling the faster delivery of features, early testing, and the implementation of ongoing refinements based on client input (Amoros, 2025).

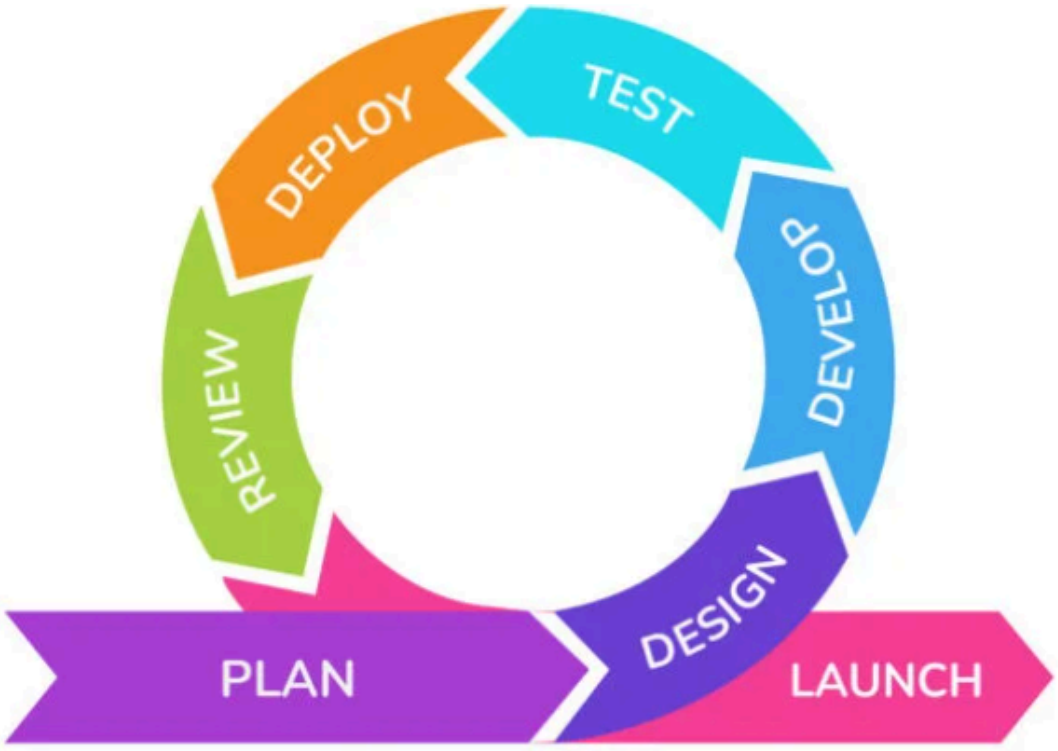


Figure 4.17. Agile Software Development Lifecycle

Plan

We will initially start with the procurement of the system's requirements through analysis and discussion of the current workflows involved in student authentication, student services, and financial transactions. This will be done through meetings and consultations with the client to understand existing pain points. Once we have gathered the requirements, we formulate the scope of the system, identifying the core modules to be developed, Authentication/RFID, Student Services, and Financial Services, and outline the overall sprint roadmap so we know what will be delivered in each iteration.

Design

Once the planning phase has come to a close, we begin the system design, wherein we structure how data will be stored and related (e.g., ISSRMS account references, transaction records, and institutional logging references) and define the overall technical architecture of the system. This phase also includes setting up the technology stack, React and Electron for the frontend/desktop application, and Node.js for the backend, establishing the environment the rest of development will build on. We maintain continuous communication with the client throughout this phase to ensure the proposed design aligns with their requirements.

Develop

After finalizing the design, we begin building the system's core features in sprints. The first sprint focuses on user authentication with role-based access control (RBAC) and RFID-based student verification, which forms the foundation for identifying users across the system. The next sprint develops the student services, namely the Institutional Logging Records reference, library, and notification modules This is followed by the financial services sprint, where we build the canteen POS, transaction history, and electronic wallet features. We use a modular approach throughout, with each member assigned to a specific module and expected to show progress every sprint until that module is complete.

Test

To ensure the reliability of the system, we conduct testing after each sprint to evaluate and validate our progress. These tests consist of unit testing, integration testing (e.g., confirming RFID identity verification, read-only logging-record retrieval, and wallet features work correctly together), user acceptance testing, and an ISO 25010 evaluation to formally assess software quality attributes such as reliability, usability, and maintainability. These tests are necessary to ensure the quality and usability of the system and allow us to identify and address bugs early on.

Deploy

Once a sprint has completed development and testing confirms the progress is up to standard, we deploy the module into the system. These deployments are introduced as incremental releases, allowing each completed module (authentication, student services, financial services) to be integrated into a controlled environment for final validation prior to full release.

Launch

Once all modules have been deployed and validated, we officially launch the product and roll it out for actual use, marking the final stage of our Agile software development process.

Maintain

After deployment, the system will be maintained through approved bug fixes, security updates, backup and recovery checks, and controlled changes that preserve the documented scope and audit controls.

4.4 Development Approach

A top-down development approach will begin with the complete ISSRMS architecture, scope boundaries, security policies, and data-zone rules. The design will then be decomposed into the web portal, School Operations desktop application, backend services, database zones, RFID interface, authentication/RBAC, Financial Services, canteen, library, notifications,

announcements, and logging-record reference. Modules will be implemented and verified independently before controlled integration.

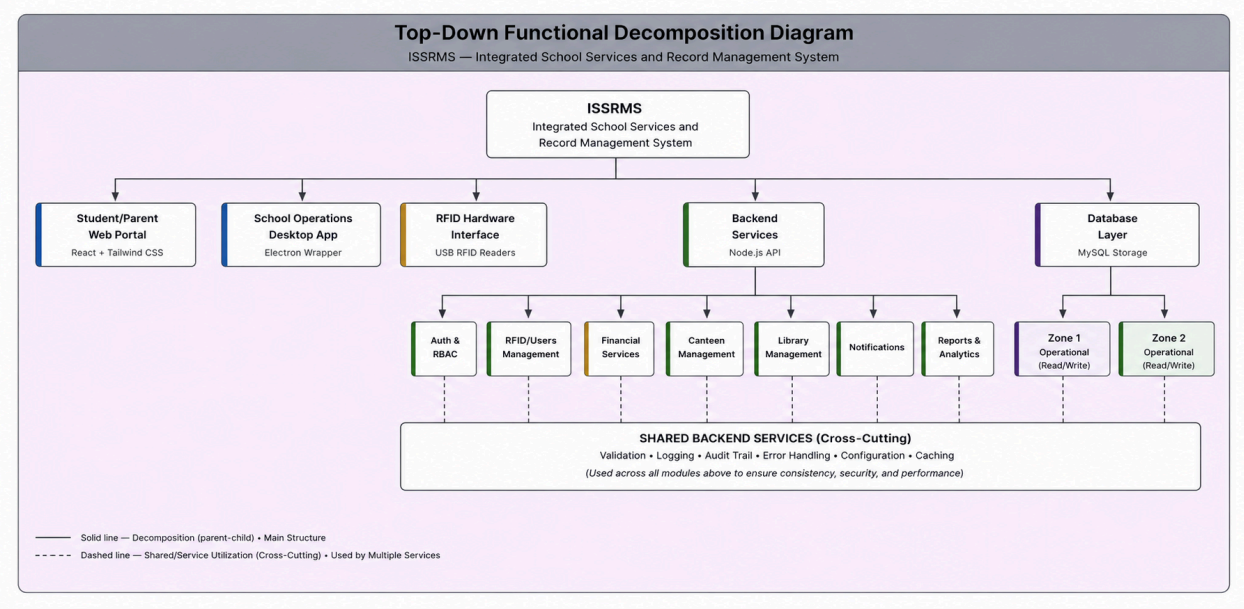


Figure 4.18. Top-Down Development Approach

4.5 Software Development Tools

React and Tailwind CSS will support the web and desktop user interfaces. The School Operations desktop application will reuse the React interface, with Electron serving as the controlled wrapper for USB RFID reader communication. Node.js will provide the locally hosted backend and authenticated REST endpoints. The approved Zone 1 integration technology will support read-only access to institutional identity and logging sources, while MySQL will support the separate Zone 2 operational database. RFID serial or device libraries will parse supported card identifiers. The security toolset includes Argon2 or bcrypt password hashing, token/session controls, backend validation, RBAC middleware, ACID database transactions, double-entry ledger validation, audit logging, backup tools, and automated testing.

The tools are selected according to maintainability, community and institutional support, interface usability, database and hardware connectivity, code reuse, licensing, learnability, and compatibility with the school's local

infrastructure. Final versions, licenses, hardware models, and deployment specifications must be confirmed before implementation approval.

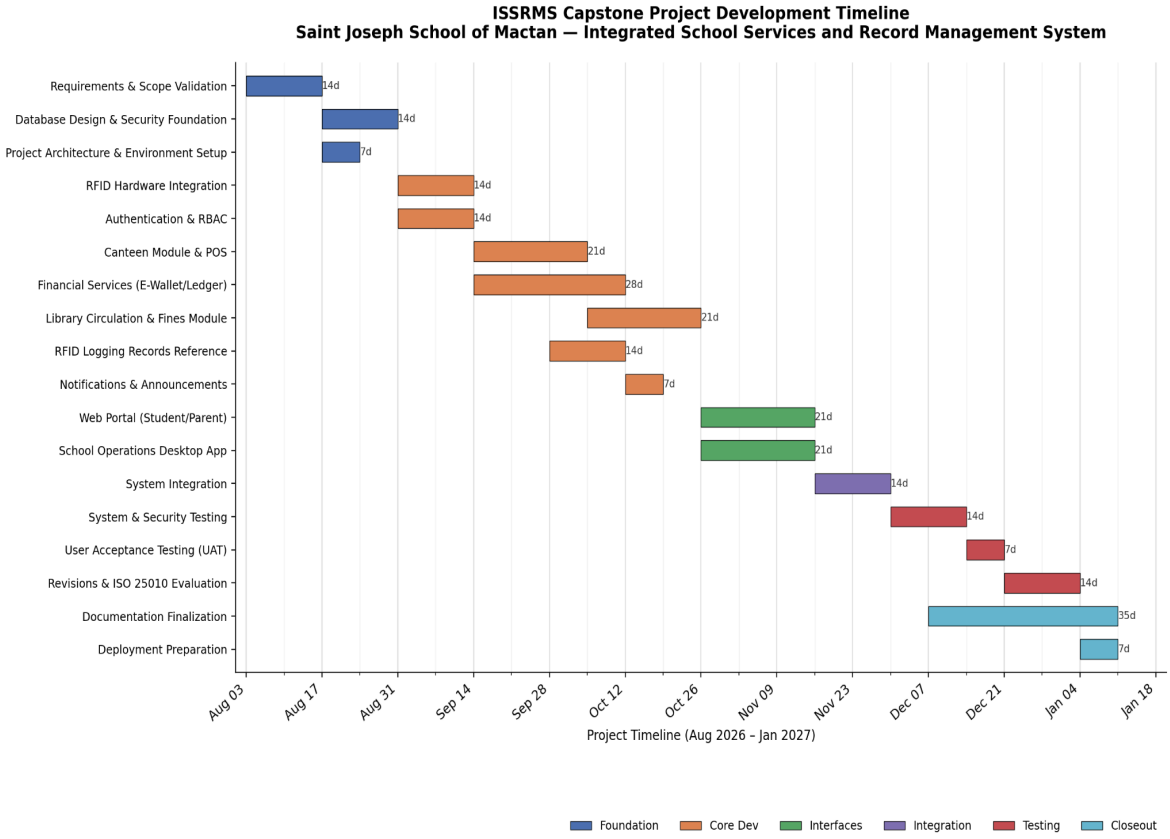
4.6 Project Management

Project management will control scope, schedule, responsibilities, costs, risks, quality, and documentation. The Technical Background and approved SRS form the scope baseline. Any proposed change shall be checked against that baseline before design or implementation, reviewed by the team and adviser, and recorded with its effect on requirements, diagrams, code, tests, schedule, and budget.

4.6.1 Schedule and Timeline

The schedule shall include requirements validation, design completion, database preparation, security foundation, module implementation, hardware integration, integration testing, user acceptance testing, revision, documentation, and deployment preparation. Dependencies, milestones, responsible members, planned dates, actual dates, and status shall be shown.

Table 4.3. *Project Schedule and Timeline*



4.6.2 Responsibilities

The four proponents are Kyle Christian A. Zenarosa, Joel June Fabillar, Kyle Jayverson Baguio, and Jay Niño Clark Basilgo. The responsibility matrix shall assign each member a primary module or technical area, supporting activities, required deliverables, review partner, target date, and completion status. Security, testing, documentation, and presentation responsibilities must be distributed rather than left implicit.

Table 4.4. *Project Team Responsibilities*

Name	Frontend	Backend	Design	Deployment	Techlead	Project Manager
Zenarosa, Kyle Christian	Yes	Yes	Yes	Yes	Yes	Yes
Basilgo, Jay Niño Clark	Yes	Yes	Yes	Yes	Yes	Yes
Baguio, Kyle Jayverson	Yes	Yes	Yes	Yes		
Fabillar, Joel June	Yes	Yes	Yes	Yes		

4.6.3 Budget and Cost Management

The budget shall itemize required RFID readers or replacements, server or storage upgrades, backup media, messaging or communication costs, testing materials, printing and binding, transportation, and contingency. Each row shall show quantity, quoted unit cost, planned total, funding source, responsible member, approval status, actual expenditure, and variance. Receipts and quotations shall support reconciliation.

Table 4.5. *Project Budget and Cost Monitoring*

Category	Item	Cost
RFID Hardware	USB RFID Reader (canteen & library stations, incl. 1 spare)	1,200.00
RFID Hardware	Test RFID cards/tags (for dev & UAT simulation)	15.00
Server/Infrastructure	Local server / workstation storage upgrade (SSD)	3,500.00
Server/Infrastructure	External backup drive (2TB, for backup/recovery testing)	4,200.00
Server/Infrastructure	Domain/local network testing accessories (cables, switch)	1,500.00
Communication & Testing	SMS gateway credits (temp. password / OTP delivery testing)	1,000.00
Communication & Testing	Mobile data / internet allowance for remote testing & meetings	300.00
Documentation	Printing & binding (proposal, manuscript, final drafts)	550.00

Documentation	Bond paper, ink, and office supplies	1,200.00
Logistics	Transportation (school site visits, interviews, consultations)	250.00
Logistics	Communication load/allowance for school coordination	150.00
Server/Infrastructure	USB RFID Reader (canteen & library stations, incl. 1 spare)	1,200.00
Total		Php 15,065

4.7 Verification, Validation, and Testing

Verification, validation, and testing are conducted to ensure that the proposed Integrated School Services and Record Management System (ISSRMS) meets the approved system requirements, supports the intended school operations, and functions correctly, securely, and reliably before deployment.

All verification and validation activities shall follow a unified evaluation metric:

Success Rate (%) = (Number of Passed Test Cases ÷ Total Executed Test Cases) × 100

A 100% pass rate shall be required before proceeding to the next testing phase. Any failed test cases shall be documented, corrected, and re-tested until they pass successfully.

4.7.1 Verification

Verification focuses on ensuring that the proposed Integrated School Services and Record Management System (ISSRMS) is developed according to the approved Software Requirements Specification (SRS), system architecture, database design, and software design documents. Verification activities shall include reviewing the Software Requirements Specification (SRS), system architecture, database design, software design documents, API and RFID parsing tests, Role-Based Access Control (RBAC) verification, financial calculations, ledger rules, and ACID transaction controls. Verification also includes unit testing, integration testing, and system testing, which are discussed in the following sections.

Unit Testing

Unit testing shall verify that each individual module works correctly before it is combined with the other modules of the system. Each module—including Authentication and Account Management, Institutional Logging Records Reference, Financial Services, Library Management, Canteen Management, Notifications, and Home and Announcements—shall be tested separately to ensure that its functions, calculations, database operations, and input validations work correctly using normal, boundary, and invalid test inputs. The proponents assigned to each module shall prepare and perform unit test cases to identify and fix errors early in the development process.

Integration Testing

Integration testing shall verify that the different ISSRMS modules work properly when connected. Since the system consists of a web-based Student/Parent Portal, a School Operations desktop application, a Node.js backend, RFID reader integration, and a MySQL operational database, this testing shall ensure that data is transferred correctly between these components. Test cases shall include RFID identification, wallet top-ups, canteen purchases,

inventory updates, library transactions, notification generation, and retrieval of authorized institutional logging records through the approved read-only integration mechanism.

System Testing

System testing shall evaluate the ISSRMS as a complete and fully integrated system. The proponents shall perform complete user workflows to verify that all major system functions operate correctly when the modules are used together. Test scenarios shall include user login, role-based access control, RFID identification, canteen e-wallet transactions, library borrowing and returning, inventory management, institutional logging-record viewing, notifications, announcement management, and user account management. The purpose of system testing is to ensure that all modules work together correctly and reliably before the system proceeds to User Acceptance Testing (UAT).

The prepared test-case tables are presented in this subsection and will be used to execute and document unit, integration, system, security, and role-based acceptance tests. Actual execution records and final results shall be completed only after implementation and approved testing.

Test Case Tables

The prepared test-case tables below define the scenarios, sequential steps, and expected results for unit, integration, system, security, and planned role-based acceptance testing. They are proposal-stage preparation artifacts; no execution results are claimed at this stage.

Table 4.7. Proposed Test Case Tables

Test Case ID	TC-001
Module	Authentication
Test Scenario	Valid login
Test Steps	1. Open the login page. 2. Enter a valid ID number and password. 3. Select Login.
Expected Result	The backend authenticates the account and shows only the functions authorized for the assigned role.

Test Case ID	TC-002
Module	Authentication
Test Scenario	Invalid login
Test Steps	1. Enter a valid ID number and an incorrect password. 2. Select Login.
Expected Result	Access is denied, a safe error message is shown, and protected data is not exposed.

Test Case ID	TC-003
Module	Financial Services
Test Scenario	Cash top-up
Test Steps	1. Registrar confirms the student and wallet. 2. Enter the approved amount. 3. Confirm the top-up.
Expected Result	An immutable transaction header and balanced debit/credit ledger entries

	commit together; the derived balance updates and an audit event is recorded.

Test Case ID	TC-004
Module	Canteen Management
Test Scenario	Purchase using the e-wallet
Test Steps	1. Identify the student by RFID or approved manual ID. 2. Select items and quantities. 3. Confirm payment.
Expected Result	The purchase is approved only when P ≤ B + C_avail. Cash is used first; any uncovered amount updates credit_used. Sale, stock, credit, header, ledger, and audit records commit atomically.

Test Case ID	TC-005
Module	Canteen Management
Test Scenario	Insufficient funds or credit
Test Steps	1. Select items whose total exceeds B + C_avail. 2. Confirm payment.
Expected Result	The purchase is rejected and no sale, stock, wallet, credit, or ledger changes persist.

Test Case ID	TC-006
Module	Library Management

Test Scenario	Borrow material
Test Steps	1. Identify the borrower. 2. Select an available material. 3. Confirm the borrowing transaction.
Expected Result	A circulation record is created and the material status becomes Borrowed. No canteen wallet or financial-ledger entry is created.

Test Case ID	TC-007
Module	Library Management
Test Scenario	Return material and assess a fine
Test Steps	1. Open the circulation record. 2. Record the return. 3. Assess an overdue fine when applicable.
Expected Result	The material becomes available, and fine_amount and fine_status are updated in Library_Circulation only. Library fines do not affect the canteen wallet or financial ledger.

Test Case ID	TC-008
Module	Canteen Inventory
Test Scenario	Update stock information
Test Steps	1. Open the inventory module. 2. Select an item. 3. Update the stock quantity or approved price. 4. Save the change.

Expected Result	Server validation accepts the authorized update, and the new values are reflected in subsequent canteen sales.

Test Case ID	TC-009
Module	Institutional Logging Records
Test Scenario	View read-only entry and exit history
Test Steps	1. Open the logging-record view. 2. Select an authorized student and date range. 3. View the records.
Expected Result	Existing institutional identity and logging records are displayed through the approved read-only mechanism. No write, edit, or delete operation is available.

Test Case ID	TC-010
Module	Account and Role Administration
Test Scenario	Limited Administrator access
Test Steps	1. Administrator provisions or deactivates a Zone 2 account or assigns an approved role. 2. Attempt a financial, canteen, or library operational transaction.
Expected Result	The account or role change is authorized and audited. The operational transaction is denied; logging and audit views remain read-only.

Test Case ID	TC-011
Module	Announcements
Test Scenario	Publish an announcement
Test Steps	1. Open the announcements module. 2. Enter approved content. 3. Publish the announcement.
Expected Result	The announcement is stored and shown to permitted users, and the action is audited.

Test Case ID	TC-012
Module	Notifications
Test Scenario	View notifications
Test Steps	1. Log in. 2. Open the notifications module.
Expected Result	Permitted alerts are displayed, including configured due-date, fine, and wallet-confirmation notices. The system does not generate official SOA or billing data.

Test Case ID	TC-013
Module	Financial Services
Test Scenario	Record an adjustment or reversal
Test Steps	1. Select the finalized transaction. 2. Enter the reason and reference. 3. Authorize the adjustment or reversal.
Expected Result	The original record remains immutable. A certified new transaction with balanced

	ledger entries and an audit trail is created; duplicate reversal is rejected.

Test Case ID	TC-014
Module	Financial Services
Test Scenario	Settle canteen credit to zero
Test Steps	1. Registrar verifies the approved payment record. 2. Select the student's credit balance. 3. Confirm Settle-to-Zero.
Expected Result	A new auditable settlement transaction records the payment and credit_used resets to 0.00. No finalized ledger row is edited or deleted.

4.7.2 Validation

UAT shall be conducted through stakeholder demonstrations and task-based User Acceptance Testing (UAT) with representative users of the proposed Integrated School Services and Record Management System (ISSRMS).

For UAT, participants shall be selected through purposive, role-based sampling. This ensures that task completion, permissions, and workflow coverage are verified for each system role. The required UAT size shall use the following role-coverage formula:

$n_{UAT} = \Sigma m_r$, where m_r is the minimum number of participants assigned to each distinct user role. The approved UAT plan sets $n_{UAT} \geq 30$.

n_{UAT} = 1 Administrator + 1 Registrar + 1 Library Personnel + 1 Canteen Personnel + 26 Student/Parent users = 30 respondents.

The final participant list shall be confirmed with the school before UAT. Additional participants may be added when a role has multiple workflows or when the school requests broader coverage. The participant count is intended to cover the system's required user roles and workflows.

Each respondent shall perform test scenarios appropriate to their assigned role. The Administrator shall validate Zone 2 account provisioning and deactivation, approved role assignment, announcements, approved non-financial settings, read-only Institutional Logging Records viewing, and read-only system-wide Audit Logs. The Administrator shall not execute financial, canteen, or library operational transactions. The Registrar shall validate wallet management, approved credit operations, financial services, and read-only library-fine viewing. Library Personnel shall validate library circulation, inventory management, and library fine processing. Canteen Personnel shall evaluate canteen sales, inventory management, and wallet top-up transactions. Student respondents shall validate authentication, RFID-based identification, institutional logging-record viewing, wallet transaction viewing, notifications, and other Student/Parent Portal functions.

Test scenarios shall cover RFID and manual-ID identification, institutional logging-record lookup, canteen purchases, wallet top-ups, approved credit usage, library circulation and fines, role-based access restrictions, notifications, and recovery from invalid or unavailable inputs.

After completing the assigned tasks, respondents shall evaluate the system using a structured questionnaire based on the ISO/IEC 25010 Software Quality Model and a five-point Likert scale. The collected responses shall be analyzed using the weighted mean to determine the level of user acceptance and identify improvements before final deployment.

Validation shall confirm that each user role can access only the records and functions permitted by the approved requirements. Feedback shall be documented, prioritized, and traced to approved system changes.

4.7.3 Security and Quality Testing

System testing shall cover functional correctness, module integration, transaction atomicity, concurrent processing, rollback behavior, ledger immutability, authorization controls, session handling, input validation, audit completeness, data-zone isolation, backup and restore, RFID-reader failure, performance, usability, and compatibility.

Financial test cases shall verify that partial transactions do not persist, finalized financial records cannot be edited or deleted, debit and credit ledger entries remain balanced, reversals preserve the original transaction, and manual settlements require authorization, payment verification, and an audit record.

Library test cases shall verify that library fines remain separate from the canteen e-wallet and financial ledger. They shall also confirm that the Registrar may view unpaid library fine information for reference only, while only authorized Library Personnel may assess, adjust, reduce, waive, and settle library fines.

Zone 1 integration tests shall verify that institutional identity and logging records are retrieved only through the approved read-only integration mechanism and that ISSRMS cannot create, update, or delete source records.

4.8 ISO 25010 Evaluation Criteria

The proposed Integrated School Services and Record Management System (ISSRMS) will be evaluated using the ISO 25010 Software Quality Model. This international standard provides a framework for assessing software quality based on characteristics such as functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Evaluation will use a role-based User Acceptance Testing (UAT) questionnaire administered to the planned minimum of 30 participants identified through the role-coverage formula in Chapter IV Section 4.7.2. Participants will represent the School Administrator, Registrar, Library Personnel, Canteen Personnel, and Student/Parent user groups. The questionnaire measures perceived quality after participants complete their assigned workflows.

Table 4.9 *ISO 25010 Software Quality Evaluation Criteria*

Quality Characteristic	Description	Evaluation Indicators
Functional Suitability	Measures whether the system provides functions that satisfy user requirements.	The system performs RFID identity verification, logging entry/exit, library management, electronic wallet transactions, canteen operations, notifications, and user account management accurately and completely.
Performance Efficiency	Measures the response time and resource utilization of the software.	The system responds quickly during login, RFID identity verification, authorized entry/exit logging-record retrieval, library transactions, canteen purchases, and electronic wallet transactions.
Compatibility	Measures the ability of the software to operate with other systems and hardware.	The system operates correctly with RFID readers, the school's existing student data source, desktop computers, supported web browsers, and the approved secured network environment.
Usability	Measures how easy the software is to learn and operate.	Students, parents, Library Personnel, Canteen Personnel, the Registrar, and Administrators can navigate their permitted interfaces and complete assigned tasks with minimal assistance.

Reliability	Measures the consistency and stability of the software during operation.	The system consistently processes entrance and exit logging, library transactions, canteen purchases, and electronic wallet transactions without crashes, errors, or data loss.
Security	Measures how well the software protects data and prevents unauthorized access.	The system implements secure authentication, Role-Based Access Control (RBAC), password hashing, and protects sensitive user and transaction data from unauthorized access.
Maintainability	Measures how easily the software can be modified or updated.	Developers can efficiently maintain, update, and enhance the system modules without affecting existing functionalities.
Portability	Measures how easily the software can be deployed to another environment.	The system can be deployed and operated on supported computers within the school's local network with minimal installation and configuration requirements.

Table 4.10 ***Likert Scale for Evaluation***

Scale	Range	Verbal Interpretation
5	4.21 – 5.00	Excellent
4	3.41 – 4.20	Very Good
3	2.61 – 3.40	Good
2	1.81 – 2.60	Fair
1	1.00 – 1.80	Poor

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APPENDIX A TRANSMITTAL LETTER

June 26, 2026

The Principal
Saint Joseph School of Mactan
Mactan, Lapu-Lapu City, Cebu

Dear Sir/Madam,

Peace and all good!

We are third-year Bachelor of Science in Information Technology (BSIT) students from the University of San Carlos currently undertaking our Capstone Project requirement.

As part of our study, we are considering your school as one of our prospective clients. In view of this, we are requesting to conduct an interview and consultation with you and your authorized representatives. This interview aims to identify existing operational challenges, current processes, and technology-related needs that may benefit from technological improvement and may serve as the basis for our capstone project.

The information gathered will be used solely for academic purposes and will be treated with utmost confidentiality. Should you have any questions or require further information, you may get in touch with us through the email details provided below.

We would greatly appreciate a positive response to this request. Thank you so much.

Respectfully yours,

Kyle Christian A. Zenarosa
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Noted by:

06/26/26

Glenn Pepito
Adviser

Approved by:

Saint Joseph School of Mactan

June 26, 2026

The Principal
Saint Joseph School of Mactan
Mactan, Lapu-Lapu City, Cebu
Dear Sir/Madam,
Peace and all good!
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Respectfully yours,

Kyle Christian A. Zenarosa
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Approved by:

Noted by:

Glenn Pepito, MBA,MSIT
Mactan
Adviser

Saint Joseph School of

APPENDIX B INTERVIEW GUIDE

Respondent Information

Name (Optional): _____

Position/Designation: _____

Years of Service: _____

Date of Interview: _____

A. Office Background and Structure

1. Could you briefly introduce your office or department and explain its primary functions within the school?
2. Could you briefly describe your role and the key responsibilities your office handles on a daily basis?
3. Could you describe the organizational structure of your office or department?
4. How many modules, functions, or major processes does your office currently handle?

B. Current Operations and Processes

5. How are records, information, and transactions currently managed in your department, and how do you coordinate with other offices when information needs to be shared?
6. What kinds of reports or information are most frequently requested from your office, and what difficulties do you encounter in preparing or providing them?
7. Are there any concerns regarding the security, accuracy, or accessibility of important records in your office?

C. Challenges and Their Impact

8. What are the most common problems, errors, or delays you encounter in your day-to-day operations, and which of these consume the most time and effort?

9. Have you or your team developed any workarounds to deal with these recurring problems? If so, what are they?
10. How do these challenges affect your staff, and what complaints or concerns do you most frequently receive from students, parents, or other departments?

D. Priorities and Recommendations

11. If you could address just one operational problem in your office right now, what would it be and why?
12. What do you think is the single most significant problem the school is currently facing that needs immediate attention?
13. Is there anything else about your office's challenges or operations that you think is important for us to understand?

APPENDIX C SOFTWARE REQUIREMENTS SPECIFICATION

Integrated School Services and Record Management System (ISSRMS)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, data, interface, security, and quality requirements for the Integrated School Services and Record Management System (ISSRMS) of Saint Joseph School of Mactan. It provides the baseline for design, implementation, testing, acceptance, and change control.

1.2 Document Conventions

The word “shall” identifies a mandatory requirement, while “may” identifies permitted or conditional behavior. Each requirement has a stable identifier for traceability. Functional features are assigned High, Medium, or Low priority. Text inside square brackets is an insertion or confirmation item that must be completed using approved project evidence. B, T_total, P_total, C_limit, C_used, C_avail, and P are defined in Section 2.8.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for the proponents, faculty adviser and panel, school representatives, developers, testers, database administrators, the Registrar, Canteen Personnel, Library Personnel, Administrators, and authorized acceptance-test participants. Readers seeking a high-level understanding should begin with Sections 1 and 2. Developers should focus on Sections 3 through 6, while testers should use Sections 4 through 6 to derive verification, validation, security, and acceptance test cases. The proposed test-case tables is provided in Chapter IV Section 4.7.1 for the requirements and workflows described in this SRS.

1.4 Product Scope

ISSRMS is a role-controlled platform with on-premises backend and data services and secure web access for authorized Student/Parent users. Its scope includes RFID-assisted identity verification; read-only viewing of authorized Gate Entry, Gate Exit, Library Entry, and Library Exit records; canteen e-wallet transactions; canteen sales and inventory; library catalog, circulation, inventory, and separately managed fines; notifications; announcements; a Student/Parent web portal; and a School Operations desktop application.

The canteen e-wallet supports confirmed cash top-ups, canteen purchases, approved credit use, authorized manual credit settlement, and certified adjustment or reversal transactions. Library fines remain in the Library Management Module and never change the e-wallet balance or credit. The school's official debts, billing records, and Statement of Account remain entirely outside ISSRMS. The system shall not generate, modify, query, consolidate, export to, or integrate with those institutional financial records.

Feature Priority Classification

The SRS distinguishes mandatory features from optional or conditional elements. Mandatory features are required for the approved ISSRMS scope and acceptance. Optional or conditional elements may be enabled only when separately approved or when supporting information is available; their absence must not violate the core scope.

Feature	Classification	Scope / SRS Reference
Authentication and RBAC	Mandatory	Core access control (Section 4.1; REQ-4.1-1 to REQ-4.1-5)

RFID/manual-ID verification and read-only logging lookup	Mandatory	Approved identity and logging reference only (Section 4.2)
Canteen financial services and e-wallet	Mandatory	Top-up, purchase, approved credit, settlement, adjustment/reversal, double-entry, and ACID processing (Section 4.3)
Canteen sales and inventory	Mandatory	Sales, item lines, pricing, stock, and atomic posting (Section 4.4)
Library catalog, circulation, inventory, and fines	Mandatory	Independent library workflow; fines never create wallet or ledger entries (Section 4.5)
Student/Parent web portal	Mandatory	Authorized views and records; protected staff actions remain unavailable (Section 4.6)
School Operations desktop application	Mandatory	Role-authorized staff workflows and controlled RFID-reader bridge (Section 4.7)
Notifications and internal announcements	Mandatory	**Included within** approved role permissions (Section 4.8)
Security and reliability controls	Mandatory	Server-**side vali**dation, audit logging, immutable financial records, backups, recovery, and ACID safeguards (Sections 5–6)
SMS-based temporary-password and OTP delivery	Optional / Conditional	May use the school-approved SMS infrastructure when available; otherwise use the approved

		administrator-assisted recovery procedure.

1.5 References

Chapter III, Technical Background, is the authoritative scope and policies shall remain consistent with those chapters.

Ref#	Document Title	Version/Date	Source
[1]	Chapter III – Technical Background	July 2026	ISSRMS Capstone Documentation
[2]	Chapter IV – Design and Methodology	July 2026	ISSRMS Capstone Documentation
[3]	Capstone Overview – Proposed Capstone Direction	July 2026	ISSRMS Capstone Documentation
[4]	Republic Act No. 10173 – Data Privacy Act of 2012	2012	Republic of the Philippines
[5]	ISO/IEC/IEEE 29148:2018 – Systems and	2018	ISO, IEC, and IEEE

	Software Engineering—Life Cycle Processes—Requirements Engineering		

2. Overall Description

2.1 Product Perspective

ISSRMS is a new integrated school-service platform that coordinates selected canteen, library, wallet, logging reference, notification, and announcement workflows through a centralized backend. It uses an HTTPS-accessible Student/Parent web portal, an internal School Operations desktop application, an on-premises Node.js backend, supported USB RFID readers, and two isolated data zones. External portal requests pass through the school-approved firewall and reverse proxy or web server before reaching authenticated backend endpoints.

Zone 1 contains the school's institutional student registry and logging-record sources. ISSRMS accesses permitted Zone 1 data through one school-approved read-only integration mechanism. Depending on the existing architecture and access approved by the school, the selected mechanism may use authenticated read-only API endpoints, including an approved adapter API, or a dedicated least-privilege database connection restricted to approved SELECT operations. Only one method shall be selected and implemented after school-approved technical validation; the two methods are not simultaneous requirements. Zone 2 is the ISSRMS operational database and stores project-generated accounts, wallets, ledger entries, canteen records, library records, notifications, announcements, and audit records.

A common institutional ID is verified by backend application logic before an operational account or transaction is accepted. Cross-database physical foreign keys are not created between Zone 1 and Zone 2. Client applications do not connect directly to either data zone or institutional source; all protected access is mediated by authenticated backend services.

2.2 Product Functions

ID	Major Function
PF-01	Authenticate users and enforce role-based permissions.
PF-02	Validate supported RFID card identifiers and permit authorized manual-ID fallback.
PF-03	Display authorized Institutional Logging Records in read-only form, including Gate Entry, Gate Exit, Library Entry, and Library Exit records.
PF-04	Process confirmed cash-only wallet top-ups by the Registrar or authorized Canteen Personnel; canteen purchases by Canteen Personnel; approved credit use and credit-limit management by the Registrar; and Registrar-authorized manual settlement, certified adjustments, or reversals.

PF-05	Record canteen sales, items, prices, stock changes, and transaction references.
PF-06	Manage library catalog, circulation, inventory, fine amounts, and fine status independently of the wallet.
PF-07	Display authorized wallet, canteen, library, logging, notification, and announcement records through the appropriate client.
PF-08	Produce standard operational and administrative audit events.

2.3 User Classes and Characteristics

User Class	Characteristics	Privileges / Access
Student/Parent	Primarily non-technical; uses a supported browser and accesses the portal as needed.	Read-only access to the authorized student's permitted wallet, canteen, library, logging, notification, and announcement records.

Registrar	Authorized school personnel responsible for financial-service workflows.	Operates the Financial Services Module; processes approved top-ups, credit limits, settlements, adjustments, and reversals; reviews financial records; may view unpaid library fine information for reference only.
Canteen Personnel	Frequent operational user at a school workstation and point-of-sale station.	Performs permitted canteen sales, manages authorized canteen inventory, pricing, and sales records, and processes confirmed over-the-counter cash-only wallet top-ups. Does not manage credit limits, settlements,

		adjustments, reversals, or library fines.
Library Personnel	Frequent operational user responsible for library workflows.	Manages catalog, circulation, material inventory, and independent library-fine assessment, adjustment, waiver, and settlement.
Administrator	Technical or supervisory user responsible for system administration.	Manages Zone 2 account and role administration, approved non-financial settings, announcements, read-only Institutional Logging Records, and read-only system-wide Audit Logs. It cannot execute financial, canteen, or library operational transactions.

		Also manages accounts, roles and permissions, announcements, system settings, Institutional Logging Records viewing, and the read-only system-wide Audit Logs interface. All Administrator actions remain authenticated, server-authorized, validated, and audited.

2.4 Operating Environment

Students and parents shall access the Student/Parent Portal through an approved HTTPS address using the route Student/Parent device → Internet → school perimeter firewall → secure reverse proxy or web server → internal Node.js backend. Authorized campus personnel shall use the School Operations desktop application on approved Windows workstations connected to the secured School LAN. The desktop application shall reuse the React and Tailwind CSS interface within Electron, with supported USB RFID readers connected locally to authorized staff workstations. The Node.js backend, Zone 1 integration,

Zone 2 database, and protected administration services shall remain hosted within the school's secured on-premises environment.

The system shall support the latest stable versions of Google Chrome, Microsoft Edge, and Mozilla Firefox. Development shall utilize the latest stable Long-Term Support (LTS) version of Node.js together with the current stable releases of React, Tailwind CSS, Electron, and MySQL available at the start of implementation. The application server shall provide sufficient processing power, memory, and storage to support concurrent internal School Operations users and authenticated external Student/Parent Portal users. The RFID subsystem shall support USB-connected RFID readers compatible with the selected RFID cards and required drivers.

2.5 Design and Implementation Constraints

ID	Constraint
CON-01	Access to institutional identity and logging data within Zone 1 shall be restricted to school-authorized read-only integration methods. ISSRMS shall not create, update, or delete records residing in this zone.
CON-02	Project-generated records shall be stored only in the isolated Zone 2 operational database.
CON-03	Finalized Financial_Transactions and Financial_Ledger_Entries shall not be edited or deleted.

CON-04	Financial corrections shall create a linked adjustment or reversal entry.
CON-05	Library fines shall remain in Library_Circulation and shall not be deducted from E_Wallet.
CON-06	ISSRMS shall not expose a use case, interface, entity, calculation, or data flow for the school's official Statement of Account or institutional billing records.

2.6 Assumptions and Dependencies

The system depends on accurate institutional identity and logging data; authorized technical access to the Zone 1 infrastructure; approved interface or schema documentation; school approval of the read-only integration mechanism; functioning local-network and server services; compatible RFID cards and readers; valid ISSRMS user accounts and role assignments provisioned from authorized school identity records; correct canteen and library master data; and written school policies for financial authorization, data retention, backup, incident handling, and acceptance testing.

2.7 Security Architecture

Security shall use defense in depth. Credentials shall be salted and hashed with an approved algorithm such as Argon2 or bcrypt. Authenticated sessions or tokens shall be validated by the backend. Least-privilege RBAC shall be enforced on both the client interface and every protected endpoint. The desktop renderer shall not receive unrestricted operating-system or database privileges. Integration credentials, database accounts, service accounts, and access tokens shall be separated by purpose and restricted according to least privilege. Any credential used for Zone 1 integration shall permit only approved read operations. External Student/Parent requests shall pass through the school-approved perimeter firewall and reverse proxy or web server. The backend, Zone 1 integration, Zone 2 database, RFID interfaces, and protected administration services shall not be directly accessible from the Internet.

Financial security shall use server-side validation, ACID transactions, a double-entry accounting model, immutable Financial_Transactions headers, immutable Financial_Ledger_Entries lines, debit-credit balance validation, certified adjustment and reversal transactions, transaction references, and detailed audit logs. Every financial transaction shall maintain balanced debit and credit ledger entries. A canteen sale, inventory movement, credit update, transaction header, and corresponding ledger-entry lines shall either commit together or roll back together. Error messages shall not expose credentials, connection details, stack traces, or sensitive record contents.

2.8 Financial Transaction and Mathematical Model

The following equations apply only to the canteen e-wallet, Financial_Transactions, and Financial_Ledger_Entries.

$$B = T_{total} - P_{total}$$

$$C_{avail} = C_{limit} - C_{used}$$

$$P \leq B + C_{avail}$$

$$Cash_{Used} = \min(P, B)$$

$$Credit_{Added} = P - Cash_{Used}$$

$$C_{used(new)} = C_{used(old)} + Credit_{Added}$$

$$C_{used(new)} = 0, \text{ only after authorized manual settlement}$$

B is the active spendable cash balance. T_total is the cumulative value of confirmed cash top-ups and positive financial adjustments. P_total is the cumulative cash-funded portion of canteen purchases posted as debits against the student wallet after valid reversals. Purchase amounts covered by approved credit increase credit_used and are represented through the corresponding balanced ledger entries rather than being deducted from the active cash balance. The balanced immutable ledger-entry lines are the audit source for the calculation.

C_avail is the remaining approved credit, C_limit is the approved credit ceiling, and C_used is the current unsettled canteen credit. credit_limit and credit_used shall be stored separately. A purchase P shall be approved only when it does not exceed active cash plus available credit. Cash shall be consumed before credit_used increases.

At each grading or payment period, the Registrar may reset C_used only after verifying, through an official receipt, payment reference, or other school-approved record, that the student's outstanding canteen credit has been paid. The system shall create an auditable settlement transaction with balanced ledger-entry lines and shall not edit or delete an earlier finalized record. Library

fines, official school debts, billing records, and the institutional Statement of Account are not variables in this model.

3. External Interface Requirements

3.1 User Interfaces

ID	Requirement
UI-01	The interface shall display only role-authorized modules and records.
UI-02	Protected financial actions shall display the affected account, amount, action type, confirmation prompt, and resulting reference.
UI-03	Adjustment, reversal, and manual-settlement screens shall require a reason and shall identify the original or related transaction when applicable.
UI-04	Logging-record screens shall clearly state that the displayed institutional records are read-only.
UI-05	Library fine screens shall remain separate from wallet screens and shall not offer wallet deduction.
UI-06	Validation and error messages shall be readable and shall not expose sensitive implementation details.

3.2 Hardware Interfaces

Interface	Purpose	Requirement / Model
USB RFID Reader	Captures card identifiers for supported staff workflows.	HW-01–HW-02

ID	Requirement
HW-01	The desktop wrapper shall communicate only with approved USB RFID readers through the selected local serial or device interface.
HW-02	Card-read events shall be parsed, normalized, validated, and passed to the active authorized workflow.
HW-03	Reader timeout, disconnect, unreadable-card, and reconnect conditions shall be handled without corrupting a transaction.

3.3 Software Interfaces

Interface	Purpose	Data Exchanged
ISSRMS Backend API	Provides authenticated services to the web portal and desktop application.	JSON or approved API payloads over authenticated requests.
Zone 1 Read-Only Integration	Retrieves authorized institutional identity and logging records.	Approved identity fields and logging-record fields through API or SELECT-only integration.
Zone 2 Operational Database	Stores project-generated operational and financial records.	Accounts, wallets, ledger entries, canteen, library, notifications, announcements, and audits.

ID	Requirement
SI-01	The web portal and desktop application shall use authenticated backend endpoints.
SI-02	The backend shall access authorized Zone 1 data through the school-approved read-only integration mechanism and shall use a separate least-privilege read/write connection for the Zone 2 operational database.
SI-03	The backend shall reject any attempted create, update, or delete action directed to Zone 1.
SI-04	No software interface to the school's official billing or Statement of Account process shall be implemented.

3.4 Communications Interfaces

ID	Requirement
CI-01	External Student/Parent communication shall use HTTPS through the school-approved firewall and reverse proxy or web server.
CI-02	Protected requests shall include a valid session or token and shall be rejected when expired, revoked, malformed, or unauthorized.
CI-03	Timeouts and temporary network failures shall not produce partial financial, canteen, or library transactions.
CI-04	The Node.js backend, database services, RFID interfaces, backup services, and administration services shall not be directly exposed to the Internet.
CI-05	Client applications shall not connect directly to Zone 1 or Zone 2.

4. System Features

4.1 Authentication and Role-Based Access Control

4.1.1 Description and Priority

The feature authenticates accounts and restricts protected data and actions by approved role, account status, and session state. Priority: High.

4.1.2 Stimulus/Response Sequences

Stimulus / User Action	System Response
A user submits credentials.	The backend verifies the account, password hash, account status, and role permissions.
The credentials and account state are valid.	The system creates an authenticated session or token and displays only permitted modules.
The credentials, session, account state, or requested action are invalid.	The system rejects the request and records the appropriate security event.

4.1.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.1-1	The system shall authenticate every non public user before displaying protected records.	High
REQ-4.1-2	The backend shall enforce role permission on every protected endpoint, regardless of interface visibility.	High
REQ-4.1-3	Credentials shall be compared through an approved password-hash verification procedure.	High
REQ-4.1-4	Invalid, inactive, expired, revoked, or unauthorized requests shall be rejected and logged.	High
REQ-4.1-5	Security audit events shall record the actor or supplied identifier, timestamp, action, result, and available request reference.	High

4.2 RFID Identity and Institutional Logging Records Reference

4.2.1 Description and Priority

The feature validates supported RFID-card or manual identifiers and displays permitted institutional logging records without changing the source. Priority: High.

4.2.2 Stimulus/Response Sequences

Stimulus / User Action	System Response
An authorized user taps a supported RFID card or enters an institutional ID manually.	The desktop application parses the identifier and sends it to the backend for validation.
The identifier is valid and the user requests permitted logging history.	The backend retrieves approved institutional logging records through the Zone 1 read-only integration mechanism.
The identifier or request is unmatched, inactive, unreadable, or unauthorized.	The system displays a clear rejection or error result without altering source records.

4.2.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.2-1	The desktop application shall parse and validate identifiers from supported RFID readers.	High
REQ-4.2-2	Authorized workflows shall permit manual ID entry when the physical card is unavailable, subject to the same validation and role checks.	High
REQ-4.2-3	The system shall return a clear result for matched, unmatched, inactive, unreadable, or unauthorized identifiers.	High
REQ-4.2-4	Institutional logging records shall be retrieved only through the school-approved read-only Zone 1 integration mechanism.	High
REQ-4.2-5	The system shall not create, edit, delete,	High

	mark, or recompute records in the school's existing institutional logging sources.	

4.3 Financial Services and E-Wallet

4.3.1 Description and Priority

The feature manages canteen-only wallet activity through secure, auditable, ACID-compliant transactions. Cash-only top-ups may be processed by the Registrar or authorized Canteen Personnel. Credit-limit management, manual settlement, adjustment, reversal, and complete financial review remain restricted to the Registrar. Priority: High.

4.3.2 Stimulus/Response Sequences

Stimulus / User Action	System Response
The Registrar or authorized Canteen Personnel initiates a confirmed cash-only top-up; the Registrar initiates a credit-limit, settlement, adjustment, or reversal action; and authorized Canteen Personnel initiates a canteen purchase.	The backend verifies the actor, request data, wallet state, limits, and required references.

All validations and ledger-balance checks succeed.	The system creates the financial transaction header, balanced ledger entries, related updates, audit record, and transaction reference in one ACID transaction.
Validation, authorization, database posting, or balance checks fail.	The system rolls back the operation and returns a safe error without finalizing partial records.

4.3.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.3-1	The system shall support confirmed cash-only top-ups processed by the Registrar or authorized Canteen Personnel; canteen purchases processed by Canteen Personnel; approved credit use and credit-limit management by the Registrar; and Registrar-authorized manual credit settlement, certified adjustments, or reversals.	High

REQ-4.3-2	The system shall apply B = T_total − P_total when deriving active spendable cash.	High
REQ-4.3-3	The system shall apply C_avail = C_limit − C_used when deriving remaining approved credit.	High
REQ-4.3-4	The system shall approve a purchase only when P ≤ B + C_avail.	High
REQ-4.3-5	A purchase shall consume active cash first and shall increase credit_used only by the uncovered amount.	High
REQ-4.3-6	credit_limit and credit_used shall be stored as separate decimal fields with approved bounds and defaults.	High
REQ-4.3-7	Top-up, purchase, settlement, adjustment, and reversal operations shall execute inside ACID transactions.	High
REQ-4.3-8	A finalized financial transaction or ledger entry shall not be edited or deleted.	High

REQ-4.3-9	A correction shall create a linked adjustment or reversal that preserves the original entry.	High
REQ-4.3-10	Each finalized financial transaction shall have at least two Financial_Ledger_Entries; the total debit amount shall equal the total credit amount. Financial records shall not contain library_circulation_id or a Library_Fine transaction type.	High
REQ-4.3-11	Manual settlement shall be restricted to the Registrar, require verification through an official receipt, payment reference, or other school-approved payment record, require a reason or reference, set credit_used to zero, and create an audit and settlement record.	High
REQ-4.3-12	The manual settlement process shall be performed according to school policy at each grading or payment period.	High

REQ-4.3-13	Authorized Canteen Personnel may process confirmed over-the-counter cash-only wallet top-ups but shall not change credit_limit, settle credit_used, create adjustments or reversals, or edit finalized	High
REQ-4.3-14	The Registrar shall be able to review itemized canteen transactions contributing to outstanding canteen credit, including items, quantities, total amount, cash-funded amount, credit-funded amount, operator, timestamp, and transaction references.	High

4.4 Canteen Sales and Inventory

4.4.1 Description and Priority

The feature records canteen purchases and related stock movements using authorized item, quantity, pricing, and payment data. Priority: High.

4.4.2 Stimulus/Response Sequences

Stimulus / User Action	System Response

Canteen Personnel identifies a student and selects items and quantities.	The system validates the account, operator, item availability, active prices, stock, and payment support.
The purchase is financially supported and all validations pass.	The backend records the sale, inventory deduction, wallet or credit update, balanced ledger entries, and receipt reference atomically.
Any validation or posting step fails before commit.	The system rejects or rolls back the sale so no partial stock, wallet, sale, or ledger record persists.

4.4.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.4-1	The system shall calculate a sale total from active item prices and quantities.	High
REQ-4.4-2	The system shall reject unavailable items, invalid quantities, inactive accounts, unauthorized operators,	High

	and financially unsupported purchases.	
REQ-4.4-3	The completed sale, item records, inventory deduction, credit update, Financial_Transactions header, and balanced Financial_Ledger_Entries lines shall commit together.	High
REQ-4.4-4	Failure before commit shall roll back all affected canteen and financial changes.	High
REQ-4.4-5	A successful sale shall display or store a unique transaction reference suitable for audit and receipt review.	High

4.5 Library Catalog, Circulation, Inventory, and Fines

4.5.1 Description and Priority

The feature manages library operations independently of the canteen e-wallet and financial ledger. Priority: High.

4.5.2 Stimulus/Response Sequences

Stimulus / User Action	System Response

Library Personnel creates or updates permitted catalog, inventory, borrowing, return, due-date, or fine information.	The backend validates authorization and records the approved library change with an audit trail when required.
A student or parent opens the portal.	The system displays only the authorized student's permitted catalog, circulation, and fine information.
The Registrar requests unpaid fine information.	The system displays the fine amount and status for reference only and provides no fine-management controls.

4.5.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.5-1	Authorized Library Personnel shall manage permitted catalog and inventory records.	High
REQ-4.5-2	The system shall validate and record borrowing, return, due-date, and circulation status changes.	High

REQ-4.5-3	Library_Circulation shall contain fine_amount and fine_status, where fine_status is No_Fine, Unpaid, or Paid.	High
REQ-4.5-4	Authorized Library Personnel may adjust, reduce, waive, or settle a library fine with an audit record.	High
REQ-4.5-5	A library fine shall not create a wallet debit, change credit_used, or create a financial transaction or ledger entry.	High
REQ-4.5-6	The Registrar may view a student’s unpaid library-fine amount, status, and related circulation activity for informational purposes only. The view may include the material reference, circulation reference, borrow date, due date, return date, overdue duration, and	High

	latest library action. Only authorized Library Personnel may assess, adjust, reduce, waive, and settle library fines.	

4.6 Student and Parent Web Portal

4.6.1 Description and Priority

The feature provides role-restricted, primarily read-only access to approved student account records through supported web browsers. Priority: High.

4.6.2 Stimulus/Response Sequences

Stimulus / User Action	System Response
A student or authorized parent signs in through the portal.	The backend authenticates the account and returns only records associated with the authorized student.
The user selects a permitted wallet, canteen, library, logging, notification, or announcement view.	The system displays the authorized read-only information.
The user attempts a protected staff action.	The system rejects the action and does not display or process the protected workflow.

4.6.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.6-1	The portal shall display only accounts and records the authenticated user is authorized to view.	High
REQ-4.6-2	The portal may display active wallet balance, canteen credit used, wallet transaction history, canteen purchase history, library catalog/circulation/fine records, permitted logging records, notifications, and announcements.	High
REQ-4.6-3	The portal shall not permit protected staff actions such as top-up, credit-limit change, settlement, adjustment, reversal, catalog change, or fine change.	High

4.7 School Operations Desktop Application

4.7.1 Description and Priority

The feature provides authorized staff workflows through a locally installed Electron application and a controlled bridge to supported RFID-reader hardware. Priority: High.

4.7.2 Stimulus/Response Sequences

Stimulus / User Action	System Response
An authorized staff user signs in to the desktop application.	The application displays only workflows permitted by the authenticated role.
A supported RFID reader produces a card-read event.	Electron passes only the required local hardware event to the approved workflow, and the backend revalidates the protected action.
The renderer or user attempts unauthorized operating-system, file-system, database, secret, or protected-module access.	The application blocks the access and the backend rejects unauthorized protected requests.

4.7.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.7-1	The desktop application shall present only workflows allowed by the	High

	authenticated role. The Administrator role shall be limited to Zone 2 account and role management, approved non-financial settings, announcements, read-only Institutional Logging Records, and read-only system-wide Audit Logs.	
REQ-4.7-2	Electron shall function as the desktop wrapper and shall expose only the local hardware functions required by the approved RFID workflow.	High
REQ-4.7-3	The renderer shall not receive unrestricted file-system, operating-system, database, or secret access.	High
REQ-4.7-4	Every protected action shall be revalidated by the backend before processing.	High
REQ-4.7-5	The Registrar interface may display unpaid library-fine amounts, statuses, and	High

	related circulation activity for informational purposes but shall not provide controls for assessing, adjusting, reducing, waiving, paying, or settling those fines.	
REQ-4.7-6	Only authenticated Administrator accounts shall access the system-wide Audit Logs interface. Audit records shall be read-only and shall not provide edit or delete controls.	High

4.8 Notifications and Announcements

4.8.1 Description and Priority

The feature displays permitted operational notifications and school announcements according to role permissions. Priority: Medium.

4.8.2 Stimulus/Response Sequences

Stimulus / User Action	System Response
An approved library, wallet, or system event occurs.	The system creates or displays the corresponding permitted notification for the affected account or role.

Authorized personnel publish or update an internal announcement.	The backend validates the role and makes the announcement visible to the approved audience.
A user opens the notification or announcement view.	The system returns only records permitted for that user and role.

4.8.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.8-1	The system shall support notifications for approved library, wallet, and system events.	Medium
REQ-4.8-2	Authorized personnel shall be able to publish and manage internal announcements according to role permissions.	Medium
REQ-4.8-3	Internal announcements shall use Draft, Active, or Inactive status.	Medium
REQ-4.8-4	Only Active announcements shall be	Medium

	displayed to their permitted audience.	
REQ-4.8-5	A previously published announcement shall be deactivated rather than permanently deleted when its history must be preserved.	Medium

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement
REQ-5.1-1	The platform shall support concurrent use by the operational desktop workstations configured for deployment and by authenticated Student/Parent Portal sessions represented during system and acceptance testing, without service failure, unauthorized data exposure, or inconsistent records.
REQ-5.1-2	Login, record lookup, library, canteen, wallet, and logging-reference workflows shall be tested on the approved deployment infrastructure, and their observed response times shall be documented in the final test results.
REQ-5.1-3	Point-of-sale processing and financial-ledger commits shall either complete successfully as one atomic transaction or execute a complete rollback without leaving partial or inconsistent records.

5.2 Safety Requirements

These requirements protect against financial inconsistency, data loss, incomplete posting, and unrecoverable operational records caused by software, network, hardware, or database failure.

ID	Requirement
REQ-5.2-1	Financial and canteen posting shall be atomic: partial sale, stock, wallet, or ledger changes shall not persist.
REQ-5.2-2	Database constraints and transaction logic shall preserve valid balances, credit limits, transaction references, and entity relationships.
REQ-5.2-3	Concurrent transactions affecting the same wallet or stock item shall be isolated according to the approved locking or transaction strategy.
REQ-5.2-4	Committed transaction and audit records shall remain durable after process or network failure.
REQ-5.2-5	Operational backups shall run at approved frequency and be retained for approved period

REQ-5.2-6	Restore procedures shall be tested before deployment and final system acceptance.

5.3 Security and Privacy Requirements

ID	Requirement
REQ-5.3-1	Clear-text passwords shall never be stored.
REQ-5.3-2	Protected records and functions shall require authentication and authorization. The Administrator role shall be authorized only for account and role management, approved non-financial settings, announcements, read-only Institutional Logging Records, and read-only system-wide Audit Logs; it shall not execute financial, canteen, or library operational transactions.
REQ-5.3-3	Input values shall be validated and parameterized before database use.

REQ-5.3-4	Financial actions shall record actor, timestamp, action, wallet or account reference, amount, reason when applicable, result, and transaction reference.
REQ-5.3-5	Zone 1 institutional identity and logging data shall be accessed only through one school-approved read-only integration mechanism. The selected method may use authenticated read-only API endpoints, including an approved adapter API, or a dedicated least-privilege database connection, depending on the existing architecture and access approved by the school. Only one method shall be implemented after school-approved technical validation.
REQ-5.3-6	Credentials, service accounts, database accounts, or access tokens used for Zone 1 integration shall be granted only the minimum permissions required to retrieve approved identity and logging records.
REQ-5.3-7	Access to the Zone 2 operational database shall use a separate least privilege service account with only the permissions required by the application.

REQ-5.3-8	Secrets shall not be embedded in client code, logs, exported data, or error messages.
REQ-5.3-9	Sessions shall expire or be revoked according to the approved policy.
REQ-5.3-10	The system shall follow the school's approved privacy, retention, incident-response, and access-review policies and applicable Philippine data-protection requirements.

5.4 Software Quality Attributes

ID	Requirement
REQ-5.4-1	Labels, confirmations, validation messages, and transaction references shall be consistent across the web and desktop interfaces.
REQ-5.4-2	Modules shall expose clear interfaces and be testable independently.
REQ-5.4-3	Temporary RFID-reader, network, or Zone 1 unavailability shall be handled without corrupting Zone 2 records.
REQ-5.4-4	The release shall satisfy the approved functional test cases, security requirements, response-time requirements, and accepted ISO/IEC 25010 evaluation criteria.

5.5 Business Rules

ID	Business Rule
BR-01	Confirmed cash-only wallet top-ups may be processed by the Registrar or authorized Canteen Personnel.
BR-02	Library fines remain in the Library Management Module and are never automatically or manually deducted through the wallet.
BR-03	Finalized Financial_Transactions and Financial_Ledger_Entries are immutable.
BR-04	A financial correction requires a certified adjustment or reversal entry.
BR-05	A purchase uses active cash before increasing credit_used and shall not exceed B + C_avail.
BR-06	Manual Settle-to-Zero is restricted to the Registrar and requires verification through an official receipt, payment reference, or other

	school-approved record that the outstanding canteen credit has been paid.
BR-07	Manual settlement shall be performed by the Registrar in accordance with school policy at every grading or payment period and shall create an auditable record.
BR-08	Library fine assessment, adjustment, reduction, waiver, and settlement are restricted to authorized Library Personnel and shall remain auditable.
BR-09	Institutional identity and logging records are read-only references.
BR-10	ISSRMS has no official billing or Statement of Account function or interface.
BR-11	The Registrar may view a student's unpaid library fine amount and status for informational purposes only. Assessment, adjustment, reduction, waiver, and settlement of library fines shall

	remain restricted to authorized Library Personnel and shall remain auditable.
BR-12	Credit-limit management, manual settlement, adjustment, and reversal actions remain restricted to the Registrar.
BR-13	The system-wide Audit Logs interface is restricted to Administrator accounts and is read-only.
BR-14	Internal announcements shall use Draft, Active, or Inactive status. Previously published announcements shall be deactivated rather than permanently deleted when retention is required.

6. Other Requirements

6.1 Data Requirements

ID	Requirement
DATA-01	E_Wallet shall include wallet_id, nullable account_id for controlled system-owned wallets, wallet_type, credit_limit, credit_used, and last_updated. wallet_type shall support Student, Cash_Collection, Canteen_Revenue, Credit_Receivable, and Adjustment_Reserve. Active cash balance shall be derived from Financial_Ledger_Entries.
DATA-02	Financial_Transactions shall include financial_transaction_id, transaction_type, transaction_code, optional supporting_reference, optional reversal_of_transaction_id, initiated_by_account_id, transaction_status, optional reason, created_at, and optional posted_at.
DATA-03	Financial_Ledger_Entries shall include ledger_entry_id, financial_transaction_id, wallet_id, entry_side, amount, and posted_at. Each finalized transaction shall contain at least two entries with equal debit and credit totals.
DATA-04	Financial transaction types shall be limited to the approved canteen wallet actions and shall exclude Library_Fine.
DATA-05	Library_Circulation shall include circulation identifiers, account and material references, borrowing, due, and return data, status, fine_amount, and fine_status.

DATA-06	The Zone 1 integration shall provide a logical representation of authorized Institutional Logging Records. Approved log_type values shall include Gate_Entry, Gate_Exit, Library_Entry, and Library_Exit. Approved location values shall include Gate_1, Gate_2, and Library.
DATA-07	Retention periods for wallet, ledger, sales, inventory, library, announcement, audit, authentication, notification, and cached reference data shall be defined in the approved retention policy.
DATA-08	System_Announcements shall include announcement status, creation timestamp, optional publication timestamp, optional update timestamp, and permitted audience scope.

6.2 Verification and Acceptance

ID	Requirement
ACC-01	The approved functional and nonfunctional requirements shall be verified through applicable system, integration, security, and acceptance tests, with the results documented in the project testing records.
ACC-02	Financial tests shall prove ACID rollback, cash-first

	credit logic, credit-limit enforcement, transaction-header and ledger-entry immutability, debit-credit equality for every finalized transaction, certified corrections, manual-settlement authorization, payment verification, and audit completeness.
ACC-03	Isolation tests shall prove that Zone 1 is read-only and that library fine actions do not change wallet or ledger data.
ACC-04	Security tests shall cover authentication, RBAC, endpoint authorization, session handling, input validation, secret exposure, and restricted integration, service-account, access-token, and database privileges.
ACC-05	Stakeholder acceptance shall use approved scenarios and record the expected result, actual result, evidence, tester, date, and status.

The prepared scenarios in Chapter IV Section 4.7.1 provide the initial test-case identifiers for ACC-01 through ACC-05. Actual test results and supporting test records shall be completed only after implementation and approved testing.

6.3 Legal, Policy, and Localization Requirements

ISSRMS shall follow the school's approved privacy, retention, financial-authorization, backup, incident response, and access-review policies, as well as applicable Philippine data-protection requirements. The primary interface language shall be English unless additional school-approved language support is added through change control.

Appendix A: Glossary

Term	Definition
ACID	Atomicity, Consistency, Isolation, and Durability properties applied to database transactions.
Adjustment/Reversal	A new certified financial transaction with balanced ledger-entry lines that corrects a prior finalized transaction without editing or deleting it.
C_avail	Remaining approved canteen credit, calculated as C_limit − C_used.

Double-Entry	An accounting method in which every finalized transaction produces equal debit and credit entries across at least two ledger lines.
Financial Services Module	The Financial Services Module is overseen by the Registrar. The Registrar or authorized Canteen Personnel may process cash-only top-ups, while credit management, manual settlement, adjustment, reversal, and complete financial review remain restricted to the Registrar. The Administrator has no financial, canteen, or library operational controls.
Immutable Ledger	An append-only financial record in which finalized entries cannot be directly edited or deleted.
Institutional Logging Records	Authorized records produced by the school's existing logging infrastructure and displayed by ISSRMS in read-only form. Approved log types are Gate Entry, Gate Exit, Library Entry, and Library Exit. Approved locations are Gate 1, Gate 2, and Library.
Registrar	The authorized ISSRMS actor responsible for operating the Financial

	Services Module. The Registrar may view unpaid library fine information for reference but may not assess, adjust, reduce, waive, or settle library fines.
School Operations Desktop Application	The staff application; Electron is its controlled desktop wrapper for supported local hardware communication.
Settle-to-Zero	An authorized manual action that resets canteen credit_used after payment verification and records the settlement and audit event.
Zone 1 Read-Only Integration	The school-authorized mechanism for retrieving permitted institutional identity and logging records. Depending on the approved architecture, the selected method may use authenticated read-only API endpoints, including an approved adapter API, or a restricted database connection limited to approved SELECT operations. Only one method is implemented after school-approved technical validation.

Appendix B: Analysis Models

Model ID	Analysis Model
SRS-B1	Revised UML Use Case Diagram using Registrar as the financial actor.
SRS-B2	Context Diagram / Data Flow Diagram Level 0.
SRS-B3	Data Flow Diagram Level 1.
SRS-B4	Entity-level ERD showing only Zone 2 operational entities and their relationships; Zone 1 remains documented in the architecture and integration descriptions.
SRS-B6	Authentication and Role-Based Access Control Activity Diagram.
SRS-B7	Financial Services Activity Diagram showing the Registrar, balanced ledger posting, ACID commit or rollback, and audit events.
SRS-B8	Institutional Logging Records Reference Activity Diagram.

APPENDIX D CURRICULUM VITAE



CONTACT INFORMATION

1. Name

Kyle Christian A. Zenarosa

2. Address

BLK 9 Lot 27 Lily Cor Dama De Noche ST Camella Homes Pajac, Lapu- Lapu City, Cebu

3. Telephone number – Office & Cell/Mobile

09310010340

4. Email address – Office & Personal/Private

kczenarosa2003@gmail.com

PERSONAL INFORMATION

1. Date of Birth

07/10/2003

2. Age

22

3. Citizenship/Nationality

Filipino

4. Gender

Male

5. Marital Status (Single / Married / Divorce)

Single

6. Number of Children & Age

NA

7. Language proficiency

English - Verbal and Written

Cebuano - Verbal and Written

Filipino - Verbal and Written

8. Computer PMS and Software Skills

- Programming Languages: JavaScript, Python, C, SQL
- Web Development:
    - React,
    - HTML5,
    - CSS3,
    - Tailwind CSS,
    - Node.js,
    - Express.js,
    - Vite
- Mobile Application Development:
    - .NET MAUI
    - React Native
    - Android Studio
    - XML
- Databases:
    - MySQL
    - PostgreSQL
    - SQLite
    - Firebase
    - Supabase
- Database Management Tools: phpMyAdmin, XAMMP
- Cloud & Deployment:
    - Docker
    - Vercel
    - AWS (EC2, S3, Lambda)
- Game Development: Godot Engine
- Version Control: Git, GitHub

- Networking:
    - Cisco Packet Tracer
    - TCP/IP
    - IP Addressing
    - Subnetting
    - Routing & Switching
    - VLAN Configuration
    - Inter-VLAN Routing
    - Firewall Configuration
    - Network Troubleshooting
    -
- Cybersecurity & Virtualization:
    - Penetration Testing Fundamentals
    - Linux System Administration
    - Security Hardening
    - VirtualBox
    - VMware

- API Development & Testing: REST APIs, Postman
- UI/UX Design: Figma
- Productivity Software:
    - Microsoft Office (Word, Excel, PowerPoint)
    - Google Workspace (Docs, Sheets)
- Operating Systems: Windows, Linux

EDUCATIONAL BACKGROUND

Education Level	Year	School
Tertiary	2022 – Present	University of San Carlos, Cebu City, Philippines Bachelor of Science in Information Technology
Secondary (Senior High School)	*2022*	*Saint Joseph School Of Mactan*
Secondary (Junior High School)	*2020*	*Saint Joseph School Of Mactan*
Elementary	2016	*Saint Joseph School Of Mactan*

2. Certifications & Accreditations

Zuitt Learning Institute – Certificate of Completion, Intensive Tech Career Program (Python, Django, AWS) (May 2025)
Cisco Networking Academy – CCNA: Introduction to Networks
Cisco Networking Academy – CCNA: Switching, Routing, and Wireless Essentials
Cisco Networking Academy – Linux Essentials
Cisco Networking Academy – CyberOps Associate

STRENGTHS/TRAITS & SKILLS

- High degree of initiative and willingness to learn
- Hands-on experience in:
    - Full-stack Web Development
    - Mobile Application Development
    - Game Development
    - Database Design & Management
    - Computer Networking & System Configuration
    - Network Security & Penetration Testing
- Strong analytical and problem-solving skills
- Strong presentation and communication skills
- Good interpersonal and teamwork skills
- Able to work effectively under pressure and within tight schedules

CAREER OBJECTIVES

To obtain a position in software development or cybersecurity where I can apply my technical knowledge, continuously develop my skills, and contribute to building secure, reliable, and innovative technology solutions.

Resume updated on 22/07/2026

CONTACT INFORMATION



1. Name

Kyle Jayverson D. Baguio

2. Address

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09624505778

4. Email address – Office & Personal/Private

kylebaguio1803@gmail.com

PERSONAL INFORMATION

1. Date of Birth

06/18/2003

2. Age

23

3. Citizenship/Nationality

Filipino

4. Gender

Male

5. Marital Status (Single / Married / Divorce)

Single

6. Number of Children & Age

NA

7. Language proficiency

English - Verbal and Written

Cebuano - Verbal and Written

Filipino - Verbal and Written

8. Computer PMS and Software Skills
    - Programming Languages: JavaScript, Python, C, SQL
    - Web Development:
        - React,
        - HTML5,
        - Tailwind CSS,
        - Node.js,
    - Mobile Application Development:
        - .NET MAUI
        - Android Studio
        - XML
    - Databases:
        - MySQL
        - PostgreSQL
        - SQLite
        - Firebase
        - Supabase
    - Database Management Tools: phpMyAdmin, XAMMP
    - Cloud & Deployment:
        - Docker
        - Vercel
        - AWS (EC2, S3, Lambda)
    - Game Development: Godot Engine
    - Version Control: Git, GitHub
    - Networking:
        - Cisco Packet Tracer
        - TCP/IP
        - IP Addressing
        - Subnetting
        - Routing & Switching
        - VLAN Configuration

    - Inter-VLAN Routing
    - Firewall Configuration
    - Network Troubleshooting
- API Development & Testing: REST APIs, Postman
- UI/UX Design: Figma
- Productivity Software:
    - Microsoft Office (Word, Excel, PowerPoint)
    - Google Workspace (Docs, Sheets)
- Operating Systems: Windows, Linux

EDUCATIONAL BACKGROUND

Education Level	Year	School
Tertiary	2022 – Present	University of San Carlos, Cebu City, Philippines Bachelor of Science in Information Technology
Secondary (Senior High School)	*2022*	*University of San Carlos*
Secondary (Junior High School)	*2020*	*Saint Joseph School Of Mactan*
Elementary	2016	*Saint Joseph School Of Mactan*

2. Certifications & Accreditations

Zuitt Learning Institute – Certificate of Completion, Intensive Tech Career Program (Python, Django, AWS) (May 2025)
Technical Education and Skills Development Authority (TESDA) – Computer Systems Servicing NC II (February 2025)

STRENGTHS / TRAITS & SKILLS

- Willing to learn new skills and technologies
- Experience in:
    - Full-Stack Web Development
    - Mobile Application Development
    - Game Development
    - Database Management
    - Computer Networking and System Setup
- Good problem-solving and critical thinking skills
- Able to communicate clearly and work well with others
- Can work independently or as part of a team
- Responsible and able to finish tasks on time
- Can work well under pressure and meet deadlines
- Flexible and quick to adapt to new challenges

CAREER OBJECTIVES

To start my career in software development, cybersecurity, or computer networking, where I can apply my knowledge and skills, gain hands-on experience, and contribute to the success of the organization while continuing to learn and grow as an IT professional.

Resume updated on 22/07/2026

CONTACT INFORMATION



1. Name

Jay Nino Clark B. Basilgo

2. Address

Earnestine Homes Babag II, Lapu-Lapu City Cebu

3. Telephone number – Office & Cell/Mobile

09478525031

4. Email address – Office & Personal/Private

clarkbasilgo@gmail.com

PERSONAL INFORMATION

1. Date of Birth

01/21/2002

2. Age

24

3. Citizenship/Nationality

Filipino

4. Gender

Male

5. Marital Status (Single / Married / Divorce)

Single

6. Number of Children & Age

NA

7. Language proficiency

English - Verbal and Written

Cebuano - Verbal and Written

Filipino - Verbal and Written

8. Computer PMS and Software Skills
    - Programming Languages: JavaScript, Python, C, SQL
    - Web Development:
        - React,
        - HTML5,
        - ASP.NET Core Web API
        - Node.js,
    - Mobile Application Development:
        - .NET MAUI
        - Android Studio
        - XML
    - Databases:
        - MySQL
        - PostgreSQL
        - SQLite
        - Firebase
        - Supabase
    - Database Management Tools: phpMyAdmin, XAMMP
    - Cloud & Deployment:
        - Docker
        - Vercel
        - AWS (EC2, S3, Lambda)
    - Game Development: Godot Engine
    - Version Control: Git, GitHub
    - Networking:
        - Cisco Packet Tracer
        - TCP/IP
        - IP Addressing
        - Subnetting
        - Routing & Switching
        - VLAN Configuration

    - Inter-VLAN Routing
    - Firewall Configuration
    - Network Troubleshooting
- API Development & Testing: REST APIs, Postman
- UI/UX Design: Figma
- Productivity Software:
    - Microsoft Office (Word, Excel, PowerPoint)
    - Google Workspace (Docs, Sheets)
- Operating Systems: Windows, Linux

EDUCATIONAL BACKGROUND

Education Level	Year	School
Tertiary	2022 – Present	University of San Carlos, Cebu City, Philippines Bachelor of Science in Information Technology
Secondary (Senior High School)	*2021*	*Science and Technology Education Center*
Secondary (Junior High School)	*2020*	*Science and Technology Education Center*
Elementary	2016	*Science and Technology Education Center*

2. Certifications & Accreditations

Cisco Networking Academy – Linux Essentials
Cisco Networking Academy – CyberOps Associate
Cisco Networking Academy – Introduction to CyberSecurity

STRENGTHS / TRAITS & SKILLS

- Willing to learn new skills and technologies
- Experience in:
    - Full-Stack Web Development
    - Mobile Application Development
    - Game Development
    - Database Management
    - Computer Networking and System Setup
- Good problem-solving and critical thinking skills
- Able to communicate clearly and work well with others
- Can work independently or as part of a team
- Responsible and able to finish tasks on time
- Can work well under pressure and meet deadlines
- Flexible and quick to adapt to new challenges

CAREER OBJECTIVES

To obtain a position in networking or cybersecurity where I can apply my technical knowledge, continuously develop my skills, and contribute to building secure, reliable, and innovative technology solutions.

Resume updated on 22/07/2026

CONTACT INFORMATION



1. Name

Joel June Fabillar

2. Address

13A Borbajo St. Talamban, Cebu City

3. Telephone number – Office & Cell/Mobile

09981660377

4. Email address – Office & Personal/Private

joeljunefabillar@gmail.com

PERSONAL INFORMATION

1. Date of Birth

06/29/2004

2. Age

22

3. Citizenship/Nationality

Filipino

4. Gender

Male

5. Marital Status (Single / Married / Divorce)

Single

6. Number of Children & Age

NA

7. Language proficiency

English - Verbal and Written

Cebuano - Verbal and Written

Filipino - Verbal and Written

Chinese - Sign Language

8. Computer PMS and Software Skills
    - Programming Languages: JavaScript, Python, C, SQL
    - Web Development:
        - React,
        - HTML5,
        - CSS3,
        - Tailwind CSS,
        - Node.js,
        - Express.js
    - Mobile Application Development:
        - .NET MAUI
        - Android Studio
        - XML
    - Databases:
        - MySQL
        - PostgreSQL
        - SQLite
        - Firebase
        - Supabase
    - Database Management Tools: phpMyAdmin, XAMMP
    - Cloud & Deployment:
        - Docker
        - Vercel
        - AWS (EC2, S3, Lambda)
    - Game Development: Godot Engine
    - Version Control: Git, GitHub
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    - Routing & Switching
    - VLAN Configuration
    - Inter-VLAN Routing
    - Firewall Configuration
    - Network Troubleshooting
- API Development & Testing: REST APIs, Postman
- UI/UX Design: Figma
- Productivity Software:
    - Microsoft Office (Word, Excel, PowerPoint)
    - Google Workspace (Docs, Sheets)
- Operating Systems: Windows, Linux

EDUCATIONAL BACKGROUND

Education Level	Year	School
Tertiary	2022 – Present	University of San Carlos, Cebu City, Philippines Bachelor of Science in Information Technology
Secondary (Senior High School)	*2022*	*Cebu Mary Immaculate College*
Secondary (Junior High School)	*2020*	*Cebu Mary Immaculate College*
Elementary	2016	*Talamban Elementary School*

2. Certifications & Accreditations

Zuitt Learning Institute – Certificate of Completion, Intensive Tech Career Program (Python, Django, AWS) (May 2025)
Technical Education and Skills Development Authority (TESDA) – Computer Systems Servicing NC II (February 2025)

STRENGTHS / TRAITS & SKILLS

- Eager to acquire new knowledge and continuously improve technical skills
- Knowledge of web application design and development
- Familiar with mobile app development concepts
- Basic understanding of game design and development
- Experienced in handling and organizing databases
- Knowledgeable in computer networking, system configuration, and troubleshooting
- Strong analytical and decision-making abilities
- Effective verbal and written communication skills
- Works efficiently both independently and in collaborative environments
- Dependable with a strong commitment to completing assigned responsibilities
- Performs effectively in fast-paced and deadline-driven situations
- Adaptable to changing technologies, tasks, and work environments

CAREER OBJECTIVES

To obtain a position as a Game Developer where I can apply my programming knowledge, creativity, and problem-solving skills while continuously learning and contributing to the development of engaging and high-quality games.

Resume updated on 22/07/2026

APPENDIX E

NON-DISCLOSURE AND CONFIDENTIALITY AGREEMENT

ISSRMS Capstone Project

Integrated School Services and Record Management System (ISSRMS)

for Saint Joseph School of Mactan

This Non-Disclosure and Confidentiality Agreement (the “Agreement”) is entered into by and between Saint Joseph School of Mactan (the “School” or “Disclosing Party”) and the BSIT Capstone Group of the University of San Carlos, composed of Kyle Christian A. Zenarosa, Kyle Jayverson Baguio, Jay Niño Clark Basilgo, and Joel June Fabillar (the “Capstone Group” or “Receiving Party”). The School and the Capstone Group are collectively referred to as the “Parties.”

The Parties are undertaking the Integrated School Services and Record Management System (ISSRMS) capstone project. In the course of requirements gathering, analysis, design, development, testing, documentation, demonstration, and evaluation, the School may make limited disclosures of non-public institutional, technical, operational, security, and financial information to the Capstone Group.

These disclosures are made solely to support the approved software development activities. The Capstone Group agrees to use the information only for those activities and to protect it in accordance with this Agreement.

NOW, THEREFORE, the Parties agree as follows:

1. Purpose and Project Scope

This Agreement applies to information and activities reasonably necessary for the approved ISSRMS capstone project and does not expand the Capstone Group’s access rights.

The School is not required to provide production data or unrestricted access to existing systems. Access, if approved, must be limited to the minimum information and system functions necessary for the task, for the approved period, and through the School-approved access method.

2. Confidential Information

"Confidential Information" means non-public information disclosed by or on behalf of the School in connection with the project, whether spoken, written, displayed, stored, or transmitted electronically. It includes, without limitation:

- institutional identity information, student identifiers, contact details, and authorized entry or exit logging records;
- system workflows, business rules, operating procedures, architecture, interfaces, source code, credentials, tokens, network details, and security controls;

- database schemas, data dictionaries, sample or production data, exports, backups, audit records, and integration information;
- canteen sales, inventory, e-wallet, credit, settlement, adjustment, reversal, and immutable ledger information;
- library catalog, circulation, inventory, fine amount, and fine-status information;
- screenshots, wireframes, reports, forms, templates, meeting notes, and other administrative or technical material not publicly released by the School; and
- any information marked confidential or that a reasonable person would understand to be confidential because of its content or the circumstances of disclosure.

3. Security and Confidentiality Obligations

The Receiving Party shall:

- keep Confidential Information strictly confidential and protect it using reasonable organizational, physical, and technical safeguards;
- access and use Confidential Information only for approved ISSRMS tasks and only to the extent necessary for those tasks;
- limit access to the named Capstone Group members when access is necessary and authorized;
- not upload Confidential Information to public repositories, public file-sharing services, personal cloud storage, or other unapproved third-party systems;
- not use identifiable student or staff data in demonstrations, screenshots, presentations, manuscripts, or other academic outputs when de-identified or synthetic data is sufficient;
- store credentials, exports, screenshots, and other sensitive material securely; never share passwords, access tokens, or database credentials;
- not copy, reproduce, publish, sell, distribute, or disclose Confidential Information to a third party without the School's prior written consent, except as permitted by Section 4; and
- notify the School without undue delay after discovering or reasonably suspecting unauthorized access, use, loss, disclosure, or compromise of Confidential Information.

4. Permitted Use and Required Disclosures

Confidential Information may be used only for the approved ISSRMS capstone activities described in Section 1. The Receiving Party may include de-identified, redacted, or synthetic material in capstone manuscripts, defense materials, demonstrations, and other academic submissions when the material is necessary and does not identify a person or expose a protected School system.

Any proposed use of identifiable data, production screenshots, School credentials, or other material that could expose the School or a data subject requires the School's prior written approval. If disclosure is required by law or a valid government order, the

Receiving Party shall, when legally permitted, notify the School before disclosure and disclose only the portion legally required.

5. Exclusions

Confidential Information does not include information that the Receiving Party can demonstrate: (a) was lawfully known before disclosure without a confidentiality duty; (b) becomes publicly available through no breach of this Agreement; (c) is independently developed without using or referring to Confidential Information; or (d) is lawfully received from a third party without a confidentiality duty owed to the School.

6. Return, Destruction, and Retention

Upon completion of the project, termination of access, or the School's written request, the Receiving Party shall promptly return or securely destroy Confidential Information and all copies, extracts, exports, and credentials within its possession or control. Upon request, the Receiving Party shall confirm completion in writing.

The Receiving Party may retain only the minimum de-identified or redacted material required for University of San Carlos academic evaluation or archival purposes, provided the School has approved the retention in writing and the retained material remains protected by this Agreement.

7. Ownership and No License

The School retains all rights and ownership in its existing systems, institutional records, workflows, documentation, databases, forms, reports, and other Confidential Information. This Agreement grants no ownership interest, license, or right to commercialize School systems or data, except for the limited, revocable right to use approved information for the purpose stated in Section 1.

Ownership, licensing, publication, and permitted reuse of the newly developed ISSRMS application and project deliverables, if applicable, must be addressed in a separate written agreement. This Agreement does not decide those matters.

8. ISSRMS Data and System Boundaries

The Parties acknowledge the following project boundaries, which do not expand the Receiving Party's access rights:

- Institutional identity and logging sources are read-only references. ISSRMS must not create, edit, delete, mark, or recompute records in those institutional sources.
- Project-generated records are stored in the separate ISSRMS operational database. Any cross-zone identity reference is application-validated and does not grant a database write privilege to institutional sources.

- The financial services scope is limited to approved canteen e-wallet operations, including cash top-ups, purchases, approved credit, manual settlement, adjustments, reversals, and balanced immutable ledger records.
- Library circulation and library fines remain operationally separate from the canteen e-wallet and financial ledger. Library fines must not be deducted from wallet balances or posted as wallet transactions.
- ISSRMS does not generate, modify, integrate with, or write to the School's official billing or Statement of Account system.

9. Personal Data and Privacy

When Confidential Information contains personal data, the Parties shall handle it in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173), its implementing rules, applicable regulations, and School or University policies. The Receiving Party shall:

- process personal data only on the School's documented instructions and only for the approved project purpose;
- apply access control, secure authentication, least-privilege access, appropriate encryption or protected storage, and secure deletion procedures;
- avoid collecting or retaining more personal data than the approved task requires; and
- notify the School without undue delay of an actual or suspected personal-data incident and cooperate with reasonable containment, investigation, and remediation steps.

This Agreement does not by itself authorize data sharing, production access, or deployment. Those activities require the School's separate written approval and any additional data-sharing, data-processing, or institutional agreement required by the School or University.

10. Term and Survival

This Agreement takes effect on the date of the last signature and remains in effect for the duration of the project and for _____ (___) year(s) thereafter, unless terminated earlier by written agreement of the Parties. The obligations covering trade secrets, security credentials, personal data, and non-public institutional records survive expiration or termination for as long as the information remains protected by law, School policy, or its confidential nature.

11. Governing Law and Dispute Resolution

This Agreement is governed by the laws of the Republic of the Philippines. The Parties shall first attempt in good faith to resolve any dispute through discussion between their authorized representatives before seeking any remedy available under applicable law.

12. General Provisions

Entire Agreement. This Agreement contains the Parties' understanding regarding confidentiality for the project and supersedes prior discussions on that subject.

Amendments. Any amendment must be in writing and signed by authorized representatives of the Parties.

Severability. If a provision is held invalid or unenforceable, the remaining provisions remain effective.

No Waiver. Failure to enforce a provision is not a waiver of that provision or any other provision.

Independent Relationship. Nothing in this Agreement creates employment, partnership, joint venture, or agency between the Parties.

No Obligation to Disclose. The School is not obligated to disclose any particular information or to proceed with development, deployment, or implementation of ISSRMS.

13. Notices

Notices under this Agreement shall be in writing and delivered personally, by email, or by registered mail to the addresses stated below, or to another address later designated in writing by a Party.

School notice address/email: _____

Capstone Group notice address/email: _____

IN WITNESS WHEREOF, the Parties have executed this Agreement on the date and at the place first written above.

FOR THE DISCLOSING PARTY

SAINT JOSEPH SCHOOL OF MACTAN

Authorized representative	
Position	
Signature	
Date	

FOR THE RECEIVING PARTY

BSIT CAPSTONE GROUP

UNIVERSITY OF SAN CARLOS

Proponent	Signature	Date
Kyle Christian A. Zenarosa		
Kyle Jayverson Baguio		
Jay Niño Clark Basilgo		
Joel June Fabillar		

WITNESSES

Role	Printed name	Signature / Date
Disclosing Party witness		
Receiving Party witness		